QRify: A CLOUD-BASED HOTEL AND RESTAURANT MANAGEMENT SYSTEM FOR ANAHAW ISLAND VIEW RESORT

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by

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A.M.D

J.S.B.D

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EXECUTIVE SUMMARY

The primary goal of this study is to create a hotel and restaurant management system to optimize operations and elevate guest experience using an online web application. The system aims to deliver efficient and dependable services, allowing customers to make room reservations, view menus, place orders, and access essential information about the hotel and restaurant. The system will provide all the clients with features that will assist them and upgrade their traditional way of reserving rooms and ordering food. It is also accompanied by the admin system to manage the information and be accountable for every user's information. It is also utilized to meet customers' requirements for the ISO 25010 criteria. To conduct this study, they prepared eight (8 categories of research questions and employed the survey questionnaire. Fifty people (50) participated in the study, including IT experts, administrators, staff, and customers.

The evaluation results show the system's effectiveness as it functions according to the proposed and suggested features. Functional Suitability got an overall mean of 3.97 which is described as "Very Good," Reliability with an overall mean of 3.85 which is described as "Very Good," Portability with an overall mean of 4.07, which was described as "Very

Good," Usability with overall mean 3.97 which described as "Very Good", Performance Efficiency with an overall mean of 3.94 which described as "Very Good" Security with an overall mean of 4.04 which described as "Very Good" Compatibility with an overall mean of 4.01 which described as "Very Good" and lastly Maintainability with an overall mean of 4.04 which described as "Very Good".

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INTRODUCTION

Project Context

The primary objective of the hotel and restaurant management system was to ensure streamlined operations and enhance guest experience through an online web application. This system provided efficient and reliable services to customers, enabling them to book rooms and tables, view menus, place orders, and access information about the hotel and restaurant's amenities (Leung, R., 2019).

The website's overall quality played a crucial role in delivering the intended message to potential customers. The website's usability was also a significant factor that provided a better user experience, ensuring that users could easily navigate and use the website's features. The website was designed to be user-friendly, interactive, and accessible to all ages and genders (Pathak, et al., 2021).

The hotel and restaurant management system allowed users to interact with the website by creating, modifying, and making changes anytime needed. This meant the system was not static but provided real-time services, ensuring customer satisfaction (Buhalis, et al., 2019).

The development of the hotel and restaurant management system was systematic and relative to current technological advancements. The system's structure was based on its intended purpose: streamline operations and enhance guest experience (Law, et al., 2019).

This project aimed to create an online web application as a one-stop shop for all hotel and restaurant services. The system provided various features such as online reservations, menu viewing, order placement, and access to hotel and restaurant amenities. The system aimed to streamline operations, provide efficient and reliable services, and enhance the overall guest experience.

Objectives of the Study

The general objective of the study was to develop a web application for a Hotel and Restaurant Management System that would streamline operations and enhance the guest experience.

Specifically, the study aimed to:

1. develop a user-friendly web application with features such as online reservations, room and table management,

chat room, inventory and order management, and billing and payment processing;

2. enhance guest experience by providing personalized services such as room preferences, special requests, and QR-code-enabled menu ordering;
3. QRify operations by integrating various departments, such as the front desk, cashier, and management, to improve communication and coordination; and
4. evaluate the effectiveness of the developed web application in QRify operations and enhancing the overall guest experience.

Scope and Limitation of the Study

The study focused on developing an online web application for streamlining operations and enhancing guest experience for a hotel and restaurant management system. The scope of the study included the development of a web application for a hotel and restaurant management system with features such as reservation, room service, and billing. The web application allowed guests to make reservations, view menu items, order food and drinks, make payments online, and leave comments and ratings. The hotel and restaurant management had access to the web application's backend, which included features for managing orders, inventory, and customer data. As a young developer, the web application features were yet to expand but were not limited to its current features and were expected to scale in the future.

The delimitation of the study included the following:

The study was exclusively for the hotel and restaurant, and no other establishments were included. All the exclusive features stated in the study were only for the guests and management of the hotel and restaurant. The web application was designed only for customers who had made a reservation with the hotel and restaurant, and other persons could still visit the website with limited features. The study did not cover the development of a mobile application or other software systems beyond the web application. Furthermore, the focus was specifically on enhancing the digital experience within the confines of the hotel and restaurant domain.

Significance of the Study

The study was conducted to develop a system that provided valuable insights into the significance of streamlining operations and enhancing guest experience through an online web application for hotel and restaurant management systems.

The study was beneficial to the following:

Hotel and Restaurant Owners and Managers. The results of this study helped them streamline their operations and enhance the guest experience. They managed their hotel and restaurant operations effectively through the online web application. The study also provided insights into the effectiveness of online tools in managing hotel and restaurant businesses.

Employees of the Hotel and Restaurant Industry. The study helped hotel and restaurant industry employees understand the importance of online web applications in streamlining operations and enhancing guest experience. They also learned how to use the online tools effectively to improve their work efficiency.

Customers. The study provided insights into how online web applications could enhance their experience as guests. They could access hotel and restaurant services online, book reservations, and provide feedback on their experience. The study helped them enjoy better service and convenience during their stay.

Technology Providers. The study provided insights into the hotel and restaurant industry's needs regarding online web application development. Technology providers learned how to design and develop a system that met the industry's specific needs, thus improving their market competitiveness.

Researchers. The study allowed the researcher to gain practical experience, which was essential for growth as a developer. This lets them learn new things, improve their skills, and better understand the technologies they are working with.

Future Researchers. This study served as a reference for future researchers on how online web applications could streamline operations and enhance the guest experience in the hotel and restaurant industry. They also learned how to develop a similar system that could be applied to other businesses. The experience provided a foundation for the researchers to innovate and contribute to the broader landscape of technological advancements in various industries.

Conceptual Framework

This section presented the central concept in a visual format, illustrating the input, process, and output of the system's categorized functions. The visual representation effectively conveyed the intricate interplay between these components, providing a comprehensive overview of the system's functionality.

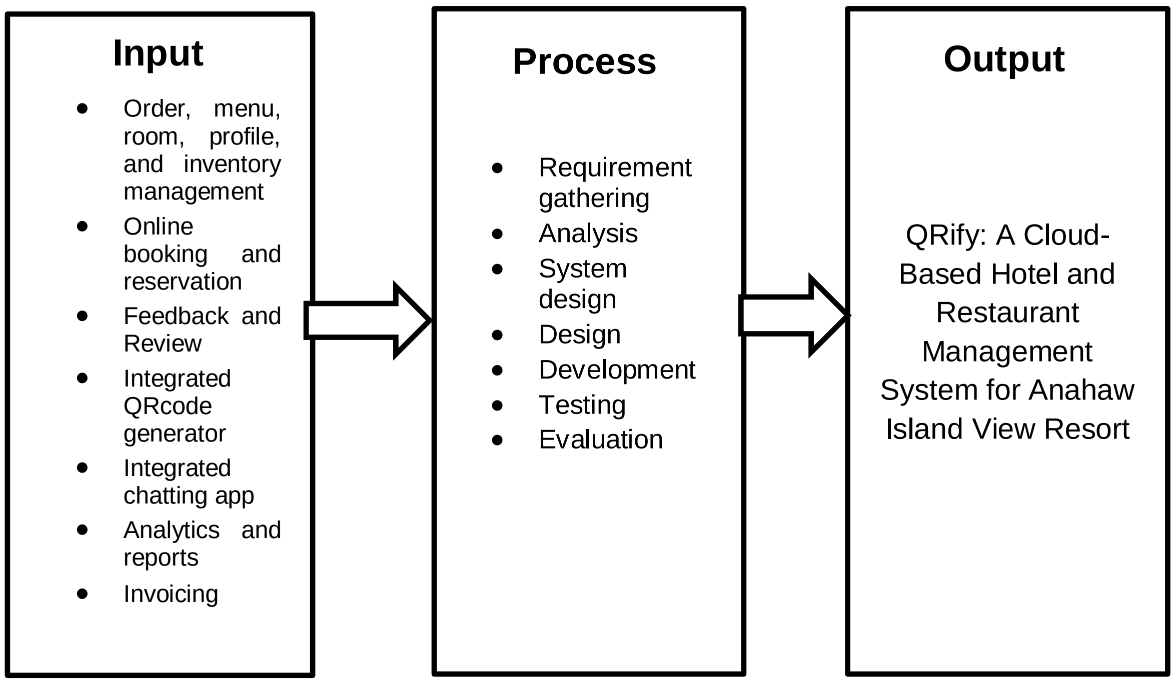

Figure 1. Conceptual Framework of the Study

The concept of this study was to develop an online web application for streamlining operations and enhancing guest experience in the Hotel and Restaurant industry. The input included the various functions and features that were required to be incorporated into the web application to facilitate efficient operations management and enhance guest satisfaction. The process emphasized the comprehensive and structured approach to developing the web application to ensure its quality and effectiveness. The researchers developed the Hotel and Restaurant Management System based on the development process outlined in the methodology. They evaluated the system to identify areas for improvement before its implementation.

Definition of Terms

To quickly understand the terms used in the system, the following terms were operationally defined:

Customer satisfaction. The extent to which customers were satisfied with the products or services provided by a business, measured by their willingness to return and recommend the business to others.

Guest experience. The overall experience of a customer during their stay or visit to a hotel or restaurant, including interactions with staff, the quality of services, amenities, and facilities.

Hotel and restaurant management system. A computerized system that helped hotels and restaurants manage their day-to-day operations, such as reservations, bookings, menu management, and customer service.

Online web application. A computer program designed to run on a web server and accessed via a web browser, providing users with a graphical interface to interact with the software.

Real-time. The ability of a system to process data and deliver results instantly, with minimal or no delay.

Streamlined operations. The process of optimizing hotel and restaurant management activities to ensure efficient and effective performance.

Usability. The measure of how easy and effective a website or application is to use, navigate, and accomplish tasks.

User experience. A user's overall experience when interacting with a website or application, including ease of use, navigation, and accessibility.

Chapter II

REVIEW OF RELATED LITERATURE/SYSTEM

This chapter presents the review of related literature that provides the researchers with a strong foundation for the study. The researchers gathered data from trusted scholarly sites to ensure the credibility and quality of the information collected. These will make a path for the researchers to develop a comprehensive and well-structured capstone project anchored to the given objectives of this study.

Related Literature

According to Man et al. (2021); and Prasetyo et al. (2020), the more determined someone is to use a QR code, the more probable they will do so. QR codes are frequently used if the program offers a variety of benefits. Evidence from earlier studies suggests that behavioral intention to use can spur actual action.

Zhang et al. (2020) claim that patients who believe their COVID-19 condition is severe and vulnerable have a favorable attitude toward adopting technology. According to these results, Covid 19 can be harmful to a person's health by causing symptoms like fever, colds, coughing, asthma, and even death. Customers adopt an adaptive behavior by scanning a QR code to examine the menu to reduce the risk of transmission. A person can avoid transmitting Covid 19 since the app can lessen direct contact for patrons at eateries where patrons can virtually order menus without queuing.

Prasetyo et al. (2020) incorporated PMT and the Theory of Planned Behavior (TPB) to examine the efficacy of Covid 19 prevention measures. The result shows how perceptions of a person's fragility and seriousness can indirectly influence behavioral intentions and result in actual behavior. In Indonesia, in particular, no research has ever been done on the relationship between PMT and TPB in applying QR codes in the restaurant industry. Therefore, this study aims to evaluate consumer perceptions of the use of QR codes by integrating PMT and TPB theories to shed light on the restaurant industry, particularly in Indonesia.

Zhao and Bacao (2020), by fusing the UTAUT with the task-technology fit model and the expectancy confirmation model, extended the UTAUT in the setting of OFDS. The study findings demonstrated that performance expectancy, social influence, trust, and perceived task-technology fit were significant predictors of the continuous intention to use OFDS. Specifically, trust, perceived task-technology fit, and satisfaction were added to the original UTAUT. The three main factors from UTAUT, namely performance expectancy, effort expectancy, and social influence, are used in this study along with two additional constructs (i.e., trust and food safety risk perception) to improve the prediction of the factors influencing the intention to purchase OFDS.

Liang (2020) that hotel restaurants have been under pressure to draw in and keep guests due to intense competition, decreased expenditure per person, and rising prices. A global public health emergency's unexpected shock has worsened the situation. Non-essential human mobility and business activity were halted in China from late January to April 2020 due to the COVID-19 epidemic.

Parulis-Cook (2020), for opulent hotels and their restaurants, survival has taken precedence amid the pandemic. Offering an ordering and home delivery service with online-to-offline (O2O) food delivery companies like Meituan and Eleme—two O2O services heavyweights in China—has been one of the tactics used by luxury hotel restaurants in China to stay alive.

In the study conducted by Polizzi et al. (2020), to the best of our knowledge, no study has looked at upscale eating options in an O2O setting. Additionally, people's attitudes and behaviors have changed due to the COVID-19 outbreak. It is vital to look at O2O dining experiences with upscale hotel restaurants during the pandemic in light of the issues posed by these inquiries.

Ma and Hsiao (2020) state that these environmental signals are especially crucial for luxury restaurants since they promote favorable feelings in patrons and have a greater influence on patrons' behavioral intentions. Most researchers concur that ambient conditions, facility aesthetics, spatial arrangement, and seating comfort are all sub-dimensions of physical environment quality.

Wen et al. (2020), customers who need confidence and risk-free food consumption prioritize hygiene and sanitation. Due to COVID-19, extreme care must now be taken to maintain cleanliness and hygiene in order to reduce the risk of infection.

According to Cho et al., (2020), in a restaurant where illnesses like coughs and colds may be common, Covid 19 is highly contagious when in direct contact with another person. Exposure to COVID-19 results in a stronger impression of threat, which suggests that adaptive behavior is more likely.

According to Ong et al. (2020); Wu, 2020; Yang et al., (2019), scan QR codes to stop transmission when looking at menus. Numerous studies have shown that a person's attitude will be more upbeat the more serious something is perceived.

According to Kocaman (2020), more people are likely to visit because of the application's excellent performance in reducing direct contact and lowering the risk of transmission. Supply chain workers claim that by enhancing this service system, it is possible to raise sales and service quality standards.

Köhler and Pizzol (2020), point out that blockchain is a component of a larger system of technologies rather than a standalone technology and say that further research is necessary to determine whether blockchain-based technologies will improve food supply networks' sustainability.

According to Lee et al., 2019; and Zhao & Bacao (2020), recent studies in the OFDS literature have shown no significant correlation between effort expectancy and the continuous usage intention of OFDS, indicating that customers experience little difficulty using new apps, including OFDS, as the use of smartphones and apps has matured. The interface of smartphone apps has stabilized due to the development of information and communication technologies over time.

According to Ray et al. (2020) and Yeo et al. (2019), many clients are driven to O2O by lower pricing, as the platforms make it easier for price comparison, is another negative effect of using O2O meal delivery services. This is not good for upscale restaurants. This raises the question of whether upscale hotel restaurants and online marketplaces for food delivery can coexist harmoniously. Further difficulties affect this, such as whether upscale restaurants on O2O platforms target the wrong customers and whether the challenges brought on by the platforms negatively impact the restaurants, which then harms the restaurants' reputations and the hotels' reputations.

According to Pillai et al. (2019), "QR codes" are posted in the Kempinski Hotel in Ajman rooms so that visitors can place meal orders in four different languages. The kitchen receives these orders right away. Hotels like the Fontainebleau Hotel Miami Beach, the Mirage Hotel and Casino, and the Best Western Hotel Le Montparnasse utilize "QR codes" as well. Using "QR codes" that contain PDFs with extensive descriptions of the amenities found in hotel rooms can enhance the guest experience.

Li et al. (2019), due to the maturity of the field and the requirement to keep track of its development, continual efforts to examine and assess the body of hospitality research are especially crucial. Despite the significance of the restaurant industry to the hospitality sector, the bibliometric approach has not yet been used in a study in this area. By addressing the need for more information on the state of the art in restaurants and illustrating its thematic evolution, the current study aims to close this gap in the academic literature. The study uses a bibliometric approach using co-word analysis and science mapping to analyze the scientific literature on restaurants in-depth, focusing on the years 2000–2019. It examines the scientific articles on restaurants included in the "Hospitality, Leisure, Sports, and Tourism" category of the Web of Science (WoS) citation index.

Ray et al. (2019) state that online food delivery services (OFDS) connect customers with partner restaurants via their websites or mobile applications for meal ordering and delivery. The COVID-19 pandemic, proclaimed in March 2020, increased the OFDS market's quick expansion (EHL Insights, n.d.). The OFDS market has been expanding significantly over the previous few years.

According to Lilyquist (2019), "QR codes" are a useful tool for improving interactive marketing and streamlining various operational tasks. With interactive marketing, businesses hope to increase customer and business interaction, increasing demand for their goods and services. Interactive marketing is also known as trigger-based marketing, which occurs when a consumer activity triggers the addition of more marketing initiatives.

In the study conducted by Kim et al.and Maimaiti et al. (2019), OFDS has helped restaurants by giving them a new way to serve customers. However, because OFDS adds delivery to the standard restaurant service process, there are potential risks for restaurant owners and clients regarding temperature control during delivery, delivery drivers' hygiene, and even food tampers. In addition, because customers using online/mobile platforms frequently struggle with interface design, communication speed, privacy, and security of the service interfaces, including payment processors, OFDS customers are not free.

In the study conducted by Roh & Park (2019 and Troise et al. (2020), research explained that customers are more likely to believe that adopting new technology will boost their life's productivity the greater their effort expectations are. Most researchers who conducted OFDS-related studies discovered that once customers experience ease in using OFDS, they tend to consider the service useful. A recent study in the context of OFDS argued that ease of OFDS no longer had a meaningful impact on discovering the service's usefulness in case of repeated usage.

According to Nguyen et al. (2019), specifically, it was found that customers' trust in an online food shopping website influences their intention to use the site favorably.

Similarly, one of the main factors influencing the continued use of online banking services is trust in the service. Customers are more likely to use OFDS when they think the service will go smoothly, according to existing OFDS studies.

According to Xiao et al. (2019), it is a growing e-commerce paradigm that combines offline consumption of goods and services with online marketing, sales, and assessment. O2O systems have become increasingly important in various consumer life scenarios, particularly for locally focused life services items like food delivery, tickets, and auto rentals.

According to Kapoor and Vij (2019); and Cho et al. (2019), the Stream of research focuses on understanding the variables that affect consumers' opinions of food delivery services. It has been proven that app features like convenience, design, dependability, and the diversity of food options have a significant impact on how valuable a meal delivery app is, which increases the likelihood that users would use the service.

According to Zhang et al. (2019) and He et al. (2019), the quantity of reviews and total ratings is especially important for high-sale restaurants to boost sales volume, whereas the delivery service effect is more significant for low-sale restaurants. The adaptive behaviors of restaurants have also been investigated using an agent-based methodology, with the results showing that a crucial element in boosting sales volume is the timely modification of food quality to account for changes in consumer preferences.

Suhartanto et al. (2019), food quality, including presentation of flavors, variety, and accessibility of healthy options, attitude of the delivery team, order conformance, packaging, and food condition upon receipt, as well as delivery costs, have a significant impact on customer loyalty.

Kiatkawsin and Han (2019), some comments also revealed that when the restaurants failed to deliver as expected, their brand credibility was damaged. Many customers complained about the ordinariness of the food, noting that the taste was just "too normal and not distinctive enough." When there are discounts or special offers, these criticisms become more serious. A probable explanation is that lowering the price is viewed as a problematic promotion because a high price is one of the defining characteristics of upscale dining establishments.

Gagarin (2019), restaurant staff members from Spain, Germany, France, Italy, and Japan range from managers and owners to chefs. Employees from a variety of various food establishments, including fully functional restaurants, cafes/bars, fast food restaurants, canteens, and even commercial kitchens that solely serve take-out ready meals, were questioned at the same time.

According to ProHotelia (2019), the Henn administration thinks that some technologies are developing too quickly and others are becoming obsolete. For almost four years, the hotel has employed many robots; during that period, several technologies have "gone forward," like Siri or Google Assistant. The hotel administration understood that the complex still needs people but did not abandon the idea of an autonomous hotel.

Yang et al. (2019) that people are deterred from using QR codes if their use is hard and full of difficulties. They will, however, accept the technology if mastering QR codes doesn't take a lot of time, effort, or money. The results showed that people are less likely to adopt new technologies when the response cost increases.

Synthesis

The studies above on the impact of false news and misinformation highlight the need for reliable platforms to disseminate accurate information. In the hotel and restaurant industry context, an online web application with QR code technology can play a crucial role in promoting trust, reducing misinformation, and enhancing operational efficiency.

By leveraging QR code technology, the web application provides a trustworthy platform for hotels and restaurants to communicate with guests. QR codes placed strategically throughout the establishment enable guests to access reliable and up-to-date information about menus, services, and promotions. This eliminates the risk of misinformation that may arise from social media or other unreliable sources.

Furthermore, the web application can incorporate interactive systems to create a user-friendly interface with multiple pages and user accounts. This enables establishments to distribute advertisements such as job postings and news updates to their target audience. The application helps ensure that accurate and reliable information reaches the intended audience by centralizing information dissemination.

Chapter III

METHODOLOGY

This chapter discussed the preparation of the Online Web Application for the Hotel and Restaurant Management System. The research methodology and method were explained to justify the data gathered further, along with the system’s structural models, to expand the presentation and understanding of the development of the website.

Development Method

The research method used in this study was the "Questionnaires / Survey" method. It aimed to collect data from the respondents by answering questions with pre-made answers, where they selected their best response. The researchers chose this method because it can combine and quantify data easily. Figure 2 shows the researchers' Agile Method as their guide in developing the project. The study utilized the System Development Life Cycle, in which the software development process was divided into six (5) phases that played dynamic roles in defining the tasks that must be accomplished at each stage.

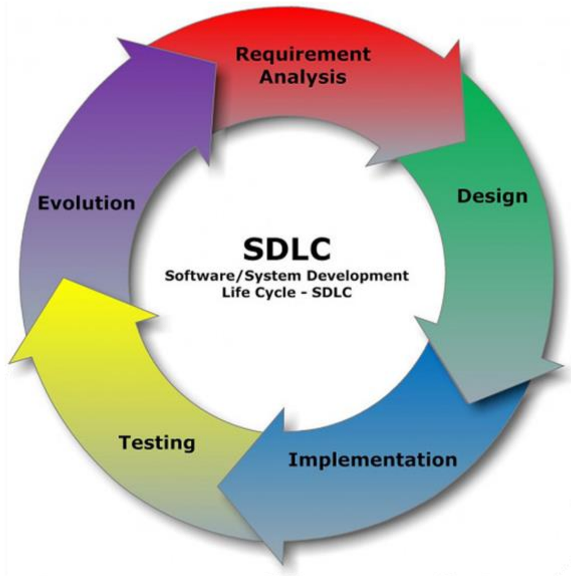

Figure 2. Agile Model

The sequential phases in the Agile model are:

1. Requirement Analysis

In the requirements phase of developing QRify, an agile approach was adopted to meticulously plan and research the objectives of creating a cloud-based hotel and restaurant management system for Anahaw Island View Resort. Project prioritization was a key focus during this stage, involving strategic planning executed through stakeholder meetings, requirements collection, and market research. Identifying key stakeholders was crucial to achieve a comprehensive understanding of the system's scope, and their needs and potential challenges were nuancedly explored. The outcome of this phase was a high-level project plan that outlined tasks, timelines, and resource requirements. Moving into the Analysis phase, a deeper exploration of requirements unfolded. This encompassed a thorough physical analysis of tangible elements, logical analysis defining the system's structure and data models, and architectural considerations outlining the high-level system architecture. Technologies, frameworks, and platforms were prioritized to align with the overarching goals of creating a cloud-based management system for Anahaw Island View Resort. The process ensured that QRify would be finely tuned to meet the specific needs and aspirations of the resort.

2. Design

In the design stage of QRify, we created a user-friendly graphical interface for Anahaw Island View Resort's cloud-based hotel and restaurant management system. This involved two key phases: UI Concept and UI Designing. In the UI Concept stage, we focused on refining project goals, ensuring alignment with the resort's requirements, and defining specific user interactions and aesthetics. Visual aids like wireframes were prepared to facilitate effective communication.

Moving to UI Designing, an iterative process was adopted for collaboration among designers, developers, and stakeholders. The goal was to prioritize usability, consistency, and a cohesive user experience. Interactive prototypes were crafted for stakeholder feedback, ensuring QRify's interface met the resort's unique needs.

3. Implementation

In the implementation stage of QRify, the researcher executed the development of a cloud-based hotel and restaurant management system for Anahaw Island View Resort. This phase encompassed coding the front-end user interface, writing back-end code, and integrating third-party APIs. The implementation phase was structured into smaller tasks or sprints, each focused on delivering specific features. Agile methodologies were embraced, ensuring adaptability and regular feedback loops. Collaboration tools like Jira facilitated task management, while version control systems like Git ensured seamless code integration. This iterative approach allowed for continuous refinement, addressing issues promptly and aligning the implementation with the evolving needs of Anahaw Island

View Resort. The result was a system that met the specific requirements of the resort's hotel and restaurant operations, providing a tailored and efficient solution.

4. **Testing**

In the testing stage of QRify, our focus was on ensuring the reliability and performance of the cloud-based hotel and restaurant management system designed for Anahaw Island View Resort. Various testing methodologies were employed, including functional, performance, and security testing. Collaborating closely with QA specialists, we executed functional tests to verify that QRify's features met the specified requirements. Performance testing tools like JMeter were employed to assess system responsiveness and scalability. Security testing addressed potential vulnerabilities, ensuring data protection. An iterative testing approach allowed for prompt identification and resolution of issues. Regular feedback loops with stakeholders ensured QRify's testing phase aligned with the resort's expectations. The result was a thoroughly tested system, meeting high-quality standards and ready to deliver a secure and seamless experience for Anahaw Island View Resort.

5. **Evolution**

In the evolution stage of QRify, our team continues to enhance and refine the cloud-based hotel and restaurant management system to meet the evolving needs of Anahaw Island View Resort. This phase involves incorporating user feedback, fixing bugs, and adding new features to ensure the system remains up-to-date and aligned with the latest industry standards. Regular feedback loops with resort stakeholders help identify areas for improvement and guide the development team in prioritizing enhancements. The agile development methodology is maintained, allowing flexibility and responsiveness to changing requirements. The evolution phase is about fixing issues and actively seeking ways to enhance QRify's functionality. New features and improvements are planned and implemented, ensuring the system grows in tandem with the resort's operational requirements. This ongoing process ensures QRify remains a dynamic and efficient solution for Anahaw Island View Resort's hotel and restaurant management needs.

Gantt Chart

The plan and schedule for developing the system are shown in the Gantt Chart presented below. It shows the segregation of the whole process until the system is finished.

Table 1. Gantt Chart

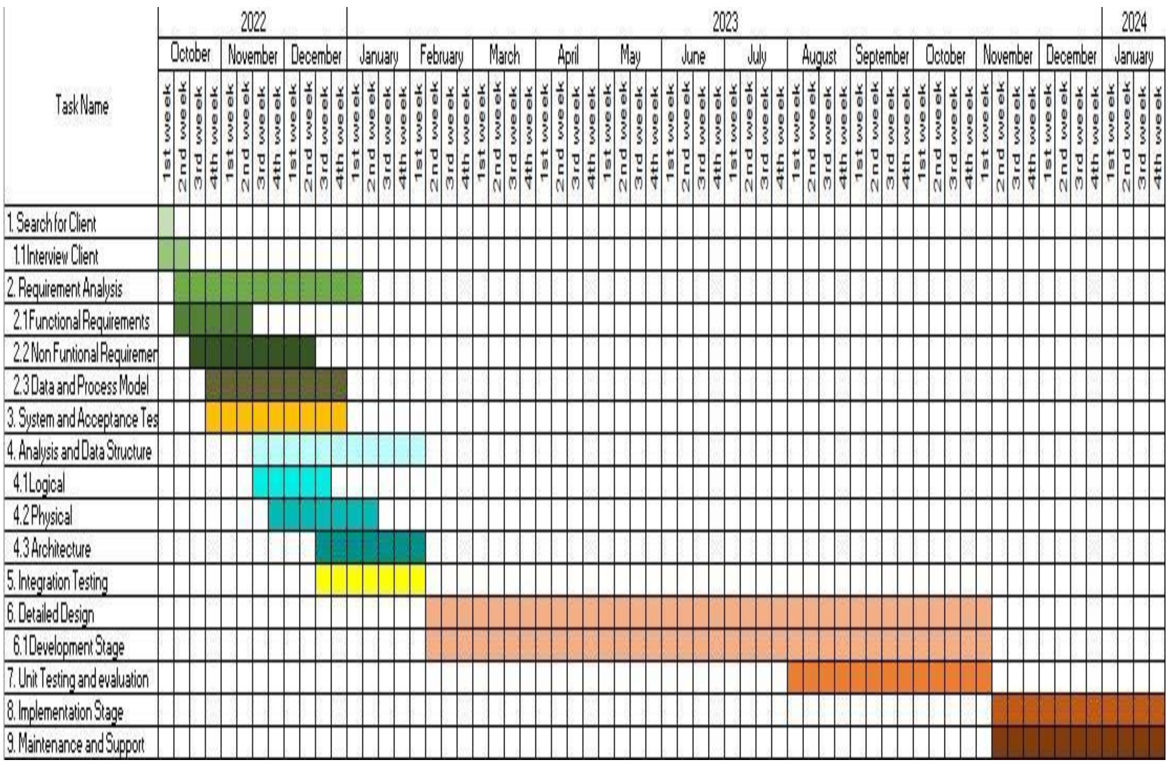

Table 1 meticulously outlines the chronological progression of events, providing a visual roadmap for the website's development.

The seamless integration with the Gantt chart not only ensures a well-structured execution of plans but also facilitates efficient monitoring and adjustment to project timelines, reinforcing a strategic and agile approach throughout the development process. Additionally, this alignment enhances project management by fostering adaptability, allowing the team to respond proactively to evolving requirements and challenges.

Requirements Specifications

To maximize the use of the proposed project, the users needed to be knowledgeable and correctly oriented to the system; this involved understanding the functional requirements, user interface, hardware interface, software interface, and security interface, as well as familiarizing themselves with the processes and procedures of implementing the website to utilize the project fully. Additionally, a comprehensive training program should be implemented to ensure users can navigate and leverage the system's functionalities effectively.

Functional Requirement

An online web application designed for hotel and restaurant management aimed to QRify operations and enhance the guest experience. The application had a robust room and table reservation system that allowed guests to book and manage their reservations easily. It also had an online ordering system for food and drinks, enabling guests to place orders and track their status in real-time.

Table 2. Functional Requirement of the System

Features	**Description**
Online Reservations	Streamlines room and table bookings with real-time availability checks and instant confirmations, ensuring a seamless reservation process.
Order Management	Empowers guests with an advanced online ordering system, providing real-time order tracking for customers and staff.
Inventory and Order Management	Enhances efficiency in inventory control, offering insights into stock levels, finished product values, and event-specific management through comprehensive reports.
Payment Processing	Customers enjoy flexible payment options, including Cash on Delivery (COD) and secure online payments. The system seamlessly integrates these payment methods, providing users a convenient and secure transaction experience.
QR Code-enabled Ordering	Enhancing the dining experience, QRify introduces a contactless ordering option. Customers can scan a QR code with their smartphones, gaining access to the restaurant's website.
Transaction Monitoring	Provides owners with a centralized dashboard for comprehensive transaction monitoring, covering order progress, total transaction values, and event performance.

Table 2 indicates that QRify features streamlined online reservations with real-time availability, an advanced order management system, efficient inventory control, flexible payment options, and innovative QR Code-enabled functionalities for contactless dining and reservations. The system includes a centralized dashboard for owners to monitor transactions comprehensively. QRify offers a modern and efficient solution aligned with the operational and experiential needs of the resort.

User Interface

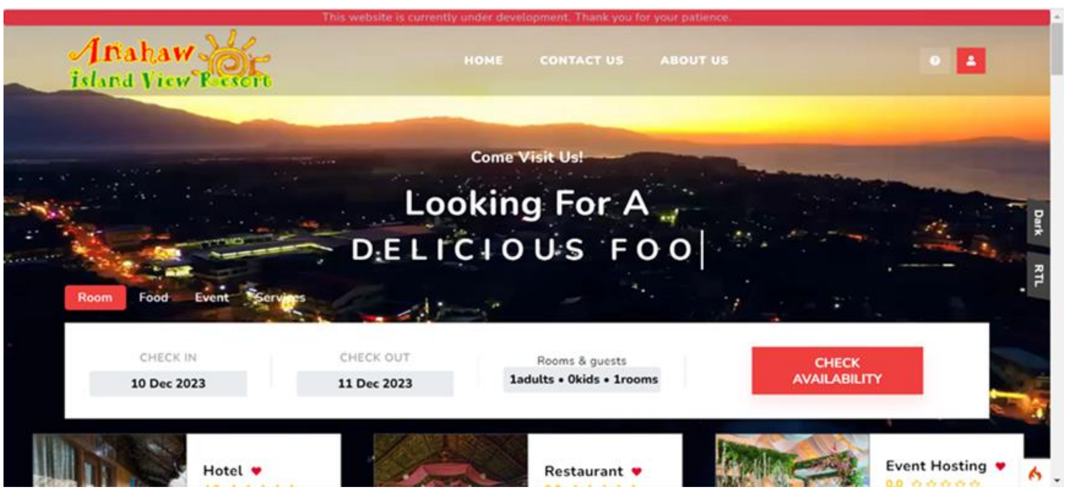

Figure 3. User Interface

The user interface for an online web application designed for hotel and restaurant management was user-friendly and efficient. It allowed staff and guests to navigate the system easily, providing relevant information at a glance. The interface was also responsive to different devices and featured a visually appealing design. The interface provided staff with easy management of reservations, orders, and other tasks with notifications to keep them up-to-date. The interface offered guests a seamless and personalized experience for booking rooms, making restaurant reservations, ordering food and drinks, and requesting services. Overall, the user interface was designed to streamline operations, enhance guest experience, and ultimately drive revenue and profitability for the hotel and restaurant.

Hardware Interface

The hardware interface for an online web application for a hotel and restaurant management system with QR code technology and data-driven insights requires devices supporting QR code scanning, such as smartphones, tablets, or dedicated QR code scanners. Additionally, the system must be compatible with standard hardware such as desktop computers, laptops, and servers for data storage and processing. Proper integration with these hardware devices and systems was necessary to ensure a seamless and efficient user experience.

Software Interface

The software interface for an online web application for a hotel and restaurant management system with QR code technology and data-driven insights was designed to allow seamless integration with the relevant databases and APIs. It provided easy-to-use interfaces for managing customer and reservation data. The system could generate QR codes for menus and orders and offer real-time data insights to help hotel and restaurant managers make data-driven decisions. The interface was user-friendly and intuitive, allowing staff to easily navigate the system and access the information needed to provide exceptional customer service.

Furthermore, the incorporation of QR code technology aimed to enhance contactless interactions and streamline ordering processes. The software's commitment to user-friendliness ensured a smooth experience for both staff and customers, fostering efficient operations in the hospitality industry.

Security Requirements

Security requirements played a pivotal role in developing the web application, particularly for a hotel and restaurant management system handling sensitive information. The security specifications for this system encompassed crucial features, including robust data encryption, robust user authentication and authorization mechanisms, the implementation of secure communication protocols, safeguards against malware and hacking attempts, and a regimen of regular security updates.

Moreover, the system adhered to a comprehensive backup and disaster recovery plan to mitigate the impact of security breaches or data loss. The integration of QR code technology also adhered to stringent security standards, preventing any unauthorized access to the system or its data.

Technical Background

The technical background not only outlined the necessary requirements for the successful execution of the website's development but also served as a foundational document elucidating the intricate technical aspects of the project. Below were the hardware and software specifications meticulously employed by the researchers, underscoring the robust foundation upon which the project was built.

Hardware Specifications

The table shows the components with the required and recommended hardware specifications that were used in developing the system. The components, minimum specifications, and recommended specifications for using the system were described in the table.

Table 3. Hardware Specifications

Component	Minimum Specifications	Recommended Specifications
Lan or Wifi	25 Mbps	100 Mbps
Memory	4GB RAM	8GB Ram
Monitor Display	13” LCD Monitor	14" LCD monitor, resolution of 1600x900
Storage	115 GB SSD or 500 GB HHD	250 GB SSD or 1 TB HHD
Processor	1.10GHz	3.80GHz

Table 3 shows all the detailed specifications consists of component’s system’s operating system, memory capacity, storage and Processor.

Software Specifications

The table showed the components with the required and recommended software specifications used to develop the system. These specifications were necessary to run and maintain the system efficiently. This section described how the software was expected to perform using software specifications. Additionally, the outlined specifications served as a crucial reference point for the development team, guiding them in precisely implementing software elements to meet the system's operational requirements.

Table 4. Software Specifications

Component	Minimum Specifications	Recommended Specifications
Operating System	Windows 7	Windows 8 or latest
Disk Space	30 GB	30 GB or 100 GB
Web Server	Apache HTTP Server	LiteSpeed
Bandwidth	100 GB	100 GB or Unlimited
Database	1 Database	2-3 Available Databases
Visual Studio Code	VS Code version 1.74.1	VS Code version 1.74.2

Table 4 shows the specifications of the software used for this study. The researchers used the most recent operating system and versions mentioned above for optimal use and production performance.

System Analysis and Design

The researchers followed the development process model to construct the website. In each stage, the researchers analyzed the design and coding to plan for the next possible steps, avoiding confusion and errors that the development stage might face. The project design presented the system overview and architecture, providing a comprehensive framework for efficient development and implementation.

System Overview

The Qrify: A Cloud-Based Hotel and Restaurant Management System for Anahaw Island View Resort with QR code technology and data-driven insights was carefully designed to provide an efficient and effective way of managing hotel and restaurant operations. The application allowed users to scan QR codes placed on tables in the restaurant to access the menu and place orders, reducing the need for physical menus and minimizing contact between staff and customers. The system also collected data on customer orders and preferences, which could be used to provide personalized recommendations and promotions.

The system featured a comprehensive dashboard that allowed hotel and restaurant managers to track and analyze data on sales, customer feedback, and other key performance indicators. The data-driven insights provided by the system could help managers make informed decisions about menu offerings, pricing, and marketing strategies.

In addition, the system included features for managing room reservations, housekeeping schedules, and other hotel operations. Guests could book rooms and access hotel services through the web application, reducing the need for in-person interactions and enhancing the overall guest experience.

The system's design prioritized user-friendliness and accessibility, with a simple and intuitive interface that both staff and customers could easily navigate. The system was also designed to be scalable and adaptable, with the ability to add new features and integrations as needed to meet the evolving needs of the hotel and restaurant industry. This emphasis on scalability and adaptability contributed to the system's sustainability and prolonged effectiveness in a rapidly changing business environment.

System Architecture

This part of the research displayed the detailed structure of how the QRify: A Cloud-based Hotel and

Restaurant Management System for Anahaw Island View Resort worked and how the process flowed for the research to be completed.

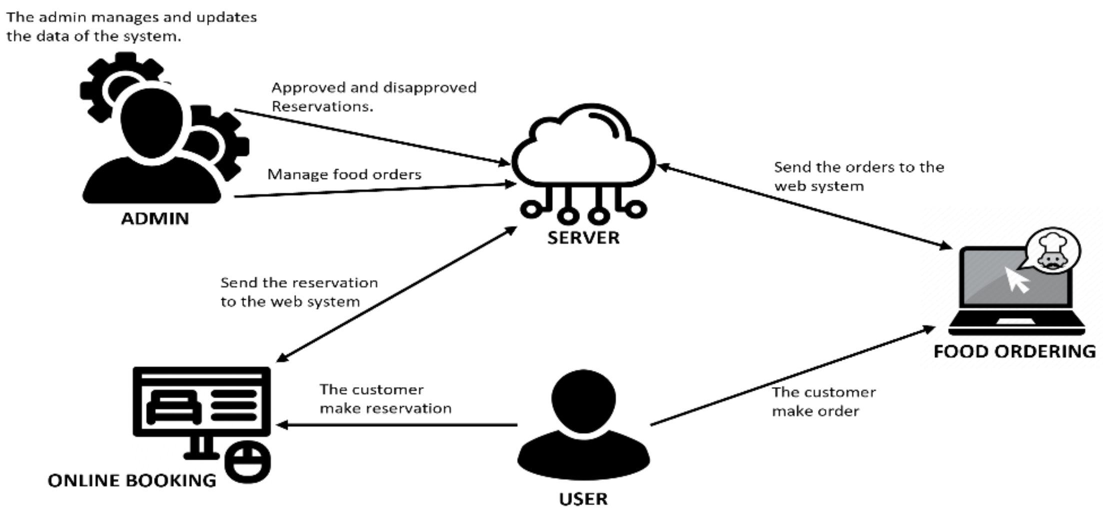

Figure 4. System Architecture

Figure 4 shows the system architecture of the Hotel and Restaurant Management System. It depicted the flow of the system, where the server or system was the center of all transactions. The customer was the one who used the system to make reservations and order food. On the other hand, the admin managed the customer's reservations and orders. The admin was also responsible for all the data saved in the system.

Use Case Diagram

This displayed the textual and visual representation of the summarized relationships among the use cases, the project, and the actors using communication links. This diagram aided the researchers in clarifying, identifying, specifying, and organizing the system requirements for the project QRify: A Cloud-Based Hotel and Restaurant Management System for Anahaw Island View Resort.

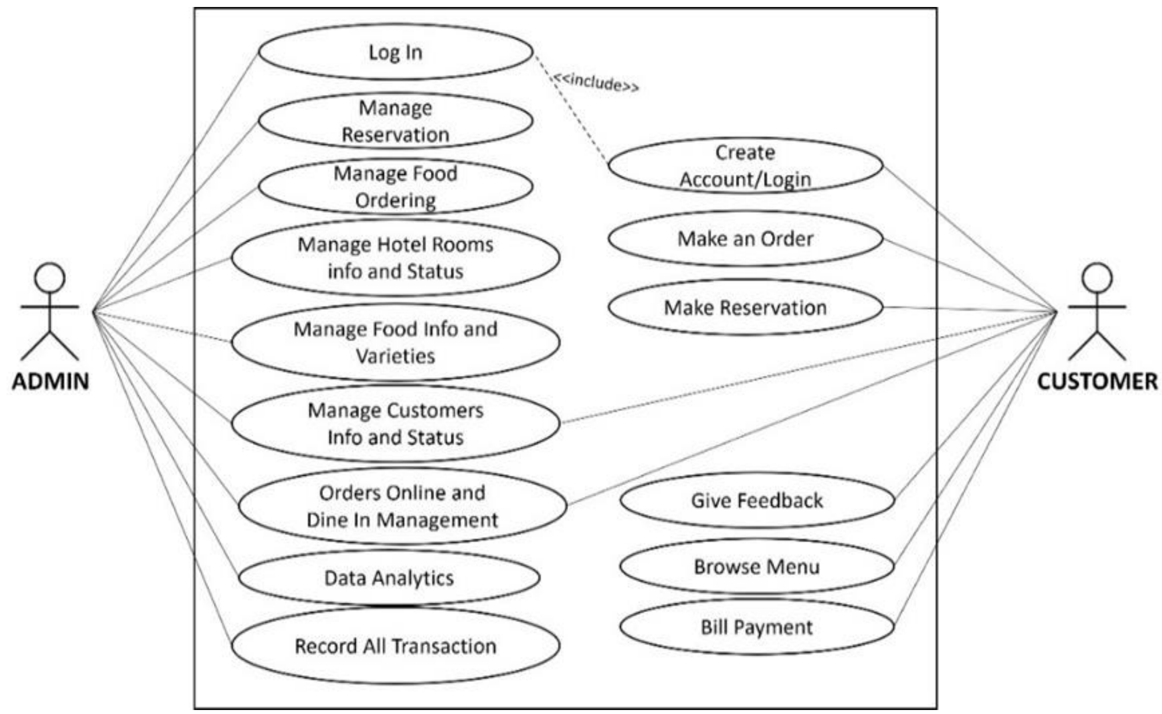

Figure 5. Use Case for Develop website

Figure 5 shows the process to be done by the users in the developed website. This view presented the users' perception of the functionality provided by the project proponents. Furthermore, the visual depiction in Figure 5 serves as a comprehensive guide, enhancing clarity regarding the anticipated user interactions and reinforcing the project's commitment to user-centric design.

Activity Diagram

This prevents the flow of the project using an object-oriented flowchart.

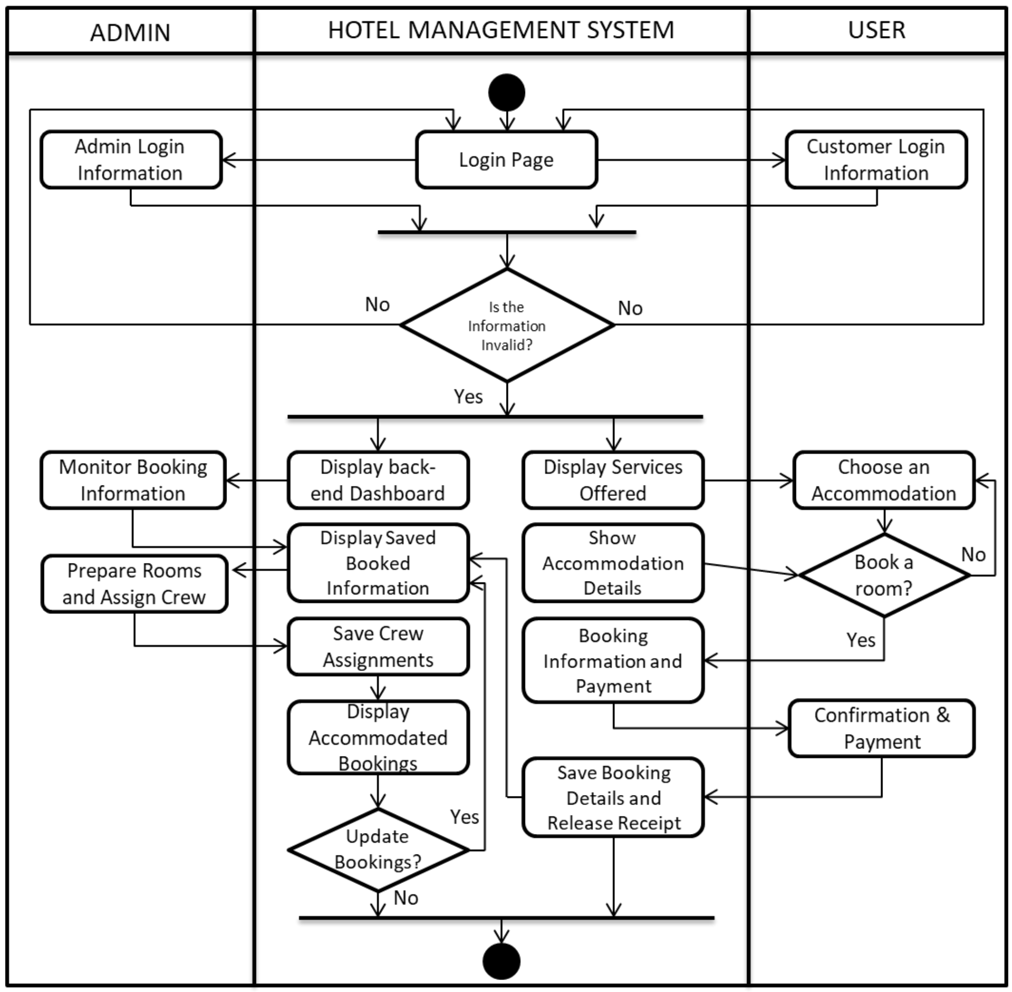

Figure 6. Activity Diagram

Figure 6 illustrates the system progression diagram of the website, beginning with the login/registration process and concluding with various outcomes.

Data Flow Diagram (DFD)

The researchers studied the flow of the manual process of the current system and made a data flow diagram to compare and determine an improvement in the system. The data flow diagrams helped identify the flow of data within an application and how the data moved between different processes in the system. Moreover, this visual representation proved instrumental in fostering a comprehensive understanding of the system's data dynamics, enabling the researchers to propose refined and optimized processes.

Context Diagram

A context diagram is a visual representation that offers a high-level view of a system and its interactions with external entities. It provides a simplified overview, focusing on the system's boundaries, external entities, and the flow of data between them. Context diagrams are valuable tools in systems analysis and design, helping stakeholders quickly grasp the scope and interactions of a system without delving into intricate details. They serve as a foundational communication tool, facilitating effective collaboration among project participants.

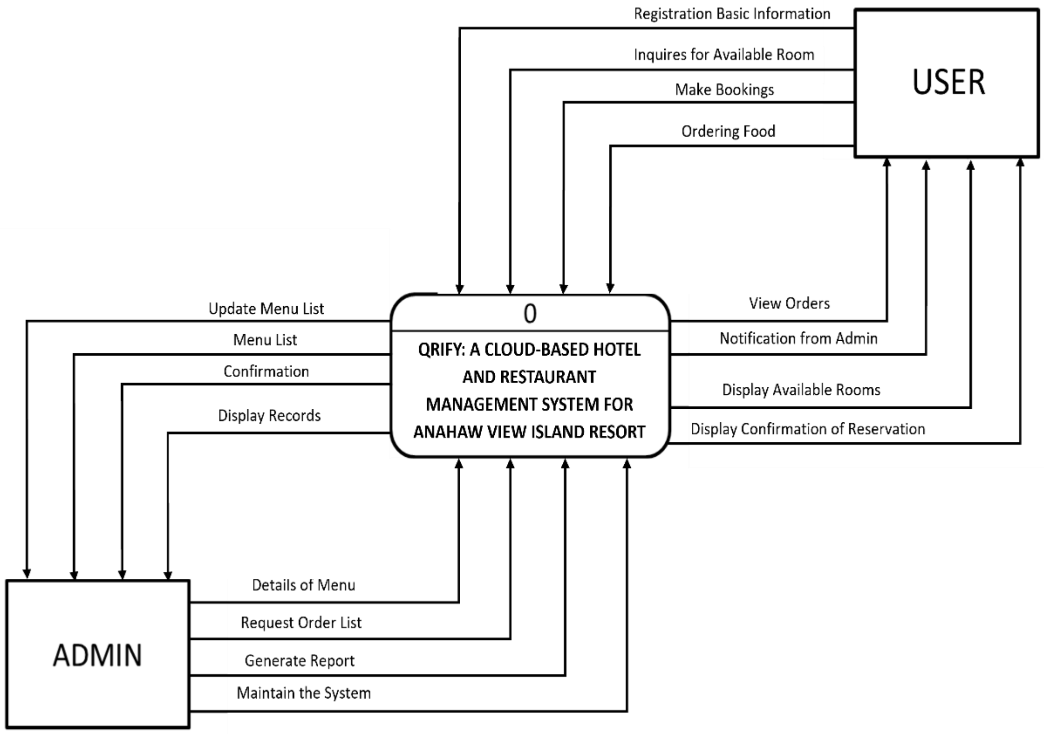

Figure 7. Context Diagram

Figure 7 illustrates the context diagram of the system. It showed the interaction between the system and the actors. The project had seven actors: the guest, rooms, time/schedule, bank, external reservation system, and bookings. The context diagram above modeled each user's activities and functions, facilitating a visual grasp of the dynamic relationships at play in the system.

Diagram 0

Diagram 0 is a serves as the foundational introduction to a project or system, offering a brief yet comprehensive overview. This initial visual representation outlines the primary elements, interactions, and system boundaries, providing a starting point for a more in-depth exploration in subsequent diagrams.

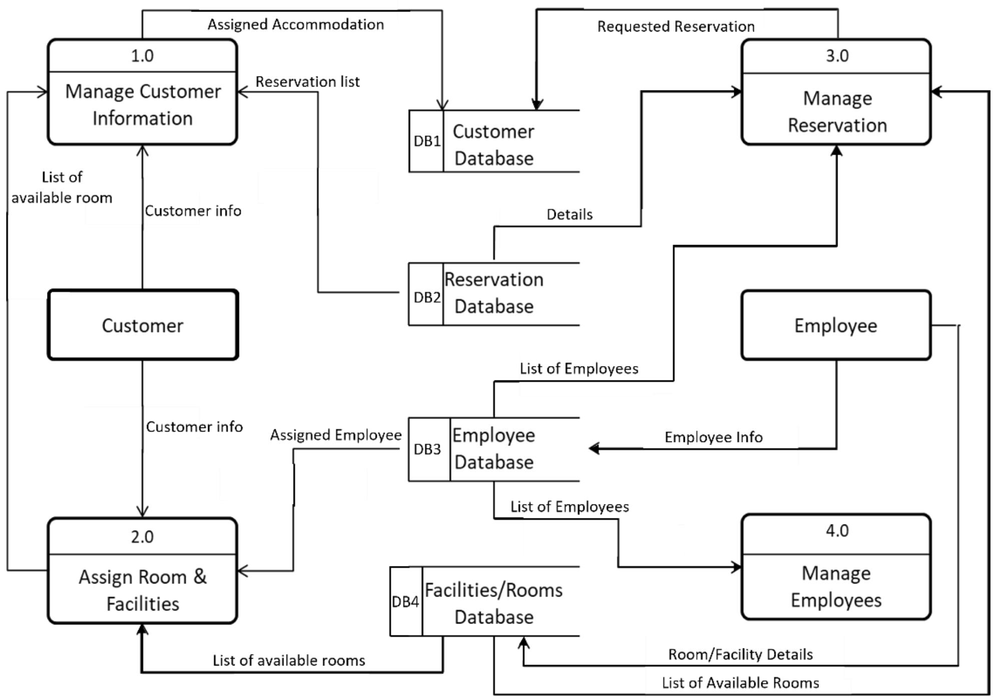

Figure 8. Diagram 0

The diagram provided a concise system overview, portraying it as a singular process with external entity relationships. Researchers described the system's operation, offering a high-level summary of the entire project.

Database Schema

Designing the website's database was crucial, as was specifying entities and attributes. This step clarified the website's purpose and functionality for users. Identifying and detailing entities and their corresponding attributes laid the foundation for the site's structure, playing a pivotal role in defining its overall purpose and functionality.

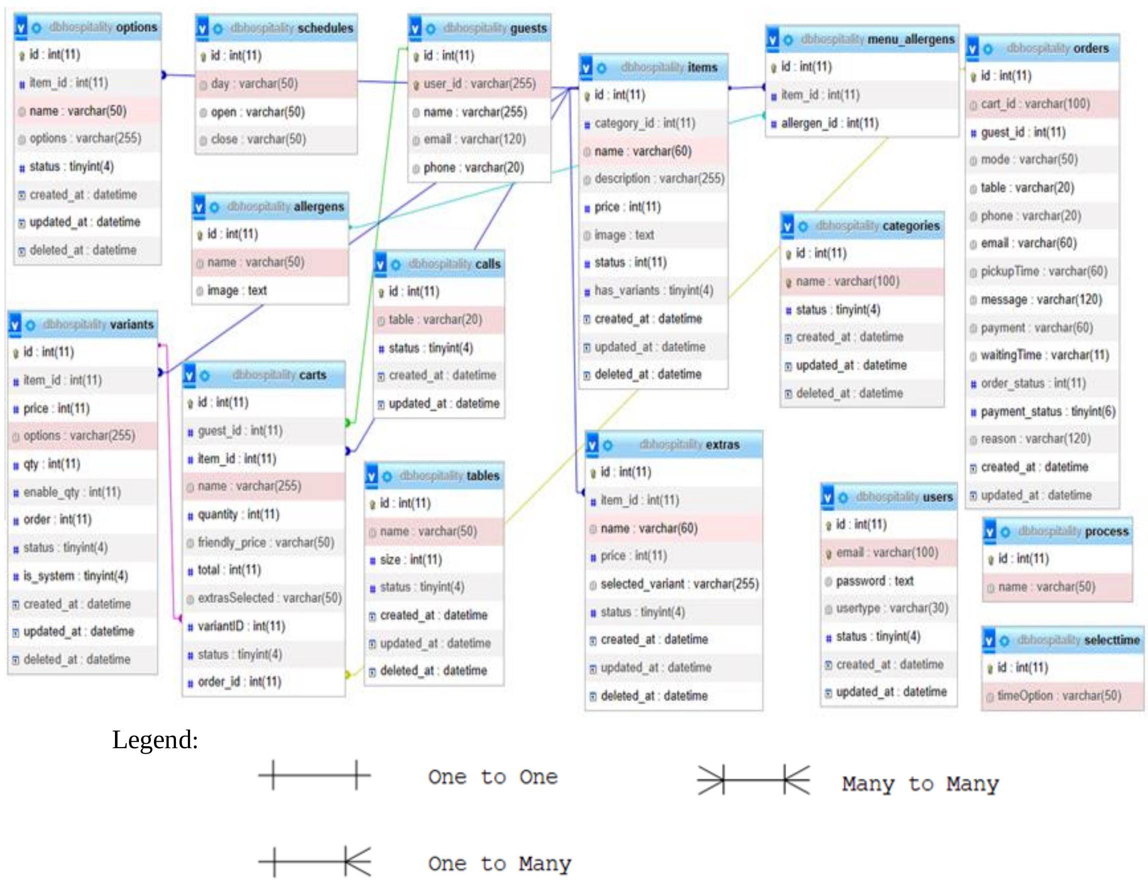

Figure 9. Database Schema

The figure showed the connection between fields and tables that had a relation using primary and foreign keys. It displayed the database schema of Qrify: A Cloud-Based Hotel and Restaurant Management System for Anahaw Island View Resort. It represented the physical layout of data on a storage system based on files and indexes. It also included the syntax for creating these data structures on disk storage.

Testing and Evaluation

QRIFY was a web-based hotel and restaurant management application. The application enabled hotel and restaurant owners to manage their businesses efficiently by providing various functionalities like online booking, inventory management, order tracking, and employee management.

QRIFY underwent software testing during the development process to ensure its quality and functionality. The testing process involved functional testing, performance testing, compatibility testing, usability testing, security testing, and maintenance testing.

After the development and testing process, the application was evaluated using ISO 25010. The admin and end-users of the application conducted the evaluation. The evaluation aimed to assess the application's performance and outcomes regarding functional stability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability.

The evaluation process involved answering a questionnaire using a 5-point Likert scale to measure the characteristics included in ISO criteria.

The Likert scale measured the following criteria:

Table 5. Likert Scale

Scale	Range	Verbal Interpretation
5	4.50 – 5.00	Excellent
4	3.50 – 4.49	Very Good
3	2.50 – 3.49	Satisfactory
2	1.50 – 2.49	Fair
1	1.00 – 1.49	Poor

The table illustrates the Likert Scale to be used in scaling responses in survey research. It serves as a valuable tool for capturing nuanced participant feedback by offering a range of response options, fostering a more comprehensive understanding of attitudes and perceptions. Furthermore, the Likert Scale's flexibility accommodates a diverse array of participant perspectives, contributing to a richer and more insightful interpretation of survey results.

Participants of the Study

The study's respondents were IT experts, administrators, staff, and customers of the hotel and restaurant management system for Anahaw View Resort. This comprehensive approach ensured that the evaluation considered the perspectives of both technical experts and end-users, contributing to a well-rounded assessment of the system's performance and user satisfaction.

Table 6. Respondents of the Study

Respondents	Number of Respondents	Percentage
IT Expert	1	2%
Admin	1	2%
Staff	27	54%
Customers	21	42%
Total	**50**	**100%**

Implementation Plan

The researchers developed an implementation plan for the client. The owner of the Hotel and Restaurant monitored and maintained the system. The project proponents handed over the system to them with the documentation. A letter of agreement was prepared for the agreement of the implementation conditions.

The implementation activities discussed the tasks that must be implemented while developing the system, emphasizing the strategic planning and detailed execution required for successful project realization.

Table 7. Implementation Activities

Activities	Target Date	Progress Notes
Discussion with client	July 03, 2023	Accomplished
Deployment Letter	October 12, 2023	Accomplished
System Deployment and Monitoring Period	October 12-December 12, 2023	Accomplished
Training for user	December 1, 2023	In Progress
System Evaluation	December, 2023	In Progres

This table showed the implementation plan for the adoption of the system. It served as a guide to the users maintaining the system. Before implementing the system, the developers provided end-users with training and lectures on using the developed system.

Chapter IV

RESULTS AND DISCUSSION

This chapter describes the procedure for developing the website. It presents the designs, approaches, and implementation plan of the study.

Presentation of System Output

This section describes the completed user interface up to the system evaluation.

The user interface is a crucial aspect of the project, especially for a website. The interface must be visually appealing and user-friendly. This is why the user interface of the created system is depicted through illustrations accompanied by descriptive captions. This presentation aims to provide a clear understanding of the project and give a better visual representation of the user interface design.

The user interface underwent rigorous usability testing to ensure a seamless and intuitive user experience. The result is a thoughtfully crafted user interface that aligns with the project's objectives and user expectations, ultimately enhancing overall user satisfaction and usability.

System Homepage

The figure below shows the system's landing page, which provides basic information about the hotel necessary for accommodating rooms. Additionally, it serves as the initial point of interaction for users, offering a user-friendly interface to access essential details seamlessly.

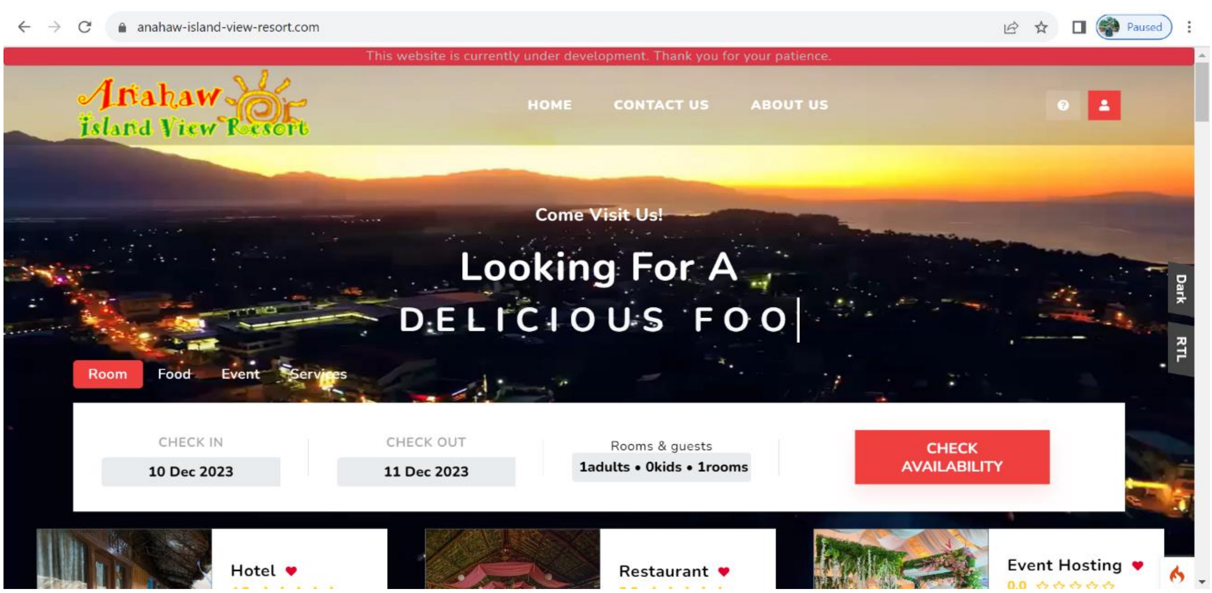

Figure 10. Homepage of the System

The figure above shows the website's landing page. Users can view the landing page without logging in to the log-in form. This page contains the Home, Contact Us, and About Us sections, offering visitors easy navigation and access to crucial information. Additionally, this design choice prioritizes user convenience and facilitates seamless exploration of the website's key features.

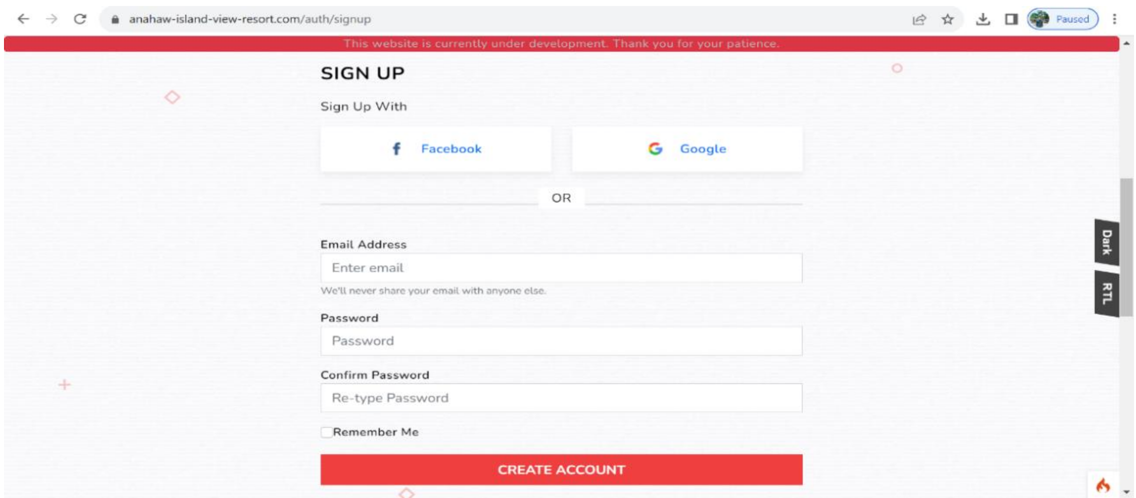

Figure 11. Register Form

The figure shows the registration form, where users are required to input their Facebook or Gmail account and their Password.

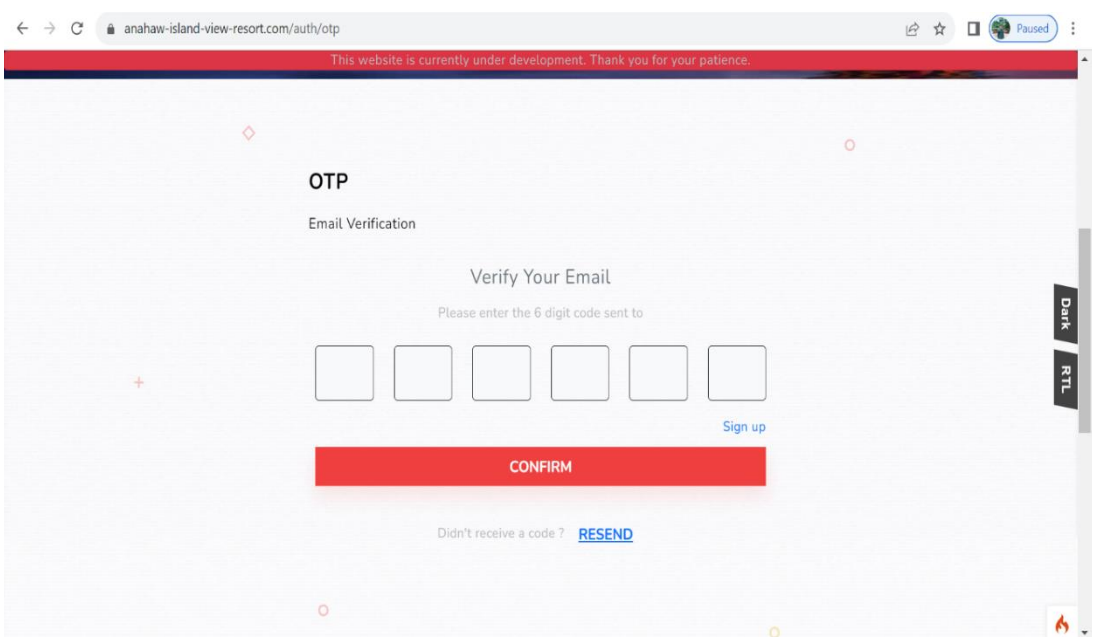

Figure 12. OTP Send

The figure above shows the creation of the account; the system will send email verification to their email, and users can now log in to the system and make reservations and order food. Upon successful email verification, users gain access to various functionalities, contributing to a seamless and secure account creation process.

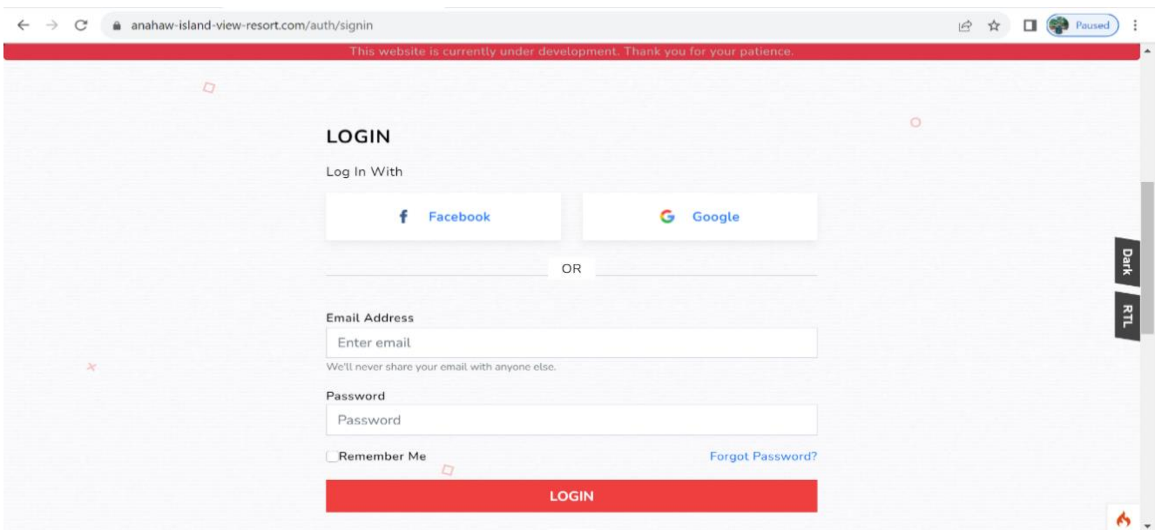

Figure 13. Login Form

The figure above shows the Login form and how the user can log in to the website after receiving the OTP Verification. This two-step authentication process adds an extra layer of security to user accounts. Once the OTP is verified, users gain access to personalized features and services, ensuring a secure and user-friendly login experience.

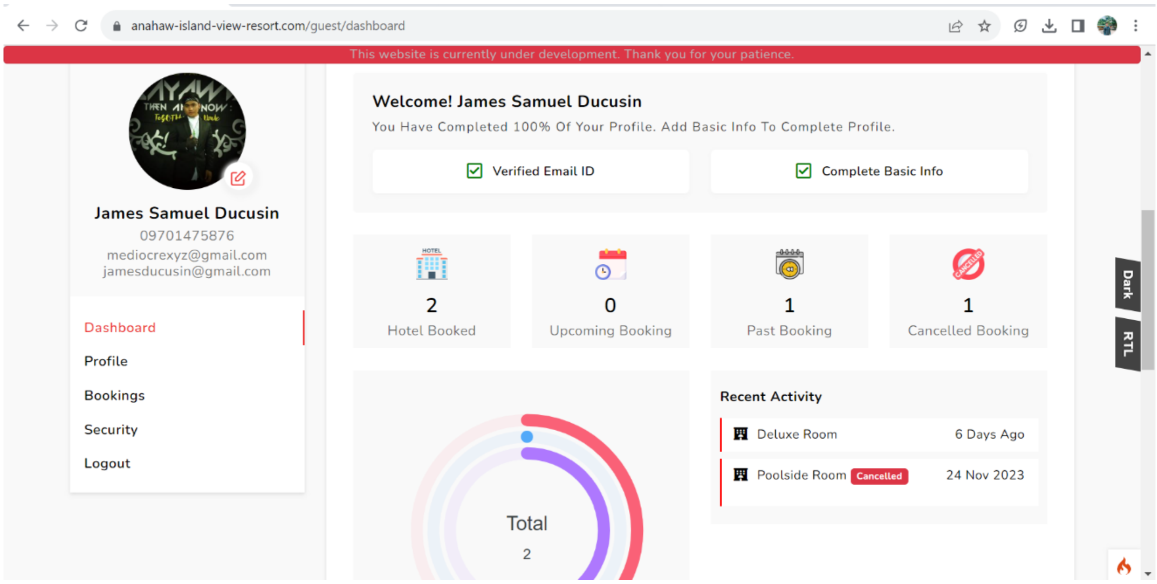

Figure 14. User Dashboard

The user dashboard is the user's landing page once logged into the system. The number of hotel bookings, upcoming reservations, past bookings, and canceled bookings is prominently displayed on the page, providing users with an instant overview of their account activity. The dashboard presents a real-time graph that dynamically visualizes data posted by the admin, offering users an interactive and insightful representation of relevant information. This feature enhances the user experience by providing a comprehensive snapshot of their engagement with the system. The intuitive design ensures users can effortlessly navigate their booking history, fostering a user-friendly interface for efficient interaction.

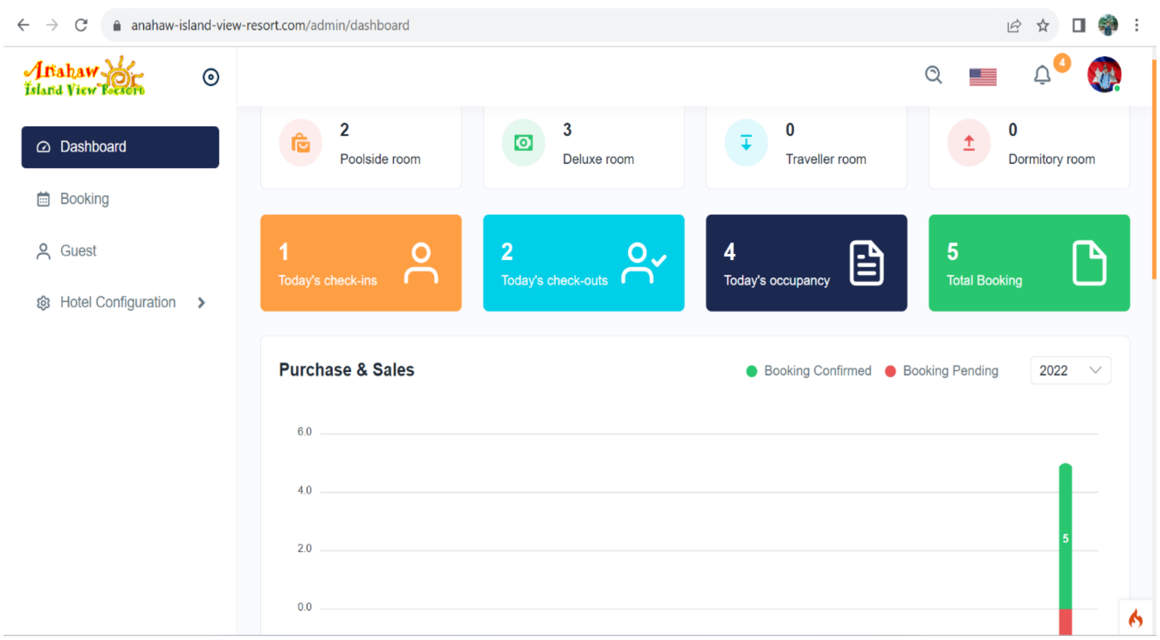

Figure 15. Admin Dashboard

The figure presents the Admin Dashboard; the admin can monitor details about the guest, including the total number of ongoing cases in different programs. Additionally, the dashboard allows the admin to track completed cases and view bookings within the system, providing a centralized hub for comprehensive oversight. This feature-rich interface empowers administrators with the tools needed to manage guest information and monitor program-related activities with ease efficiently.

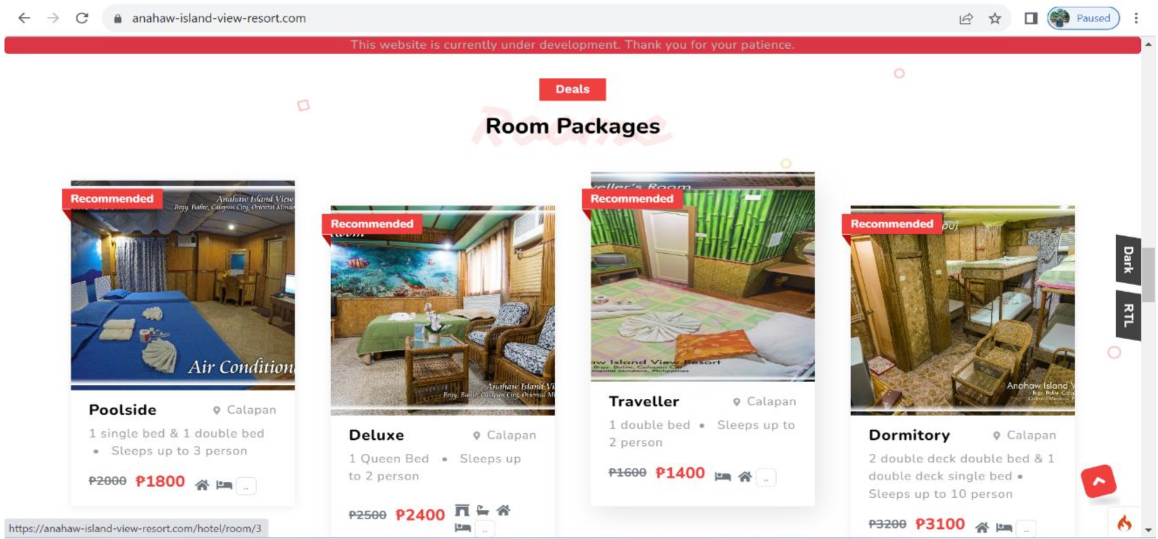

Figure 16. Room Packages

This figure shows the room packages. It presents the lists of rooms offered by the hotel; it also shows the availability status of each room under different room categories. Users can select and book based on their preferred rooms.

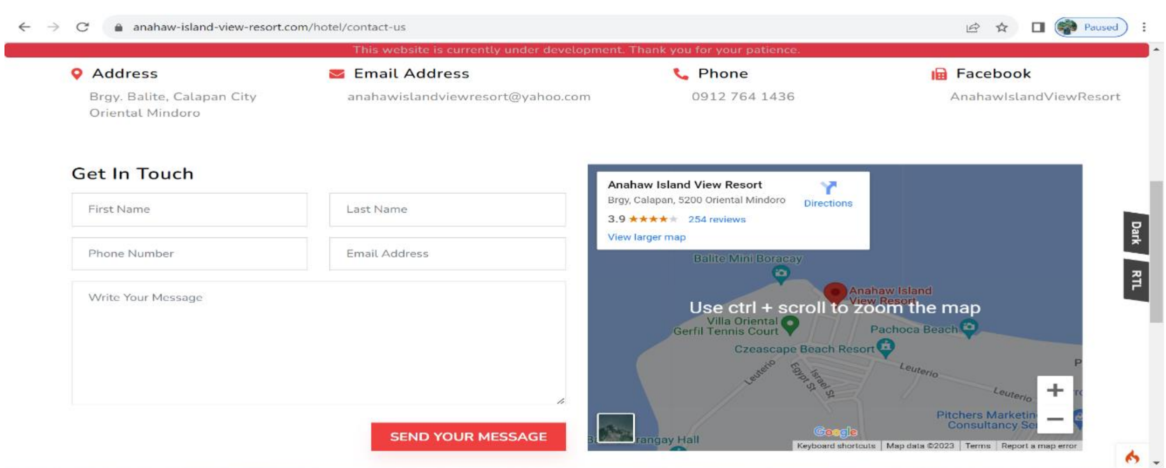

Figure 17. Contact Us

The figure above presents the Contact Us, which shows the section such as Address, Email Address, Phone Number, Facebook Page, and the location. It also has to get in touch, which can be helpful for the customer to communicate with the admin for making transactions and for other concerns.

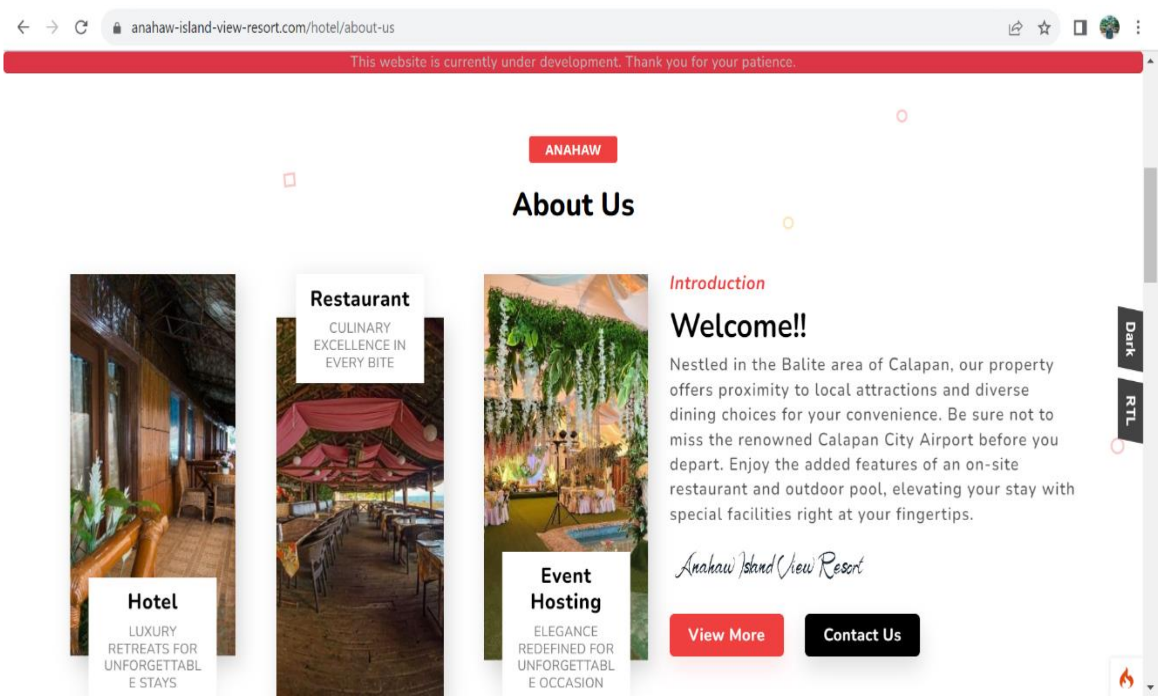

Figure 18. About Us

The figure above presents the About Us; this page provides information about the hotel and restaurant and events happening. It allows citizens to acquaint themselves with this resort further, and it continues to be patronized by the community. Notably, the About Us section serves as a vital bridge for fostering a strong connection between the establishment and the local community.

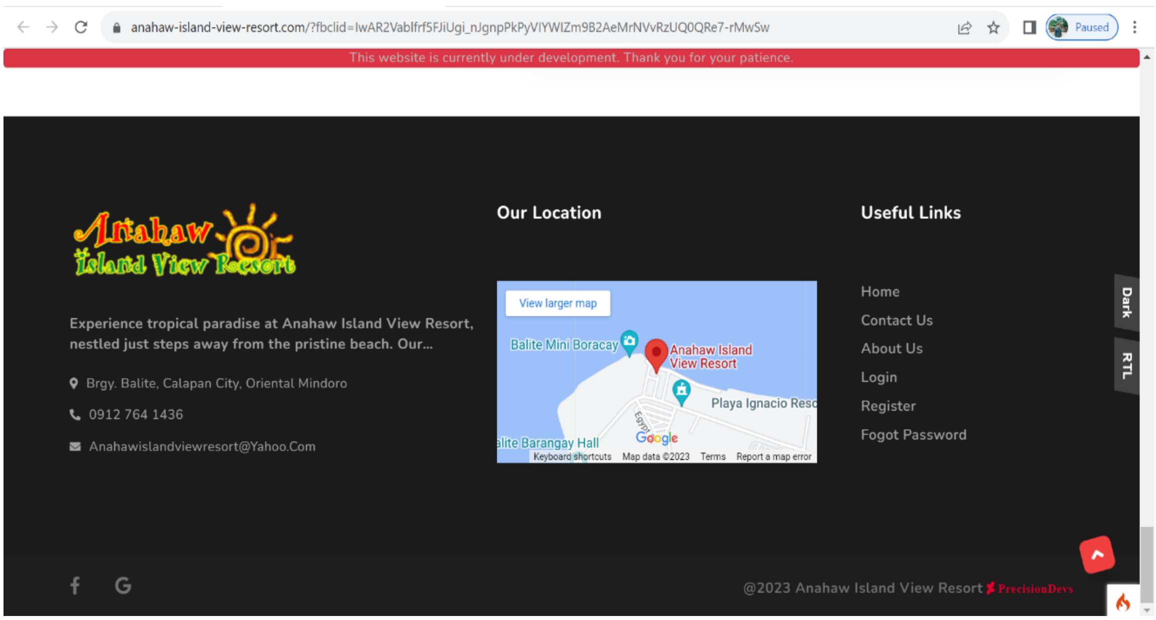

Figure 19. Website Footer

This figure presents the quick links to the website. It also contains the location and contact information of the hotel for inquiries.

Evaluation of the System

Following the development and evaluation of QRIFY, the study came to the following conclusions listed below. During the implementation of the system, an evaluation was conducted to test the system's quality in terms of functional sustainability, reliability, portability, usability, performance efficiency, security, compatibility, and maintainability.

Table 8. Functional Sustainability of the System

Functional Sustainability	Mean	Rank	Verbal Interpretation
1.1 How would you rate the system's ability to manage hotel and restaurant operations, such as reservations, check-ins, and order processing?	4.06	1	Very Good
1.2 The systems function delivers the intended result accurately.	3.98	2	Very Good
1.3 The system features able to accomplish specified tasks and objectives.	3.88	3	Very Good
Overall Mean	**3.97**		**Very Good**

The evaluations provided by the respondents are displayed as results, showing how well the system performed in terms of functional suitability. According to the feedback, users rated the website as "Very Good," with an overall mean score of 3.97. This indicated that the system effectively covered all the specified tasks and user objectives, delivering accurate results with the required precision and aligning with its intended uses. The developed system was well-suited to meet the users' needs, operating seamlessly within its intended functions. Moreover, the system's functionality aligned with the users' requirements, as indicated by the survey respondents who agreed that the website functioned as expected.

Table 9. Reliability of the System

Reliability	Mean	Rank	Verbal Interpretation
2.1 How reliable is the system regarding data accuracy and availability, especially during peak usage times?	3.9	1	Satisfactory
2.2 The system is operational and accessible whenever needed.	3.8	2	Satisfactory
Overall Mean	**3.85**		**Very Good**

According to Table 9, the respondents confirmed that the website was reliable and effective in performing its functions, which resulted in the users trusting it. The table showed the performance of the website's reliability, with an overall average score of 3.85, which was considered "Very Good" based on the verbal interpretation. The results indicated that the respondents generally agreed that the website was user-friendly and easy to operate. The system could meet the users' needs, ensure reliability in routine operations, and show resilience in managing potential component or function failures.

Table 10. Portability of the System

Portability	Mean	Rank	Verbal Interpretation
3.1 Suitable for use on any device or platform, easily accessible.	4.02	3	Very Good
3.2 The system adapts well to varying network conditions, including slow or unstable connections.	4.06	2	Very Good
3.3 The system can be utilized with modest hardware specifications.	4.14	1	Very Good
Overall Mean	**4.07**		**Very Good**

Table 10 represented the comprehensive portability performance of the website. The system achieved an overall mean of 4.07 in portability, denoting a "Very Good" rating in verbal interpretation. This favorable assessment underscores the website's effectiveness in ensuring high portability across various platforms.

Table 11. Usability of the System

Usability	Mean	Rank	Verbal Interpretation
4. 1 Users can recognize whether the system is appropriate for their needs.	3.86	6	Very Good
4. 2 The system can be used by specified users to achieve specified goals of learning to use the system effectively and efficiently, freedom from risk, and satisfaction in a specified context of use.	3.94	4	Very Good
4. 3 The system is easy to operate and control.	4.08	1	Very Good
4. 4 The system's interface and design prevent users from making errors.	3.88	5	Very Good
4. 5 The system enables a pleasing and satisfying user interface.	4.0	3	Very Good
4. 6 The system can be used by everyone, including those with disabilities or impairments.	4.04	2	Excellent
Overall Mean	**3.97**		**Very Good**

According to table 11, the website received "Very Good" ratings from users in terms of usability. The average rating was 3.97, indicating the website had a user-friendly and engaging interface. Thanks to their clear and concise labeling, the controls and buttons were easy to operate and navigate. Users also found the website easy to learn and use, making it an effective means of interaction. Overall, the website performed well in terms of usability, as per the feedback received from the respondents.

Table 12. Performance Efficiency of the System

Performance Efficiency	Mean	Rank	Verbal Interpretation
5.1 The system manages hotel and restaurant operations, in terms of speed and resource utilization.	4.06	1	Very Good
5.2 Under specific conditions, the system exhibits satisfactory performance while managing resource utilization.	3.88	2.5	Very Good
5.3 The maximum limits of the system parameter meet its requirements.	3.88	2.5	Very Good
Overall Mean	**3.94**		Very Good

The table presented the website's performance efficiency, showcasing an overall mean that indicated the system's prompt response to potential occurrences. The table illustrated the users' evaluation responses, with an overall mean of 3.94, signaling the respondents' consensus on the system's "Very Good." This was attributed to the system's ability to meet requirements regarding response and processing times during its functions. Additionally, multiple users could employ the system simultaneously without encountering any issues.

Table 13. Security of the System

Security	Mean	Rank	Verbal Interpretation
6.1 How confident are you in the system's ability to protect sensitive data and maintain the privacy of users?	4.08	1	Very Good
6.2 The system authenticates both the identity of data and its origins.	4.0	2	Very Good
Overall Mean	**4.04**		**Very Good**

The table provided an overview of the system's security performance. Respondents, on average, gave a rating of 4.04, which was interpreted verbally as "Very Good." This signified that the website boasted a robust level of security. It was evident that the system efficiently safeguarded against unauthorized data access, permitting access solely to individuals with authorized credentials.

Table 14. Compatibility of the System

Compatibility	Mean	Rank	Verbal Interpretation
7.1 The system efficiently executes its functions without causing disruptions to other systems.	4.02	1	Very Good
7.2 System seamlessly works with other existing software or hardware systems within the hotel and restaurant environment.	4.0	2	Very Good
Overall Mean	**4.01**		**Very Good**

The survey results on system compatibility revealed a strong consensus among respondents, with an overall mean of 4.1 and a verbal interpretation of "Very Good." This signified that the website consistently executed its functions seamlessly in diverse environments, encompassing software and hardware variations. Additionally, the high mean score underscores the website's robust design, instilling confidence in its ability to perform effectively across a broad spectrum of user setups.

Table 15. Maintainability of the System

Maintainability	Mean	Rank	Verbal Interpretation
8.1 How easy is it to update, maintain, and make necessary changes to the system?	4.08	1.5	Very Good
8.2 System changes, error fixes, and environmental failures require less effort.	4.06	3	Very Good
8.3 It is feasible to evaluate the effect on a product or system of a planned alteration to one or more of its components, to identify issues or causes of failures in a product, or to pinpoint components that need to be altered.	3.98	5	Very Good
8.4 The system can be improved effectively and efficiently without causing new defects or reducing the current quality of the system.	4.08	1.5	Very Good
8.5 Test standards can be set for a system, product, or component, and evaluations can be conducted to assess if those standards have been met.	4.0	4	Very Good
Overall Mean	**4.04**		**Very Good**

The presentation demonstrated the website's strong maintainability with an overall mean of 4.04, indicating a "Very Good" level. This suggests that system alterations would have minimal impact on other components.

Table 16. Summary Results of the Evaluation

Items	Mean	Rank	Verbal Interpretation
1. Functional Suitability	3.97	5.5	Very Good
2. Reliability	3.85	8	Very Good
3. Portability	4.07	1	Very Good
4. Usability	3.97	5.5	Very Good
5. Performance Efficiency	3.94	7	Very Good
6. Security	4.04	2.5	Very Good
7. Compatibility	4.01	4	Very Good
8. Maintainability	4.04	2.5	Very Good
Overall Mean	**3.99**		**Very Good**

The table presented website evaluation results based on ISO 25010 standards rated by admin, staff, customers, and IT expert. Mean scores for various characteristics were: functional suitability (M=3.97), reliability (M=3.85), portability (M=4.07), usability (M=3.97), performance efficiency (M=3.94), security (M=4.04), compatibility

(M=4.01), and maintainability (M=4.04). The overall mean of 3.99 suggests users found the system effective.

Implementation Result

This section demonstrated the overall outcome of the implementation activity's progress and how it affected the system's development and outcome.

Table 17. Implementation Result

Activities	Target Date	Progress Note	Outcome	Remarks
Discussion with client	November 30, 2022	Priority and features of the system	System functions gathered and listed as required.	Achievement discussed with a few suggestions.
Deployment Letter	November 09, 2023	Approved	The system was deployed successfully	None
System Deployment and Monitoring Period	October 13, 2023 - November 13, 2023	Few Patches	The bugs and issues were fixed	Some functionality have errors and require fixing due to incorrect results
System Evaluation	November 14, 2023	Add Features	Required features have been added	None

Implementation results in the table indicate successful system development and process based on progress notes. Researchers effectively applied system requirements and testing methods.

Chapter V

SUMMARY, CONCLUSION, AND RECOMMENDATION

This chapter summarizes this study's findings, conclusions, and recommendations. This section contains the researchers' interpretation based on the survey results and gathered data that was evaluated and analyzed.

Summary

The study aimed to develop Qrify: A Cloud-Based Hotel and Restaurant Management System for Anahaw Island View Resort. The researchers focused on designing a user-friendly user interface and incorporating technical innovation material that supported and complemented the project's ability to function and meet the needs and expectations of users.

The data collected from every respondent was used to efficiently evaluate the system's functional suitability, operating without displaying any unusual errors and functioning following user requirements. The system waBased on the acquired data, theeemed beneficial and functional based on thethat the website could continue to operate and manage updated errors.

The website underwent evaluation in compliance with the ISO 25010 standards. The system was evaluated by 50 respondents, including IT Experts, Staff, Admin, Client, and Customers. The system's performance was thoroughly tested and evaluated using ISO25010 and rated based on specific criteria. The evaluation results demonstrated the system's effectiveness, as it operated following the proposed and suggested features. Functional Suitability achieved an overall mean of 3.97, described as "Very Good"; Reliability achieved an overall mean of 3.85, described as "Very Good"; Portability achieved an overall mean of 4.07, described as "Very Good"; Usability achieved an overall mean of 3.97, described as "Very Good"; Performance Efficiency achieved an overall mean of 3.94, described as "Very Good"; Security achieved an overall mean of 4.04, described as "Very Good"; Compatibility achieved an overall mean of 4.01, described as "Very Good"; and lastly, Maintainability achieved an overall mean of 4.04, described as "Very Good."

In general, the Qrify system effectively managed hotel and restaurant operations, as evidenced by the acquired data. It significantly assisted users in disseminating information and efficiently tracking documents, empowering them to manage information seamlessly.

Conclusions

Upon the culmination of the study on Qrify: A Cloud-Based Hotel and Restaurant Management System for Anahaw Island View Resort, several key conclusions and observations were identified:

1. The successful development of a user-friendly web application with these features is expected to enhance efficiency and convenience for both staff and guests significantly.
2. Implementing personalized services is anticipated to lead to an enhanced overall guest experience, potentially resulting in increased guest satisfaction and loyalty.
3. Integrating various departments is expected to streamline operations, fostering better communication and coordination. This could reduce errors, improve workflow, and increase overall operational efficiency.
4. Assessing the web application's effectiveness in QRify operations and enhancing the guest experience will provide valuable insights. Positive outcomes, such as improved staff productivity, faster service

delivery, and positive guest feedback, suggest the system's success in meeting its objectives.

Recommendations

1. For future researchers, they should include a notification to the admin dashboard to alert staff about newly received data.
2. For future researchers, a more attractive and responsive website is recommended.
3. Future researchers in real-time chat systems should enhance user experience, prioritize security measures, and explore innovative features like advanced encryption.
4. The owner is advised to secure a reliable internet connection to support the system's optimal performance.

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APPENDIX A

Project Team Assignments Form

Team Alias	**QRIFY: A CLOUD-BASED HOTEL AND RESTAURANT MANAGEMENT SYSTEM FOR ANAHAW ISLAND VIEW RESORT**

Name and Signature	Project Role	Email Address and Number
James Samuel B. Ducusin	❖ Programmer ❖ Database Manager ❖ Researcher	jamesducusin090500@gmail.com 09701475876
Arvin M. Divinagracia	❖ Project Manager ❖ Design and Layout ❖ Researcher	arvin.divinagracia7@gmail.com 09517391746
Joshua S. Mercene	❖ System Analyst ❖ Technical writer ❖ Researcher	togzzzaldous@gmail.com 09514354485

APPENDIX B

Capstone Project Topic Proposal Form

CAPSTONE/RESEARCH PROJECT TOPIC PROPOSAL	
Proposed Title	**QRIFY: A CLOUD-BASED HOTEL AND RESTAURANT MANAGEMENT SYSTEM FOR ANAHAW ISLAND VIEW RESORT**
Name of Student/ Course, Year & Section	Arvin B. Divinagracia BSIT 4-F1 James Samuel B. Ducusin BSIT 4-F1 Joshua S. Mercene BSIT 4-F1
Introduction	In the fast-paced and dynamic realm of the hospitality industry, staying ahead required more than just impeccable service; it demanded a sophisticated approach to management. Recognizing this need for seamless operations, Anahaw Island View Resort was proud to introduce "QRIFY" – a revolutionary Cloud-based Hotel and Restaurant Management System designed to elevate the resort's efficiency and redefine guest experience. Due to fierce competition, declining spending per customer, and increasing costs, hotel restaurants had been under pressure to obtain and retain customers. This situation had been compounded by the unprecedented shock of a worldwide public health emergency. With the outbreak of COVID-19, non-essential human movements and economic activities were paused from late January to April 2020 in China (Liang, 2020). The website's overall quality played a crucial role in delivering the intended message to potential customers. The website's usability was also a significant factor to consider providing a better user experience, ensuring users could easily navigate and use the website's features. The website should

	have been user-friendly, interactive, and accessible to all ages and genders (Pathak, A., et al., 2021). The hotel and restaurant management system enabled users to interact with the website by creating, modifying, and making changes anytime they needed to. This meant that the system was not static but was able to provide real-time services, ensuring customer satisfaction (Buhalis, D., et al., 2019). In this regard, the study aimed to provide an online web application that served as a one-stop shop for all hotel and restaurant services. The system provided various features such as online reservations, menu viewing, order placement, and access to hotel and restaurant amenities. The system's goal was to streamline operations, provide efficient and reliable services, and enhance the overall guest experience.
Statement of the Problem	A Cloud-based Hotel and Restaurant Management System for Anahaw Island View Resort addressed several critical challenges. Manual information dissemination led to delays, and the current document tracking system was both time-consuming and error-prone. Data security concerns loomed over the resort's handling of sensitive information, while manual processes for reservation and inventory management hindered operational efficiency. Limited collaboration and communication among staff further impacted decision-making. Additionally, the resort faced difficulties in adapting to industry changes. QRIFY aimed to revolutionize these aspects by providing a centralized, secure, and interactive platform, streamlining operations, and ensuring adaptability to evolving industry standards.
Objectives of the Study	The general objective of the study was to develop a web application for a Hotel and

	Restaurant Management System that would streamline operations and enhance the guest experience. Specifically, the study aimed to: 1. Develop a user-friendly web application with features such as online reservations, room and table management, chat room, inventory and order management, and billing and payment processing. 2. Enhance guest experience by providing personalized services such as room preferences, special requests, and QR-code-enabled menu ordering. 3. QRify operations by integrating various departments, such as the front desk, cashier, and management, to improve communication and coordination. 4. Evaluate the effectiveness of the developed web application in QRify operations and enhancing the overall guest experience.
Scope and Limitation of the Study	The study focused on developing an online web application for streamlining operations and enhancing guest experience for a hotel and restaurant management system. The scope of the study included the development of a web application for a hotel and restaurant management system with features such as reservation, room service, and billing. The web application allowed guests to make reservations, view menu items, order food and drinks, make payments online, and leave comments and ratings. The hotel and restaurant management had access to the web application's backend, which included features for managing orders, inventory, and customer data. As a young developer, the web application features were yet to expand but were not limited to its current features and were expected to scale in the future. The delimitation of the study included the following: The study was exclusively for the hotel and restaurant, and no other establishments were included. All the exclusive features stated in the study were only for the guests

	and management of the hotel and restaurant. The web application was designed only for customers who had made a reservation with the hotel and restaurant, and other persons could still visit the website with limited features. The study did not cover the development of a mobile application or other software systems beyond the web application. Furthermore, the focus was specifically on enhancing the digital experience within the confines of the hotel and restaurant domain.
Review of Related Literature and System	The researchers have gathered the following related works regarding the proposed research topic. According to Lee et al., 2019; Zhao & Bacao, (2020), Recent studies in the OFDS literature have shown no significant correlation between effort expectancy and the continuous usage intention of OFDS, indicating that customers experience little difficulty using new apps, including OFDS, as the use of smartphones and apps has matured and the interface of smartphone apps has stabilized due to the development of information and communication technologies over time. According to Parulis-Cook (2020), for opulent hotels and their restaurants, survival has taken precedence amid the pandemic. Offering an ordering and home delivery service with online-to-offline (O2O) food delivery companies like Meituan and Eleme—two O2O services heavyweights in China—has been one of the tactics used by luxury hotel restaurants in China to stay alive. According to Kiatkawsin and Han, (2019), Some comments also revealed that when the restaurants failed to deliver as expected, their brand credibility was damaged. Many customers complained about the ordinariness of the food, noting that the taste was just "too normal and not distinctive enough." When

	there are discounts or special offers, these criticisms become more serious. A probable explanation is that lowering the price is viewed as a problematic promotion because a high price is one of the defining characteristics of upscale dining establishments. According to Prasetyo et al. (2020) incorporated PMT and the Theory of Planned Behavior (TPB) to examine the efficacy of Covid 19 prevention measures. The result shows how perceptions of a person's fragility and seriousness can indirectly influence behavioral intentions and result in actual behavior. In particular in Indonesia, no research has ever been done on the relationship between PMT and TPB in the application of QR codes in the restaurant industry. Therefore, the purpose of this study is to evaluate consumer perceptions of the use of QR codes by integrating PMT and TPB theories to shed light on the restaurant industry, particularly in Indonesia. According to Yang et al. (2019), People are deterred from using QR codes if their use is hard and full of difficulties. They will, however, accept the technology if mastering QR codes doesn't take a lot of time, effort, or money. The results showed that people are less likely to adopt new technologies when the response cost is higher.
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Prepared by:

Arvin M. Divinagracia	**James Samuel B. Ducusin**	**Joshua S. Mercene**
BSIT 4-F1	BSIT 4-F1	BSIT 4-F1

APPENDIX C

Notice of Title Acceptance

C E R T I F I C A T I O N

The undersigned members comprising the panel for oral examination hereby approve the Research/Capstone Project entitled **QRIFY: A CLOUD-BASED HOTEL AND RESTAURANT MANAGEMENT SYSTEM FOR ANAHAW ISLAND VIEW RESORT** including its team members composed of James Samuel B. Ducusin, Arvin M. Divinagracia, Joshua S. Mercene.

JESUS M. BAUTISTA, *MSIT*
Research Adviser

DEZZA MARIE M. MAGSINO, *MSIT*
Program research Coordinator

EPIE F. CUSTODIO, *DIT*
Program Chairperson

LEONEL C. MENDOZA
Research & Development Coordinator

JOHN EDGAR S. ANTHONY, *MSIT*
College Dean

APPENDIX D

Endorsement for Final Defense

CAPSTONE PROJECT FINAL DEFENSE ENDORSEMENT

This Research/ Capstone Project entitled **"QRIFY: A CLOUD-BASED HOTEL AND RESTAURANT MANAGEMENT SYSTEM FOR ANAHAW ISLAND VIEW RESORT."** prepared and submitted by: James Samuel B. Ducusin, Arvin M. Divinagracia, and Joshua S. Mercene as a partial fulfillment of the requirement for the degree Bachelor of Science in Information Technology is hereby accepted and recommended for FINAL ORAL DEFENSE by me as their Capstone Project Adviser after thorough review of their manuscript and system output. Their oral presentation may be scheduled on December 14, 2023.

JESUS M. BAUTISTA, *MSIT*
Capstone Project Adviser

Noted:

EPIE F. CUSTODIO*, DIT*
Program Chairperson, BSIT

JOHN EDGAR S. ANTHONY, *MSIT*
Dean, College of Computer Studies

APPENDIX E

GRAMMARIAN'S CERTIFICATE

Date: January/26/2024

<u>GRAMMARIAN'S CERTIFICATE</u>

This is to certify that the undersigned has reviewed the grammatical construction and the text organization of this Research/ Capstone Project entitled **"QRIFY: A CLOUD-BASED HOTEL AND RESTAURANT MANAGEMENT SYSTEM FOR ANAHAW ISLAND VIEW RESORT."**

Signed:

____

ALICE R. RAMOS, *Ph.D.*
Grammarian

Conformed:

____

JOSHUA S. MERCENE
Project Manager

APPENDIX F
Certification of Originality

CERTIFICATION OF ORIGINALITY

This is to certify that the research work presented in this Research/Capstone Project, QRIFY: A CLOUD-BASED HOTEL AND RESTAURANT MANAGEMENT SYSTEM FOR ANAHAW ISLAND VIEW RESORT for the degree Bachelor of Science in Information and Technology at the Mindoro State University embodies the result of original and scholarly work carried out by the undersigned. This Research/Capstone Project does not contain words or ideas taken from published sources or written works that have been accepted as basis for the award of a degree from any other higher education institution, except where proper referencing and acknowledgment were made.

ARVIN M. DIVINAGRACIA
Researcher

JAMES SAMUEL B. DUCUSIN
Researcher

JOSHUA S. MERCENE
Researcher

APPENDIX G

Program Listing

HotelAdmin.php

```
<?php

namespace App\Controllers;

use App\Controllers\BaseController;

class HotelAdmin extends BaseController
{

    public function __construct()
    {

        helper(['url', 'form',
'Toastr']);
        $this->room_type_model =
model('RoomType');
        $this->rooms_model =
model('RoomModel');
        $this->housekeeping_model =
model('HousekeepingModel');
        $this->user_model =
model('UserModel');
        $this->amenity_model =
model('AmenityModel');
        $this->service_model =
model('PaidServiceModel');
        $this->price_model =
model('PriceModel');
        $this->order_model =
model('OrderModel');

    }
    public function dashboard()
    {
        $data = [
            'booking' => $this-
>order_model->select('*, profiles.id as
pid, rooms.id as rid, rt.id as rtid,
orders.id as oid, concat(firstname, " ",
lastname) as fullname')
                ->join("profiles",
"orders.user_id = profiles.user_id",
"left")
                ->join('rooms',
'orders.room_id = rooms.id', 'left')
                ->join('room_types as
rt', 'orders.room_type_id = rt.id',
'left')
                ->findAll()
        ];
        return view('admin/dashboard',
$data);
    }
    public function room_type()
    {
        $data = [
            'room_type_datas' => $this-
>room_type_model->findAll(),
            'amenities_datas' => $this-
>amenity_model
            ->select('*')
            ->join('roomtypes_amenities',
'amenities.id =
roomtypes_amenities.amenity_id', 'left')
            ->findAll()
        ];
        //
var_dump($data['amenities_datas']);
        // return $this->response-
>setJson($data);
        return view('admin/room-type',
$data);
    }
    public function booking()
    {
        $data = [
            'booking' => $this-
>order_model->select('*, profiles.id as
pid, rooms.id as rid, rt.id as rtid,
orders.id as oid, concat(firstname, " ",
lastname) as fullname')
                ->join("profiles",
"orders.user_id = profiles.user_id",
"left")
                ->join('rooms',
'orders.room_id = rooms.id', 'left')
                ->join('room_types as
rt', 'orders.room_type_id = rt.id',
'left')
                ->findAll()
        ];
        return view('admin/booking',
$data);
    }
    public function room()
    {
        $room_value = $this->request-
>getPost('room_value');
        $room_numbers = $this->request-
>getPost('room_number');

        if($room_value ===
'insert_room'){

                $floor = $this->request-
>getPost('floor');
                $room_numbers = $this-
>request->getPost('room_number');
                $room_type_ids = $this-
>request->getPost('room_type_id');

                // I-check kung parehas
ang haba ng mga array
                if (count($room_numbers)
=== count($room_type_ids)) {
                    $values = [];
```

```
                    // Gamitin ang
foreach loop para ma-iterate ang mga room
numbers at room type IDs nang sabay
                    foreach
($room_numbers as $key => $room_number) {

                        $values[] = [
                            'floor' =>
$floor,
                            'roomNumber'
=> $room_number,
                            'type_id' =>
$room_type_ids[$key],
                            'status' =>
'0'
                        ];
                    }
                }

                foreach ($values as
$value){
                    $this->rooms_model-
>insert($value);
                }

                return
redirect('admin/room')->with('success',
'Room has been added');

        }
        elseif($room_value ===
'update_room'){
            $id = $this->request-
>getPost('id');

            $values = [
                'floor' => $this-
>request->getPost('floor'),
                'roomNumber' => $this-
>request->getPost('room_number'),
                'type_id' => $this-
>request->getPost('room_type_id')
            ];

            $this->rooms_model-
>update($id, $values);

            return
redirect('admin/room');
        }
        elseif($room_value ===
'delete_room_value'){
            $room_id = $this->request-
>getPost('room_id');

            $this->rooms_model-
>delete($room_id);
        }
        elseif($room_value ===
'housekeeping_status'){
            $date_limit_str  = $this-
>request->getPost('date_limit');
            $datetime =
date_create_from_format('j M Y',
$date_limit_str);
            $formatted_date_limit =
$datetime->format('Y-m-d H:i:s');
            $values = [
                'room_id' => $this-
>request-
>getPost('housekeeping_room_id'),
                'status' => ($this-
>request->getPost('status') ===
'Dirty')?0:1,
                'remarks' => $this-
>request->getPost('remarks'),
                'user_id' => $this-
>request->getPost('user_id'),
                'date_limit' =>
$formatted_date_limit
            ];
            //
var_dump($values['status']);

            $housekeeping_find = $this-
>housekeeping_model->where('id', $this-
>request->getPost('housekeeping_id'))-
>first();

            if($housekeeping_find ==
null){
                $this-
>housekeeping_model->insert($values);
            }
            else{
                $this-
>housekeeping_model-
>update($housekeeping_find['id'],
$values);
            }
            return
redirect('admin/room');
        }
        else
        {
            $data = [
                'room_type_datas' =>
$this->room_type_model->findAll(),
                'rooms' => $this-
>rooms_model
                ->select('*, rooms.id,
housekeeping.id as housekeeping_id,
housekeeping.status as
housekeeping_status,  rooms.status as
rooms_status')
                ->join('room_types',
'rooms.type_id = room_types.id', 'inner')
                ->join('housekeeping',
'rooms.id = housekeeping.room_id',
'left')
                ->join('profiles',
'housekeeping.user_id =
profiles.user_id', 'left')
                ->groupBy('rooms.id')
                ->findAll(),

                'users' => $this-
>user_model
                ->select('*, users.id')
                ->join('profiles',
'users.id = profiles.user_id', 'inner')
                ->where('usertype',
'Guest')
                ->findAll()
            ];
```

```
        return view('admin/room', $data);

        }

    }
    public function price_manager()
    {
        $price_manager = $this->request-
>getPost('price_manager');
        if($price_manager ==
"insert_data"){
            $data = [
                'room_type_id' => $this-
>request->getPost('room_type'),
                'mon' => $this->request-
>getPost('mon'),
                'tue' => $this->request-
>getPost('tue'),
                'wed' => $this->request-
>getPost('wed'),
                'thu' => $this->request-
>getPost('thu'),
                'fri' => $this->request-
>getPost('fri'),
                'sat' => $this->request-
>getPost('sat'),
                'sun' => $this->request-
>getPost('sun'),
            ];

            $this->price_model-
>insert($data);

            return
redirect('admin/price/manager')-
>with('danger', 'Room has already been
added');
            // var_dump($data);
        }
        else if($price_manager ==
"edit_data"){
            $room_name = $this->request-
>getPost('name');
            $room_type = $this-
>room_type_model->where('name',
$room_name)->first();
            $id = $this->request-
>getPost('id');

            $data = [
                'room_type_id' =>
$room_type['id'],
                'mon' => $this->request-
>getPost('mon'),
                'tue' => $this->request-
>getPost('tue'),
                'wed' => $this->request-
>getPost('wed'),
                'thu' => $this->request-
>getPost('thu'),
                'fri' => $this->request-
>getPost('fri'),
                'sat' => $this->request-
>getPost('sat'),
                'sun' => $this->request-
>getPost('sun'),
            ];

            $this->price_model-
>update($id, $data);

            return
redirect('admin/price/manager')-
>with('danger', 'Room has already been
added');

        }
        else if($price_manager ==
"delete_data"){

            $id = $this->request-
>getPost('id');

            $this->price_model-
>delete($id);
        }
        else{

            $data = [
                'price_managers' =>
$this->price_model
                ->select('*, prices.id as
room_types_id')
                ->join('room_types',
'prices.room_type_id = room_types.id',
'inner')
                ->findAll(),
                'room_types' => $this-
>room_type_model->findAll()
            ];
            return view('admin/price-
manager', $data);
        }

    }
    public function service()
    {
        $service_data = $this->request-
>getPost('service_data');
        if($service_data ==
'insert_data'){
            $data = [
                'title' => $this-
>request->getPost('title'),
                'description' => $this-
>request->getPost('short_desc'),
                'price' => $this-
>request->getPost('price'),
                'status' => '1'
            ];

            $this->service_model-
>insert($data);

            return
redirect('admin/service')-
>with('success', 'Room has been added');
        }
        else if($service_data ==
'edit_data'){
            $id = $this->request-
>getPost('id');

            $data = [
                'title' => $this-
>request->getPost('title'),
```

```
                'description' => $this-
>request->getPost('short_desc'),
                'price' => $this-
>request->getPost('price'),
            ];

            $this->service_model-
>update($id, $data);

            return
redirect('admin/service')-
>with('success', 'Room has been added');
        }
        else if($service_data ==
'delete_data'){

            $id = $this->request-
>getPost('id');

            $this->service_model-
>delete($id);
        }
        else if($service_data ==
'status_slider'){
            $id = $this->request-
>getPost('id');
            $data = [
                'status' => $this-
>request->getPost('status'),
            ];
            $this->service_model-
>update($id, $data);
        }
        else{
            $data = [
                'services' => $this-
>service_model->findAll(),
            ];
            return view('admin/service',
$data);
        }

        // var_dump($data['service']);
    }
    public function housekeeping()
    {
        $update_housekeeping = $this-
>request->getPost('update_housekeeping');

        if($update_housekeeping ==
'update_housekeeping'){
            $idd = $this->request-
>getPost('idd');

            $data = [
                'status' => ($this-
>request->getPost('housekeeping_status')
=== 'Dirty')?0:1,
                'remarks' => $this-
>request->getPost('remarks'),
                'user_id' => $this-
>request->getPost('user_id'),
                'date_limit' => $this-
>request->getPost('date_limit'),
            ];

            $this->housekeeping_model-
>update($idd, $data);

            // var_dump($this->request-
>getPost('housekeeping_status'));

            return
redirect('admin/housekeeping')-
>with('success', 'Room has been added');
        }
        elseif($update_housekeeping ==
'add_housekeeping'){

            $date_limit_str  = $this-
>request->getPost('date_limit');
            $datetime =
date_create_from_format('j M Y',
$date_limit_str);
            $formatted_date_limit =
$datetime->format('Y-m-d H:i:s');

            $data = [
                'room_id' => $this-
>request->getPost('roomNumber'),
                'status' => $this-
>request->getPost('housekeeping_status'),
                'remarks' => $this-
>request->getPost('remarks'),
                'user_id' => $this-
>request->getPost('user_id'),
                'date_limit' =>
$date_limit_str,
            ];

            $housekeeping_id_finder =
$this->housekeeping_model-
>where('room_id', $this->request-
>getPost('roomNumber'))->first();

            if($housekeeping_id_finder ==
null){
                $this-
>housekeeping_model->insert($data);
            }
            else{
                $this-
>housekeeping_model-
>update($housekeeping_id_finder['id'],
$data);
            }

            // var_dump($data);
            return
redirect('admin/housekeeping')-
>with('success', 'Room has been added');

        }
        else if($update_housekeeping ==
"delete_data"){

            $idd = $this->request-
>getPost('idd');

            $this->housekeeping_model-
>delete($idd);
        }
        else{
            $data = [
                'datas' => $this-
>housekeeping_model
                ->select('*,
housekeeping.id as housekeeping_id,
```

```
housekeeping.status as
housekeeping_status')
                ->join('rooms',
'housekeeping.room_id = rooms.id',
'inner')
                ->join('room_types',
'rooms.type_id = room_types.id', 'inner')
                ->join('users',
'housekeeping.user_id = users.id',
'inner')
                ->join('profiles',
'users.id = profiles.user_id', 'inner')
                -
>groupBy('housekeeping.room_id')
                ->findAll(),

                'room_data' => $this-
>rooms_model->findAll(),

                'users' => $this-
>user_model
                ->select('*, users.id')
                ->join('profiles',
'users.id = profiles.user_id', 'inner')
                ->where('usertype',
'Guest')
                ->findAll()
            ];
            return
view('admin/housekeeping', $data);
        }

    }

    public function guest()
    {
        $data = [
          'users' => $this->user_model-
>select('*, concat(firstname, " ",
lastname) as fullname, users.id,
concat(street, " st.,", barangay, ", ",
city, " ", zip) as address')
          ->join('profiles', 'users.id =
profiles.user_id', 'inner')
          ->where('usertype', 'Guest')
          ->findAll()
        ];
        return view('admin/guest',
$data);
    }
    public function amenity()
    {
        $edit_data = $this->request-
>getPost('edit_data');
        if($edit_data == 'edit_data'){
            $id = $this->request-
>getPost('id');

            $data = [
                'name' => $this->request-
>getPost('name'),
                'description' => $this-
>request->getPost('description'),
                'icon' => $this->request-
>getPost('icon'),
                'for_all' => $this-
>request->getPost('for_all'),
            ];
            $this->amenity_model-
>update($id, $data);

            return
redirect('admin/amenity')-
>with('success', 'Room has been added');
        }
        else if($edit_data ==
'status_slider'){
            $id = $this->request-
>getPost('id');
            $data = [
                'status' => $this-
>request->getPost('status'),
            ];
            $this->amenity_model-
>update($id, $data);

        }
        else if($edit_data ==
'insert_data'){
            $data = [
                'name' => $this->request-
>getPost('name'),
                'description' => $this-
>request->getPost('description'),
                'icon' => $this->request-
>getPost('icon'),
                'for_all' => $this-
>request->getPost('for_all'),
                'status' => '1'
            ];

            $this->amenity_model-
>insert($data);

            return
redirect('admin/amenity')-
>with('success', 'Room has been added');
        }
        else if($edit_data ==
'deleted_at'){

            $id = $this->request-
>getPost('id');

            $this->amenity_model-
>delete($id);
        }
        else{
            $data = [
                'amenities_datas' =>
$this->amenity_model
                ->groupBy('amenities.id')
                ->findAll()
            ];
            //
dd($data['amenities_datas']);
            return view('admin/amenity',
$data);
        }

        //
var_dump($data['amenities_datas']);
    }
    public function walkthrough()
    {
        return view('admin/walkthrough-
booking');
```

```
    }

}
```

HotelGuest.php

```
<?php

namespace App\Controllers;

use App\Controllers\BaseController;
use DateTime;
use DateInterval;

class HotelGuest extends BaseController
{

    protected $roomModel;
    protected $roomTypeModel;
    protected $uspModel;
    protected $serviceModel;
    protected $reviewModel;
    protected $tabModel;
    protected $promoModel;
    protected $amenityModel;
    protected $userModel;
    protected $orderModel;
    protected $profileModel;

    public function __construct()
    {

        $this->amenityModel = new
\App\Models\AmenityModel();
        $this->roomModel = new
\App\Models\RoomModel();
        $this->roomTypeModel = new
\App\Models\RoomType();
        $this->serviceModel = new
\App\Models\ServiceModel();
        $this->reviewModel = new
\App\Models\ReviewModel();
        $this->promoModel = new
\App\Models\PromoModel();
        $this->userModel =  new
\App\Models\UserModel();
        $this->profileModel =  new
\App\Models\ProfileModel();
        $this->orderModel =  new
\App\Models\OrderModel();
    }
    public function index()
    {
        $this->session-
>remove('check_availability');
        $currentday =
strtolower(date('D'));
        $data = [
            'settings' => $this-
>settings,
            'rooms' =>$this-
>roomTypeModel->select("*, rooms.id as
rid, room_types.id as rtid, rooms.status
as rstatus, room_types.status as
rtstatus,  {$currentday} as priceToday")
            ->join('rooms',
'room_types.id = rooms.type_id', 'inner')
            ->join('prices p',
'room_types.id = p.room_type_id',
'inner')
            ->groupBy('room_types.id')
            ->findAll(),
            'services' => $this-
>serviceModel->select('*, services.id as
sid, reviews.id as rid')
                ->selectAvg('rate')
                ->selectCount('rate',
'rateCount')
                ->orderBy('rate', 'desc')
                ->orderBy('rateCount',
'desc')
                ->join('reviews',
'services.id = reviews.service_id',
'left')
                ->groupBy('services.id')

                ->findAll(),

            'featured' => $this-
>promoModel->select('*, promos.name as
pname, promos.description as
pdescription')
                ->join('room_types',
'room_types.id = promos.roomtype_id',
'inner')
                ->where(["promos.status"
=> 1])->find(),
            'amenities' => $this-
>amenityModel->select('*, rt.id as rtid,
amenities.id as aid, amenities.name as
aname')
                -
>join('roomtypes_amenities rta',
'amenities.id = rta.amenity_id', 'inner')
                ->join('room_types rt',
'rta.roomtype_id = rt.id', 'inner')
                ->where('for_all', 0)

                ->findAll(),
        ];
        // var_dump($data['amenities']);
        return view('guest/mix-layout',
$data);
    }

    public function listing()
    {
        if($this->request->getPost()) {
            $checkin_str = $this-
>request->getPost("checkin");
            $checkout_str = $this-
>request->getPost("checkout");
            $rooms = $this->request-
>getPost("quant[1]");
            $adults = $this->request-
>getPost("quant[2]");
            $kids = $this->request-
>getPost("quant[3]");
            $guest = $adults + $kids;

            $checkin = date("Y-m-d",
strtotime($checkin_str));
            $checkout = date("Y-m-d",
strtotime($checkout_str));
```

```
            $date = new
DateTime($checkin);
            $date->add(new
DateInterval('P1D'));
            if($checkout < $checkin ||
$checkout == $checkin)
                $checkout = $date-
>format("Y-m-d");

            $checkinDate =
DateTime::createFromFormat('Y-m-d',
$checkin);
            $checkoutDate =
DateTime::createFromFormat('Y-m-d',
$checkout);

                if ($checkoutDate
instanceof DateTime) {
                    $interval =
$checkinDate->diff($checkoutDate);
                    $numberOfNights =
$interval->days;
                    echo "Number of
nights: $numberOfNights";
                } else {
                    echo "Invalid
checkout date.";
                }



            $data = [
                'checkin' =>
$checkin_str,
                'checkout' =>
$checkout_str,
                'rooms' => $rooms,
                'guest' => $guest,
                'nights' =>
$numberOfNights,
                'adults' => $adults,
                'kids' => $kids,
            ];

            $this->session-
>set('check_availability', $data);

            $rooms_available = $this-
>roomModel
                ->select('*, rooms.id as
rid, rt.id as rtid, o.id as oid,
count(rooms.id) as room_count')
                ->join('room_types as
rt', 'rt.id = rooms.type_id', 'left')
                ->join('orders as o',
'o.room_id = rooms.id', 'left')
                ->where("('$checkin' >=
check_out OR '$checkout' <= check_in)",
null, false) // No date overlap
                ->orWhere('o.id', null)
                ->whereNotIn('rt.id',
[4])
                ->groupBy('rt.id')
                ->findAll();


                // dd($roomTypeCounts);
            $data = [
                    "settings" => $this-
>settings,
                    'services' => $this-
>serviceModel->select('*, services.id as
sid')
                    ->findAll(),
                    "rooms" =>
$rooms_available,
                    'amenities' => $this-
>roomTypeModel->select('*')
                        -
>join('roomtypes_amenities rta',
'room_types.id = rta.roomtype_id',
'right')
                        ->join('amenities
a', 'rta.amenity_id = a.id', 'right')
                        -
>where('a.for_all', 1)->findAll(),
                    'booked_today' =>
$this->orderModel-
>select("room_type_id")-
>selectCount("room_type_id",
"booked_count")
                        -
>where("DATE(created_at)", date('Y-m-d'))
                        -
>groupBy("orders.room_type_id")
                        ->findAll(),

                    "configuration" =>
["checkin" => $checkin_str, "checkout" =>
$checkout_str, "rooms" => $rooms,
"adults" => $adults, "kids" => $kids],
                ];
                // dd($this->session-
>get(array('check_availability')['guest']
));
                return view('guest/hotel-
listing', $data);
        }

        $data = [
            "settings" => $this-
>settings,
        ];
        return view('guest/hotel-
listing', $data);
    }

    public function booking($room_type_id
= null)
    {
        if ($this->request->isAJAX()) {


            $data = [
                'firstname' => $this-
>request->getVar('fname'),
                'lastname' => $this-
>request->getVar('lname'),
                'contact' => $this-
>request->getVar('phone'),
                'email_sub' => $this-
>request->getVar('email'),


            ];
```

```
            $data['session'] = "Contact
has been updated succesfully.";
            $this->profileModel-
>where("user_id", $this->session-
>get("id"))->set($data)->update();

            return json_encode($data);
        }
        if($this->request->getPost()){
            $image = $this->request-
>getFile('image');

            if (!$image->hasMoved()) {
                $newName = $image-
>getRandomName();
                $image->move(FCPATH .
'/assets/hotelresto/images/proof_payment/
', $newName);
            }
            $data = [

                'reference_number' =>
$this->request-
>getPost('reference_number'),
                'amount' => $this-
>request->getPost('amount'),
                'proof_payment' =>
$newName,

            ];
            $data = [
                'settings' => $this-
>settings,
            ];

            //  dd($data);
            return view("guest/success",
$data);
        }
        $currentday =
strtolower(date('D'));
        $data = [
            'settings' => $this-
>settings,
            'user' => $this->userModel-
>select('*')
                ->join('profiles',
'users.id = profiles.user_id', 'inner')
                ->find($this->session-
>get("id")),
                'details' => $this-
>roomTypeModel->select("*,
room_types.name as rname, rooms.id as
rid, room_types.id as rtid, rooms.status
as rstatus, room_types.status as
rtstatus,  {$currentday} as priceToday")
                ->join('rooms',
'room_types.id = rooms.type_id', 'inner')
                ->join('prices p',
'room_types.id = p.room_type_id',
'inner')
                -
>groupBy('room_types.id')
                ->find($room_type_id),
        ];
        return view('guest/hotel-
booking', $data);
    }
```

```
    public function detail($id)
    {

        $facilityModel = new
\App\Models\FacilityModel();
        $facilityType = new
\App\Models\FacilityType();
        $photoModel = new
\App\Models\PhotoModel();
        $data = [
            'settings' => $this-
>settings,
            'details' => $this-
>roomTypeModel->select('*, rooms.id as
rid, room_types.id as rtid')
                ->join('rooms',
'room_types.id = rooms.type_id', 'inner')
                ->find($id),

            'facilities' =>
$facilityModel->select('*')
                ->findAll(),
            'facility_types' =>
$facilityType->findAll(),

            'reviews' => $this-
>reviewModel->select('*, profiles.name as
pname, reviews.id as rid, users.id as
uid, profiles.id as pid')
                ->selectAvg('rate',
'rerate')
                ->selectCount('rate',
'rateCount')
                ->groupBy('reviews.id')
                ->orderBy('rate', 'desc')
                ->join('users',
'reviews.user_id = users.id', 'inner')
                ->join('profiles',
'users.id = profiles.user_id', 'inner')
                -
>where('reviews.service_id', 1)
                ->findAll(),
            'rateAvg' =>  $this-
>reviewModel->selectAvg('rate', 'rating')
                -
>where('reviews.service_id', 1)->find(),
            'photos' => $photoModel-
>findAll(),

            'rooms' => $this-
>roomTypeModel
                ->where("id !=", $id)
                ->findAll(),
        ];
        // var_dump($data['rateAvg']);
        return view('guest/hotel-detail',
$data);
    }


    public function forecast()
    {
        $cache =
\Config\Services::cache();

        // Define a cache key
```

```
        $cacheKey = 'weather_cast';

        // Try to get the cached data
        $cachedData = $cache-
>get($cacheKey);

        if (!$cachedData) {
            // Cache miss: Fetch data
from the API
            $url = 'https://api.open-
meteo.com/v1/forecast?latitude=13.4117&lo
ngitude=121.1803&daily=weathercode,appare
nt_temperature_max,apparent_temperature_m
in&timezone=Asia%2FSingapore';
            $response =
@file_get_contents($url);

            if ($response === false) {
                return 0;
            } else {
                // Save the API response
to cache for 1 day (86400 seconds)
                $decodedResponse =
json_decode($response, true);

                if ($decodedResponse !==
null) {
                    // Save the decoded
response to cache
                    $cache-
>save($cacheKey, $decodedResponse,
43200);
                    return $cache-
>get($cacheKey);
                }
            }
        } else {
            // Use the cached data
           return $cachedData;
        }




    }



    public function room($id = null)
    {
        if($this->request->getPost()) {
            if($id == "book") {
                $checkin_str = $this-
>request->getPost("checkin");
            $checkout_str = $this-
>request->getPost("checkout");
            $rooms = $this->request-
>getPost("quant[1]");
            $adults = $this->request-
>getPost("quant[2]");
            $kids = $this->request-
>getPost("quant[3]");
            $guest = $adults + $kids;

            $checkin = date("Y-m-d",
strtotime($checkin_str));
            $checkout = date("Y-m-d",
strtotime($checkout_str));
            $date = new
DateTime($checkin);
            $date->add(new
DateInterval('P1D'));
            if($checkout < $checkin ||
$checkout == $checkin)
                $checkout = $date-
>format("Y-m-d");

            $checkinDate =
DateTime::createFromFormat('Y-m-d',
$checkin);
            $checkoutDate =
DateTime::createFromFormat('Y-m-d',
$checkout);

                if ($checkoutDate
instanceof DateTime) {
                    $interval =
$checkinDate->diff($checkoutDate);
                    $numberOfNights =
$interval->days;
                    echo "Number of
nights: $numberOfNights";
                } else {
                    echo "Invalid
checkout date.";
                }



            $data = [
                'checkin' =>
$checkin_str,
                'checkout' =>
$checkout_str,
                'room_type_id' => $this-
>request->getPost('room_type_id'),
                'rooms' => $rooms,
                'guest' => $guest,
                'nights' =>
$numberOfNights,
                'adults' => $adults,
                'kids' => $kids,
            ];

            $this->session-
>set('check_availability', $data);
            $view = $this->session-
>get('check_availability');
            return redirect()-
>to('hotel/booking/'.$this->request-
>getPost('room_type_id'));
            }

        }
        if($id == 404) {

            return redirect()->back()-
>with('danger', 'Sorry, there is no
available room at the moment. Please
choose from another room type.');
        }
        $facilityModel = new
\App\Models\FacilityModel();
        $facilityType = new
\App\Models\FacilityType();
        $photoModel = new
\App\Models\PhotoModel();
```

```
        $currentday =
strtolower(date('D'));

        $check = $this->session-
>get('check_availability');
        if(isset($check)){

            $checkin = date("Y-m-d",
strtotime($check['checkin']));
            $checkout = date("Y-m-d",
strtotime($check['checkout']));
            $rooms_available = $this-
>roomModel
            ->select('rooms.id as rid,
rt.id as rtid, o.id as oid,
count(rooms.id) as room_count')
            ->join('room_types as rt',
'rt.id = rooms.type_id', 'left')
            ->join('orders as o',
'o.room_id = rooms.id', 'left')
            ->where("rt.id", $id)
            ->where("('$checkin' >=
o.check_out OR '$checkout' <= o.check_in
OR o.id IS NULL)", null, false)
            ->groupBy('rooms.id')
            ->find();

        }



        $data = [
            'settings' => $this-
>settings,
            'details' => $this-
>roomTypeModel->select("*, rooms.id as
rid, room_types.id as rtid, rooms.status
as rstatus, room_types.status as
rtstatus,  {$currentday} as priceToday")
                ->join('rooms',
'room_types.id = rooms.type_id', 'inner')
                ->join('prices p',
'room_types.id = p.room_type_id',
'inner')
                -
>groupBy('room_types.id')
                ->find($id),
            'room_available' => $this-
>roomTypeModel->select("*, rooms.id as
rid, room_types.id as rtid, rooms.status
as rstatus, room_types.status as
rtstatus")
                ->join('rooms',
'room_types.id = rooms.type_id', 'inner')
                ->where('rooms.status',
1)
                ->find($id),



            'facilities' =>
$facilityModel->findAll(),
            'amenities' => $this-
>roomTypeModel->select('*')
                -
>join('roomtypes_amenities rta',
'room_types.id = rta.roomtype_id',
'right')
                ->join('amenities a',
'rta.amenity_id = a.id', 'right')
                ->where('a.for_all', 1)-
>orWhere('room_types.id', $id)-
>findAll(),


            'amenity_rooms' => $this-
>roomTypeModel->select('*')
                -
>join('roomtypes_amenities rta',
'room_types.id = rta.roomtype_id',
'right')
                ->join('amenities a',
'rta.amenity_id = a.id', 'right')

                ->groupStart()
                    ->where('a.for_all',
1)
                    -
>orWhere('room_types.id !=', $id)
                ->groupEnd()
                ->findAll(),

            'facility_types' =>
$facilityType->findAll(),


            'reviews' => $this-
>reviewModel->select('*, CONCAT(firstname
, " ", lastname) as pname, reviews.id as
rid, users.id as uid, profiles.id as
pid')
                ->selectAvg('rate',
'rerate')
                ->selectCount('rate',
'rateCount')
                ->groupBy('reviews.id')
                ->orderBy('rate', 'desc')
                ->join('users',
'reviews.user_id = users.id', 'inner')
                ->join('profiles',
'users.id = profiles.user_id', 'inner')
                -
>where('reviews.service_id', 1)
                ->findAll(),
            'rateAvg' =>  $this-
>reviewModel->selectAvg('rate',
'rating')->selectAvg('value_money_rate')
            ->selectCount('rate',
'rateCount')
            ->selectAvg('service_rate')
                -
>where('reviews.service_id', 1)->find(),
            'photos' => $photoModel-
>findAll(),

            'rooms' => $this-
>roomTypeModel->select("*, rooms.id as
rid, room_types.id as rtid,
{$currentday} as priceToday")
                ->join('rooms',
'room_types.id = rooms.type_id', 'inner')
                ->join('prices p',
'room_types.id = p.room_type_id', 'left')
                -
>groupBy('room_types.id')
                ->where("room_types.id
!=", $id)
                ->findAll(),
```

```
            'weather' => $this-
>forecast(),
        ];
        return view('guest/hotel-room',
$data);
        // dd($rooms_available);
    }


    public function commingsoon()
    {
        if ($this->request->getPost()) {
            $newModel = new
\App\Models\NewModel();
            $validation = $this-
>validate($newModel->validationRules);
            $this->session-
>setFlashdata('info', 'Thank you, please
wait for further notice.');
            if ($validation) {
                $newModel-
>insert(['email' => $this->request-
>getPost('email')]);
                return redirect('/');
            }
            return redirect('/');
        }
        return view('guest/comming-
soon');
    }

    public function dashboard()
    {
        $data = [
            "settings" => $this-
>settings,
            "profiles" => $this-
>userModel->select("*")
                            ->join("profiles
as p", "users.id = p.user_id", "inner")
                            -
>where("users.id", $this->session-
>get("id"))
                            ->first(),
        ];
        // dd($data['profiles']);
        return view('guest/user-
dashboard', $data);
    }
    public function contact()
    {
        $data = [
            "settings" => $this-
>settings,
        ];
        return view('guest/contact-us',
$data);
    }
    public function about()
    {
        $data = [
            "settings" => $this-
>settings,
            'services' => $this-
>serviceModel->select('*, services.id as
sid, reviews.id as rid')
            ->selectAvg('rate')
            ->selectCount('rate',
'rateCount')
            ->orderBy('rate', 'desc')
            ->orderBy('rateCount',
'desc')
            ->join('reviews',
'services.id = reviews.service_id',
'left')
            ->groupBy('services.id')
            ->findAll(),
            'reviews' => $this-
>reviewModel->select('*,
concat(profiles.firstname, " ",
profiles.lastname) as pname, reviews.id
as rid, users.id as uid, profiles.id as
pid')
                ->selectAvg('rate',
'rerate')
                ->selectCount('rate',
'rateCount')
                ->groupBy('reviews.id')
                ->orderBy('rate', 'desc')
                ->join('users',
'reviews.user_id = users.id', 'inner')
                ->join('profiles',
'users.id = profiles.user_id', 'inner')
                -
>where('reviews.service_id', 1)
                ->findAll(),
        ];
        return view('guest/about-us',
$data);
    }
    public function faq()
    {
        $data = [
            "settings" => $this-
>settings,
        ];
        return view('guest/faq', $data);
    }
}
```

RestoAdmin.php

```
<?php

namespace App\Controllers;



class RestoAdmin extends BaseController
{
    protected $itemModel;
    protected $categoryModel;
    protected $extraModel;
    protected $variantModel;
    protected $allergenModel;
    protected $optionModel;
    protected $tableModel;
    protected $orderModel;
    protected $cartModel;

    public function __construct()
    {
        $this->itemModel = new
\App\Models\ItemModel();
        $this->categoryModel = new
\App\Models\CategoryModel();
        $this->extraModel = new
\App\Models\ExtraModel();
```

```
        $this->variantModel = new
\App\Models\VariantModel();
        $this->allergenModel = new
\App\Models\AllergenModel();
        $this->optionModel = new
\App\Models\OptionModel();
        $this->tableModel = new
\App\Models\TableModel();
        $this->orderModel = new
\App\Models\OrderModel();
        $this->cartModel = new
\App\Models\CartModel();
        helper('alert');
    }


    public function items()
    {
        if ($this->request->getPost()) {
            $image = $this->request-
>getFile('image');
            $category = $this->request-
>getPost('category');
            $result = $this-
>categoryModel->find($category);

            if (!$image->hasMoved()) {
                $newName = $image-
>getRandomName();

                $image->move(FCPATH .
'/assets/images/menu/', $newName);


                $data = [
                    'category_id' =>
$result['id'],
                    'name' => $this-
>request->getPost('name'),
                    'description' =>
$this->request->getPost('description'),
                    'price' => $this-
>request->getPost('price'),
                    'image' => $newName,
                ];
                $this->itemModel-
>insert($data);
            }
            $this->session-
>setFlashdata('success', 'Item has been
added.');
        }
        $data = [
            'categories' => $this-
>categoryModel->findAll(),
            'menus' => $this->itemModel
                ->select('*, c.id as cid,
items.id as mid, c.name as cname,
items.name as mname, c.status as cstatus,
items.status as mstatus')
                ->join('categories as c',
'items.category_id = c.id', 'inner')
                ->findAll(),

        ];
        // var_dump($data);
        return view('admin/items',
$data);
    }

    public function category()
    {
        $category = $this->request-
>getPost('name');
        $id = $this->request-
>getPost('id');
        $validation = $this-
>validate($this->categoryModel-
>validationRules);

        if ($validation) {
            if (isset($id)) {
                $this->categoryModel-
>set('name', $category)->where('id',
$id)->update();
                return redirect()-
>route('admin/items')->with('success',
'Category has been updated.');
            }

            $this->categoryModel-
>insert(['name' => $category]);
            return redirect()-
>route('admin/items')->with('success',
'Category has been added.');
        } elseif ($this->request-
>isAJAX()) {
            $this->categoryModel-
>set('status',  $this->request-
>getPost('status'))
                ->where('id', $this-
>request->getPost('id'))->update();
        }
        return redirect()->back()-
>with('error', 'This category already
exists.');
    }



    public function itemEdit($id)
    {
        if ($this->request->getPost()) {
            $image = $this->request-
>getFile('image');
            if (!empty($image-
>getName())) {
                if (!$image->hasMoved())
{
                    $newName = $image-
>getRandomName();
                    $image->move(FCPATH .
'/assets/images/menu');
                    $this->itemModel-
>set('image',  $newName)->where('id',
$id)->update();
                }
            }
            $status = $this->request-
>getPost('status');
            $variant_status = $this-
>request->getPost('variant_status');
            isset($status) ? $status = 1
: $status = 0;
            isset($variant_status) ?
$variant_status = 1 : $variant_status =
0;
            $data = [
```

```
                'category_id' => $this-
>request->getPost('category_id'),
                'name' => $this->request-
>getPost('name'),
                'description' => $this-
>request->getPost('description'),
                'price' => $this-
>request->getPost('price'),
                'status' => $status,
                'has_variants' =>
$variant_status,
            ];
            $this->itemModel->set($data)-
>where('id', $id)->update();
            $this->session-
>setFlashdata('success', 'Item has been
updated.');
        }
        $allergenInfo = $this->db-
>table('menu_allergens')-
>select('allergen_id')->where('item_id',
$id);

        $data = [
            'item' => $this->itemModel-
>select('*, c.id as cid, items.id as mid,
c.name as cname,

items.name as mname, c.status as cstatus,
items.status as mstatus')
                ->join('categories as c',
'items.category_id = c.id', 'inner')
                ->where('items.id', $id)-
>first(),
            'categories' => $this-
>categoryModel->findAll(),
            'extras' => $this-
>extraModel->where('item_id', $id)-
>findAll(),
            'options' =>  $this-
>optionModel->where('item_id', $id)-
>findAll(),
            'variants' => $this-
>variantModel->where('item_id', $id)-
>findAll(),
            'allergens' => $this->db-
>table('allergens')->whereNotIn('id',
$allergenInfo)->get()->getResultArray(),
            'menuallergens' => $this-
>allergenModel->select('*,allergens.id as
aid, menu_allergens.id as maid')-
>join('allergens',
'menu_allergens.allergen_id =
allergens.id', 'right')->findAll(),
        ];

        return view('admin/items-edit',
$data);
    }

    public function extras()
    {
        $id = $this->request-
>getPost('id');
        $variants =  $this->request-
>getPost('variant[]');
        isset($variants) ?  $variants =
join(',', $variants) : $variants = 0;
        $data = [
            'item_id' => $this->request-
>getPost('menu_id'),
            'name' => $this->request-
>getPost('name'),
            'price' => $this->request-
>getPost('price'),
            'selected_variant' =>
$variants,
        ];

        if (isset($id)) {
            if ($this->request->isAJAX())
{
                $result = $this-
>extraModel->set('status', $this-
>request->getPost('status'))
                    ->where('id', $id)-
>update();

                return "extras";
            } else {
                $this->extraModel-
>set($data)->where('id', $id)->update();
                return redirect()-
>back()->with('success', 'Extras has been
updated.');
            }
        }

        $this->extraModel->insert($data);
        return redirect()->back()-
>with('success', 'Extras has been
added.');
    }
    public function allergens()
    {
        if ($this->request->isAJAX()) {
            $checkState = $this->request-
>getPost('checkState');

            $data = [
                'item_id' => $this-
>request->getPost('menuid'),
                'allergen_id' => $this-
>request->getPost('allergenid'),
            ];

            if ($checkState == "true") {
                $result = $this-
>allergenModel->insert($data);
                return
json_encode($result);
            }

            $result = $this-
>allergenModel->where($data)->delete();
            return json_encode($result);
        }
    }
    public function options($idParam =
null)
    {
        $id = $this->request-
>getPost('option_id');

        if ($this->request->getPost()) {
            $data = [
```

```
                'item_id' => $this-
>request->getPost('menu_id'),
                'name' => $this->request-
>getPost('name'),
                'options' => $this-
>request->getPost('option'),
            ];

            $ifExist = $this-
>optionModel->where($data)-
>withDeleted()->first();
            if (isset($ifExist)) {
                $this->optionModel-
>set('deleted_at', null)->where('id',
$ifExist['id'])->update();
                return redirect()-
>back()->with('success', 'Option has been
restored.');
            } else {

                if (isset($id)) {

                    $this->optionModel-
>set($data)->where('id', $id)->update();
                    return redirect()-
>back()->with('success', 'Option has been
updated.');
                }

                $this->optionModel-
>insert($data);
                return redirect()-
>back()->with('success', 'Option has been
added.');
            }
        }
        if (isset($idParam)) {
            $this->optionModel-
>where('id', $idParam)->delete();
            return redirect()->back()-
>with('success', 'Option has been
deleted.');
        }
    }

    public function variants()
    {
        $id = $this->request-
>getPost('id');
        $options = $this->request-
>getPost('options[]');
        if (isset($options)) {
            $options = join(",",
$options);
        }

        if ($this->request->getPost()) {
            $data = [
                'item_id' => $this-
>request->getPost('menuid'),
                'price' => $this-
>request->getPost('price'),
                'options' => $options,
            ];
            $variantInfo = $this-
>variantModel->where([
                'item_id' => $this-
>request->getPost('menuid'),
                'options' => $options,
```

```
            ])->first();
            if (!isset($variantInfo)) {
                if (isset($id)) {
                    $this->variantModel-
>set($data)->where('id', $id)->update();
                    return redirect()-
>back()->with('success', 'Variant has
been updated.');
                }
                $this->variantModel-
>insert($data);
                return redirect()-
>back()->with('success', 'Variant has
been added.');
            }
            return redirect()->back()-
>with('error', 'Duplicate entry for
variant.');
        }
    }


    public function cloning($menuid)
    {
        $menuInfo = $this->itemModel-
>where('id', $menuid)->get();
        $allergenInfo = $this-
>allergenModel->where('item_id',
$menuid)->get();
        $extraInfo = $this->extraModel-
>where('item_id', $menuid)->get();
        $variantInfo = $this-
>variantModel->where('item_id', $menuid)-
>get();
        $optionInfo = $this->optionModel-
>where('item_id', $menuid)->get();
        $menuRow = $menuInfo->getRow();
        $allergenRow = $allergenInfo-
>getRow();
        $extraRow = $extraInfo->getRow();
        $optionRow = $optionInfo-
>getRow();
        $id = $this->itemModel-
>insert($menuRow);


        unset($menuRow->id);
        unset($allergenRow->id);
        unset($extraRow->id);
        unset($optionRow->id);
        if (isset($allergenRow)) {
            $allergenRow->item_id = $id;
            $this->allergenModel-
>insert($allergenRow);
        }
        if (isset($extraRow)) {
            $extraRow->item_id = $id;
            $this->extraModel-
>insert($extraRow);
        }
        if ($optionRow) {
            $optionRow->item_id = $id;
            $this->optionModel-
>insert($optionRow);
        }

        return redirect()->back()-
>with('success', 'Menu master has been
cloned.');
```

```
    }


    public function tables()
    {
        if ($this->request->getPost()) {
            $id = $this->request-
>getPost('id');
            $data = [
                'name' => $this->request-
>getPost('name'),
                'size' => $this->request-
>getPost('size')
            ];
            if (isset($id)) {
                if ($this->request-
>isAJAX()) {
                    $this->tableModel-
>update($id, ['status' => $this->request-
>getPost('status')]);
                    return "tables";
                }
                $this->tableModel-
>update($id, $data);
                return redirect()-
>back()->with('success', 'Table has been
updated.');
            }
            $this->tableModel-
>insert($data);
            // var_dump($data);
            return redirect()->back()-
>with('success', 'Table has been
added.');
        }
        $data = [
            'tables' => $this-
>tableModel->findAll(),
        ];
        return view('admin/tables',
$data);
    }

    public function orders()
    {
        $data = [
            'orders' =>  $this-
>orderModel->findAll(),
            'carts' => $this->cartModel-
>findAll(),
            'process' => $this->db-
>table('process')->get()-
>getResultArray(),
        ];
        return view('admin/orders',
$data);
    }

    public function statusUpdate($id,
$status)
    {
        $this->orderModel-
>set('order_status', $status)-
>where('id', $id)->update();
        return redirect()-
>route('admin/orders');
    }
    public function dashboard()
    {
        return view('admin/dashboard');
    }
}
```

RestoGuest.php

```
<?php

namespace App\Controllers;

use App\Controllers\BaseController;
use App\Models\ScheduleModel;


class RestoGuest extends BaseController
{
    protected $scheduleModel;
    protected $categoryModel;
    protected $itemModel;
    protected $extraModel;
    protected $variantModel;
    protected $allergenModel;
    protected $cartModel;
    protected $optionModel;
    protected $orderModel;
    protected $guestModel;
    public function __construct()
    {
        $this->scheduleModel = new
\App\Models\ScheduleModel();
        $this->categoryModel = new
\App\Models\CategoryModel();
        $this->itemModel = new
\App\Models\ItemModel();
        $this->extraModel = new
\App\Models\ExtraModel();
        $this->variantModel = new
\App\Models\VariantModel();
        $this->allergenModel = new
\App\Models\AllergenModel();
        $this->cartModel = new
\App\Models\CartModel();
        $this->optionModel = new
\App\Models\OptionModel();
        $this->guestModel = new
\App\Models\GuestModel();
        $this->orderModel = new
\App\Models\OrderModel();
        helper('cookie');
    }

    public function menu($table = null)
    {
        if($this->request->getPost()) {
            $callModel = new
\App\Models\CallModel();
            $callModel->insert(['table'
=> $this->request->getPost('table_id')]);
        }

        $menuInfo = $this->db-
>table('items')->select('category_id');
        $this->session->set('table',
$table);
        $items = $this->itemModel-
>select('*,items.id as iid, e.id as eid,
o.id as oid, v.id as vid, items.name as
iname, e.name as ename, o.name as oname,
```

```
        items.price as iprice, e.price as
eprice, v.price as vprice, e.item_id as
eiid, o.item_id as oiid, v.item_id as
viid, o.options as ooptions, v.options as
voptions')
            ->join('extras as e',
'e.item_id = items.id', 'left')
            ->join('options as o',
'o.item_id = items.id', 'left')
            ->join('variants as v',
'v.item_id = items.id', 'left')
            ->groupBy('items.id')
            ->findAll();

        $data = [];

        foreach ($items as $row) {
            $checkVariant = $this-
>optionModel
                ->join('variants as v',
'options.item_id = v.item_id', 'inner')
                ->where('v.item_id',
$row['iid'])->findAll();
            count($checkVariant) > 0 ?
$checkVariant = true :  $checkVariant =
false;
            $data[(int)$row['iid']] = [
                'id' => (int)$row['iid'],
                'name' => $row['iname'],
                'priceNotFormated' =>
(float)$row['iprice'],
                'price' => '₱' .
number_format($row['iprice'], 2),
                'image' => $row['image'],
                'description' =>
$row['description'],
                'has_variants' =>
$checkVariant,
                'extras' => [],
                'options' => [],
                'variants' => []
            ];
        }

        // Collect extras data
        $extras = $this->itemModel-
>select('*, e.id as eid, e.item_id as
eiid')
            ->join('extras as e',
'items.id = e.item_id', 'left')
            ->groupBy('e.id')
            ->findAll();

        foreach ($extras as $extra) {
            $extras_variant =
$extra['selected_variant'] == 0 ?
$extras_variant = 1 :  $extras_variant =
0;

$data[(int)$extra['eiid']]['extras'][] =
[
                'id' =>
(int)$extra['eid'],
                'item_id' =>
(int)$extra['eiid'],
                'price' =>
(float)$extra['price'],
                'name' => $extra['name'],
                'extra_for_all_variants'
=>    $extras_variant,
                'priceFormated' => '₱' .
number_format($extra['price'], 2)
            ];
        }

        // Collect options data
        $options = $this->optionModel-
>select('*, options.id as oid,
options.item_id as oiid, options.options
as ooptions')
            ->join('variants as v',
'options.item_id = v.item_id', 'left')
            // ->join('options as o',
'v.item_id = o.item_id', 'right')
            ->groupBy('oid')
            ->findAll();

        foreach ($options as $option) {

$data[(int)$option['oiid']]['options'][]
= [
                'id' =>
(int)$option['oid'],
                'item_id' =>
(int)$option['oiid'],
                'name' =>
$option['name'],
                'options' =>
$option['ooptions'],
            ];
        }


        // Collect variants data
        $variants = $this->optionModel-
>select('*, v.id as vid, v.item_id as
viid, v.options as voptions')
            ->join('variants as v',
'options.item_id = v.item_id', 'inner')
            // ->join('options as o',
'v.item_id = o.item_id', 'right')
            ->groupBy('v.id')
            ->findAll();



        foreach ($variants as $variant) {

            $optionsString =
$variant['voptions'];
            $optionsArray = explode(",",
$optionsString);
            $options = [];

            foreach ($optionsArray as
$option) {
                $parts = explode(":",
$option);
                if (isset($parts[1])) {
                    $options[$parts[0]] =
$parts[1];
                }
            }

            $optionsJson =
json_encode($options);
```

```
$data[(int)$variant['viid']]['variants'][
] = [
                'id' =>
(int)$variant['vid'],
                'item_id' =>
(int)$variant['viid'],
                'name' =>
$variant['name'],
                'price' =>
(float)$variant['price'],
                'priceFormated' => '₱' .
number_format($variant['price'], 2),
                'options' =>
$optionsJson,
            ];
        }

        $db = [
            'days' => $this-
>scheduleModel->findAll(),
            'categories' => $this-
>categoryModel->whereIn('id', $menuInfo)-
>findAll(),
            'items' => $this->itemModel-
>select('*, c.id as cid, items.id as mid,
c.name as cname,
            items.name as mname, c.status
as cstatus, items.status as mstatus')
                ->join('categories as c',
'items.category_id = c.id', 'inner')
                ->findAll(),
            'allergens' => $this-
>allergenModel->select('*')
                ->join('allergens',
'menu_allergens.allergen_id =
allergens.id', 'inner')->findAll(),
            'json_data' =>
json_encode($data, JSON_NUMERIC_CHECK),
            'table' => $table,
            'orders' => $this-
>orderModel->where('order_status != 4')-
>where('guest_id', $this->request-
>getCookie('id'))->findAll(),
            'carts' => $this->cartModel-
>where('guest_id', $this->request-
>getCookie('id'))->findAll(),

        ];
        // var_dump($db['orders']);
        return view('guest/menu', $db);
    }
    public function extras($variant_id)
    {
        $data = [
            'data' => $this->extraModel-
>select('extras.name, extras.price,
extras.item_id, extras.id,
extras.status')
                -
>where("FIND_IN_SET($variant_id,
selected_variant)")
                ->findAll()
        ];

        if (count($data['data']) < 1) {
            $data = [
                'data' => $this-
>extraModel->select('extras.name,
extras.price, extras.item_id, extras.id,
extras.status')
                    ->join('variants as
v', 'extras.item_id = v.item_id',
'inner')
                    ->where('v.id',
$variant_id)
                    -
>where('selected_variant', 0)
                    ->findAll(),
            ];
            if (count($data['data']) < 1)
{
                return
json_encode(['no_extras' => true]);
            }
            return json_encode($data);
        }
        return json_encode($data);
    }
    public function checkout($table =
null)
    {

        $data = [
            'table' => $table,
            'time' => $this->db-
>table('selecttime')->get()-
>getResultArray(),
        ];
        return view('guest/cart-
checkout', $data);
    }

    public function order_notify()
    {
        if ($this->request->getPost()) {
            $cartInfo = $this->cartModel-
>select('id')->where('guest_id', $this-
>request->getCookie('id'))-
>where('status', 0)->findAll();
            $total = $this->cartModel-
>selectSum('total')->where('guest_id',
$this->request->getCookie('id'))-
>where('status', 0)->first();

            $ids =
array_column($cartInfo, 'id');
            $result = implode(',', $ids);

            $data = [
                'cart_id' => $result,
                'guest_id' =>$this-
>request->getCookie('id'),
                'mode' => $this->request-
>getPost('dineType'),
                'table' => $this-
>request->getPost('table'),
                'phone' => $this-
>request->getPost('phone'),
                'total' =>
$total['total'],
                'email' => $this-
>request->getPost('email'),
                'pickupTime' => $this-
>request->getPost('pickupTime'),
```

```
                'message' => $this-
>request->getPost('message'),
                'payment' => $this-
>request->getPost('payment'),
            ];
            // var_dump($data);
            $id = $this->orderModel-
>insert($data);
            $cartid = explode(",",
$result);
            $this->cartModel-
>set('status', 1)->set('order_id', $id)-
>whereIn('id', $cartid)->update();

            return view('guest/order-
notify', ['table' => $data['table'],
'order_id' => $id]);
        }
        return view('guest/order-
notify');


    }
    public function cart()
    {
        if ($this->request-
>getCookie('userId') ) {
            $userId = $this->request-
>getCookie('userId');
        } else {
            $expiration = time() +
31536000;
            $userId =
md5($_SERVER['REMOTE_ADDR'] .
$_SERVER['HTTP_USER_AGENT']);

            $guestInfo = $this-
>guestModel->where('user_id', $userId)-
>first();
            if(isset($guestInfo)) {
                $this->response-
>setCookie('userId', $userId,
$expiration);
                $this->response-
>setCookie('id', $guestInfo['id'],
$expiration);
            }
            else{
                $id = $this->guestModel-
>insert(['user_id' => $userId]);
                $this->response-
>setCookie('userId', $userId,
$expiration);
                $this->response-
>setCookie('id', $id, $expiration);
            }

        }

        $item_id =  $this->request-
>getVar('id');

        if (isset($item_id)) {
            $extras = $this->request-
>getVar('extras');
            $extrasConcat = "";
            $options = "";
            $extras = isset($extras) ?
$extras = join(",", $extras) : $extras =
"";
            $variant_id = $this->request-
>getVar('variantID');
            $total = $this->request-
>getVar('total');
            $num =
floatval(preg_replace('/[^0-9.]+/', '',
$total));
            $itemName = $this->itemModel-
>where('id', $item_id)->first();
            if (isset($variant_id)) {
                $itemInfo = $this-
>itemModel->select('options')-
>join('variants as v', 'items.id =
v.item_id', 'inner')
                    ->where('v.id',
$variant_id)->where('items.id',
$item_id)->first();


                $str =
$itemInfo['options'];
                $items = explode(",",
$str);
                $map = [];
                foreach ($items as $item)
{
                    list($key, $value) =
explode(":", $item);
                    $map[trim($key)] =
trim($value);
                }
                $options = implode(":",
array_values($map));
            }

            $extraID = explode(",",
$extras);
            $extraInfos = $this-
>extraModel->whereIn("id", $extraID)-
>findAll();
            foreach ($extraInfos as
$info) {
                $extrasConcat .= " + " .
$info['name'];
            }
            $name =  $itemName['name'] .
" " . $options . $extrasConcat;

            $existingCheck = [
                'guest_id' => $this-
>request->getCookie('id'),
                'item_id' => $item_id,
                'extrasSelected' =>
$extras,
                'variantID' => $this-
>request->getVar('variantID'),
                'status' => 0
            ];



            $cartExist = $this-
>cartModel->where($existingCheck)-
>first();

            if (!isset($cartExist)) {
```

```
                $data = [
                    'guest_id' => $this-
>request->getCookie('id'),
                    'name' => $name,
                    'item_id' =>
$item_id,
                    'quantity' => $this-
>request->getVar('quantity'),
                    'extrasSelected' =>
$extras,
                    'variantID' => $this-
>request->getVar('variantID'),
                    'friendly_price' =>
$this->request->getVar('total'),

                    'total' => $num *
$this->request->getVar('quantity')
                ];

                if ($this->cartModel-
>insert($data)) {
                    return
json_encode(['status' => true]);
                }
                return
json_encode(['status' => false]);
            }
            $toUpdate = [
                'quantity' =>
$cartExist['quantity'] + $this->request-
>getVar('quantity'),
                'total' =>
$cartExist['total'] + ($num * $this-
>request->getVar('quantity'))


            ];

            $this->cartModel-
>set($toUpdate)->where('id',
$cartExist['id'])->update();
            return json_encode(['status'
=> true]);
        }

        $total = $this->cartModel-
>selectSum('total')->where('guest_id',
$this->request->getCookie('id'))-
>where('status', 0)->first();
        $attributes = $this->itemModel-
>select('*, c.id as id, c.status as
status, c.name as name')->join('carts as
c', 'items.id = c.item_id', 'inner')
            ->where('guest_id', $this-
>request->getCookie('id'))-
>where('c.status', 0)->findAll();
        if (!isset($total['total'])) {
            $total['total'] = 0;
        }
        $data = [
            'data' => $attributes,
            'total' => $total,

        ];
        return json_encode($data);
    }
    public function cart_action()
    {
        $action = $this->request-
>getVar('action');
        $id = $this->request-
>getVar('id');
        $cartInfo = $this->cartModel-
>select('*, price, i.id as iid, carts.id
as cid')->join('items as i',
'carts.item_id = i.id', 'inner')-
>where('carts.id', $id)->first();
        if ($cartInfo['has_variants']) {
            $cartInfo =  $this-
>cartModel->select('*, price, v.id as
vid, carts.id as cid')->join('variants as
v', 'carts.variantID = v.id', 'inner')-
>where('carts.id', $id)->first();
        }

        $priceFormatted =
$cartInfo['friendly_price'];

        $price = (int)str_replace("₱",
"", $priceFormatted);

        if ($action == "remove") {
            $this->cartModel->where('id',
$id)->delete();
            return json_encode(['status'
=> "item has been removed"]);
        } elseif ($action == "increase")
{
            $this->cartModel-
>set(['quantity' => $cartInfo['quantity']
+ 1, 'total' => $cartInfo['total'] +
$price])->where('id', $id)->update();
            return json_encode(['status'
=> "Cart item quantity has been
increase"]);
        } else {
            if ($cartInfo['quantity'] >
1) {
                $this->cartModel-
>set(['quantity' => $cartInfo['quantity']
- 1, 'total' => $cartInfo['total'] -
$price])->where('id', $id)->update();
                return
json_encode(['status' => "Cart item
quantity has been decrease"]);
            } else {
                $this->cartModel-
>where('id', $id)->delete();
            }
        }
    }
    public function registerVisit()
    {
        return view('guest/register-
visit');
    }
    public function orders($id)
    {
        $orderInfo = $this->orderModel-
>find($id);
        $data = [
            'order_id' => $id,
            'table' => $this->session-
>get('table'),
            'order_info' => $this-
>orderModel->select('*, c.id as cid,
orders.id as oid')->join('carts as c',
```

```
'c.order_id = orders.id', 'inner')-
>where('orders.id', $id)->findAll(),
            'process' => $this->db-
>table('process')->get()-
>getResultArray(),
            'total' => $this->cartModel-
>selectSum('total')->where('order_id',
$id)->first(),
        ];
        // var_dump($data['order_info']);
        return view('guest/orders',
$data);
    }

    public function receive($id) {
        $this->orderModel-
>set('order_status', 4)->where('id',
$id)->update();

    }

}
```

Alert_helper.php

```
<?php

function alert_message($message)
{
    if (session()->has('success')) : ?>
        <div class="alert alert-success
alert-dismissible alert-label-icon
rounded-label" role="alert">
            <i class="ri-notification-
off-line label-
icon"></i><strong>Success</strong> - <?=
session()->getFlashdata('success') ?>
            <button type="button"
class="btn-close" data-bs-dismiss="alert"
aria-label="Close"></button>
        </div>
    <?php elseif (session()-
>has('error')) : ?>
        <div class="alert alert-danger
alert-dismissible alert-label-icon
rounded-label" role="alert">
            <i class="ri-error-warning-
line label-
icon"></i><strong>Failed</strong> - <?=
session()->getFlashdata('error') ?>
            <button type="button"
class="btn-close" data-bs-dismiss="alert"
aria-label="Close"></button>
        </div>
    <?php endif; ?>


<?php } ?>
```

StarRating_helper.php

```
<?php
function generateStar($rating) {
    // Ensure the rating is within the
range [0, 5]
    $rating = max(0, min(5, $rating));

    // Calculate the number of filled
stars and the percentage for the last
star
    $filledStars = floor($rating);
    $percentage = ($rating -
$filledStars) * 100;

    // Generate the HTML code for the
star rating
    $html = '<span>' .
number_format($rating, 1) . '</span>';


    for ($i = 1; $i <= 5; $i++) {
        if ($i <= $filledStars) {
            $html .= '<i class="fas fa-
star"></i>';
        } elseif ($i == $filledStars + 1
&& $percentage > 0) {
            $html .= '<i class="fas fa-
star-half-alt"></i>';
        } else {
            $html .= '<i class="far fa-
star"></i>';
        }
    }

    $html .= '';

    return $html;
  }


?>
```

Toastr_helper.php

```
<?php
function toastr()
{
    // Get flash messages from session
    $infoMsg = session()-
>getFlashdata('info');
    $successMsg = session()-
>getFlashdata('success');
    $warningMsg = session()-
>getFlashdata('warning');
    $dangerMsg = session()-
>getFlashdata('danger');
    ?>

    <script>
        $(document).ready(function() {
            "use strict";
            var infoMsg = "<?= $infoMsg
?>";
            var successMsg = "<?=
$successMsg ?>";
            var warningMsg = "<?=
$warningMsg ?>";
            var dangerMsg = "<?=
$dangerMsg ?>";

            if (infoMsg !== "") {
                toastr.info(infoMsg, {
                    closeButton: true,
                    tapToDismiss: false,
                    progressBar: true,
                    rtl: false
```

```
            });
        } else if (successMsg !== "")
{

toastr.success(successMsg, {
                    closeButton: true,
                    tapToDismiss: false,
                    progressBar: true,
                    rtl: false
                });
        } else if (warningMsg !== "")
{

toastr.warning(warningMsg, {
                    closeButton: true,
                    tapToDismiss: false,
                    progressBar: true,
                    rtl: false
                });
        } else if (dangerMsg !== "")
{
                toastr.error(dangerMsg, {
                    closeButton: true,
                    tapToDismiss: false,
                    progressBar: true,
                    rtl: false
                });
            }
        });
    </script>

    <?php
}
?>
```

Hash.php

```
<?php
namespace App\Libraries;

class Hash
{
    public static function
make($password){
        return password_hash($password,
PASSWORD_BCRYPT);
    }

    public static function
check($entered_password, $data){

if(password_verify($entered_password,
$data)){
            return true;
        }
        else{
            return false;
        }
    }
}
?>
```

Order.js

```
"use strict";
var items = [];
var currentItem = null;
var currentItemSelectedPrice = null;
var lastAdded = null;
var previouslySelected = [];
var extrasSelected = [];
var variantID = null;
var debug = true;

function debugMe(title, message) {
    if (debug) {
        console.log("#" + title);
        console.log(message);
        console.log("--------");
    }
}

/*
 * Price formater
 * @param {Nummber} price
 */
function formatPrice(price) {
    var locale = LOCALE;
    if (CASHIER_CURRENCY.toUpperCase() ==
"PHP") {
        locale = locale + "-PH";
    }

    var formatter = new
Intl.NumberFormat(locale, {
        style: 'currency',
        currency: CASHIER_CURRENCY,
    });

    var formated =
formatter.format(price);

    return formated;
}

/**
 * Load extras for variant
 * @param {Number} variant_id the variant
id
 * */
function loadExtras(variant_id) {
    //alert("Load extras for
"+variant_id);
    $.ajaxSetup({
        headers: {
            'X-CSRF-TOKEN':
$('meta[name="csrf-
token"]').attr('content')
        }
    });

    $.ajax({
        type: 'GET',
        url: '/guest/extras/' +
variant_id,
        success: function(response) {
            var data =
JSON.parse(response);

            if (data != undefined &&
!data.no_extras) {
                data.data.forEach(element
=> {
                    // $('#exrtas-area-
inside').append('<div class="custom-
control custom-checkbox mb-3"><input
onclick="recalculatePrice(' +
element.item_id + ');" class="custom-
```

```
control-input" id="' + element.id + '"
name="extra"  value="' + element.price +
'" type="checkbox"><label class="custom-
control-label" for="' + element.id + '">'
+ element.name +
'     +' +
formatPrice(element.price) +
'</label></div>');
                    $('#exrtas-area-
inside').append('<label
class="container_check">' + element.name
+ '<span>+ ' +
formatPrice(element.price) +
'</span><input
onclick="recalculatePrice(' +
element.item_id + ');" class="" for="' +
element.id + '" id="' + element.id + '"
name="extra"  value="' + element.price +
'" type="checkbox"><span
class="checkmark"></span></label>');
                });
                $('#exrtas-area').show();

            }
            else{
                $('#exrtas-area').hide();
            }
        },
        error: function(response) {
            //return callback(false,
response.responseJSON.errMsg);
        }
    })
}




/**
 *
 * Set the selected variant, set price
and shows qty area and calls load extras
 * */
function setSelectedVariant(element) {

    //console.log(formated);

$('#modalPrice').html(formatPrice(element
.price));

    //Set current item price
    currentItemSelectedPrice =
element.price;

    //Show QTY
    $('.modal-footer').show();

    //Set variantID
    variantID = element.id;

    //Empty the extras, and call it
    $('#exrtas-area-inside').empty();
    loadExtras(variantID);

}

function getTheDataForTheFoundVariable()
{
    console.log(previouslySelected);
    var comparableObject = {};
    previouslySelected.forEach(element =>
{

comparableObject[element.option_id] =
element.name.trim().toLowerCase().replace
(/\s/g, "-");
    });
    comparableObject =
JSON.stringify(comparableObject)
        //console.log("Comparable");
        //console.log(comparableObject);

currentItem['variants'].forEach(element
=> {
        //console.log("Compare to");

//console.log(JSON.stringify(JSON.parse(e
lement.options)));
        if (comparableObject ==
JSON.stringify(JSON.parse(element.options
))) {
            //console.log("This are the
options");

//console.log(element.options);

//console.log(element.optionsiconv);
            setSelectedVariant(element);
        }
    });

}


function checkIfVariableExists(forOption,
optionValue) {

    var newElement = { "option_id":
forOption, "name": optionValue };
    // console.log('NEW ELEMNGT');
    // console.log(newElement);

    var possibleSelection =
JSON.parse(JSON.stringify(previouslySelec
ted));
    possibleSelection.push(newElement);
    //console.log(possibleSelection);

    var filteredObjects = [];
    //possibleSelection.forEach(element
=> {

currentItem.variants.forEach(theVariant
=> {
        var theOptions =
JSON.parse(theVariant.optionsiconv ?
theVariant.optionsiconv :
theVariant.options);
        var ok = true;
        Object.keys(theOptions).map((key)
=> {

            // conso    le.log(key+" :
"+theOptions[key])

possibleSelection.forEach(element => {
                if (key ==
element.option_id) {
```

```
                    if (theOptions[key] + 
"" != 
element.name.trim().toLowerCase().replace
(/\s/g, "-") + "") {
                        ok = false;
                    }
                }
            });

        })

        if (ok) {

filteredObjects.push(theVariant);
            console.log("ok")
        } else {
            console.log("not ok")
        }


//comparableObject[element.option_id]=ele
ment.name.trim().toLowerCase().replace(/\
s/g , "-");
    });

    //});


    return filteredObjects.length > 0;

}

function appendOption(name, id) {
    lastAdded = id;
    $('#variants-area-
inside').append('<div id="variants-area-'
+ id + '" class="variants-area-label"><h5
class="label-variants">' + name +
'</h5><div><div id="variants-area-inside-
' + id + '" class="flex-wrap btn-group
btn-group-toggle" data-toggle="buttons">
</div></div>');
}

function optionChanged(option_id, name) {

    var newElement = { "option_id":
option_id, "name": name };
    debugMe("selected option",
JSON.stringify(newElement));


    //Append / insert the new selectioin
    var newSelectionState = [];
    var
userClickedOnAlreadySelectedOption =
false;
    previouslySelected.forEach(element =>
{

        if
(userClickedOnAlreadySelectedOption) {
            $("#variants-area-" +
element.option_id).remove();
        }

        if (element.option_id !=
newElement.option_id) {
            //If we haven't yet found the
item add this in the selection
            if
(!userClickedOnAlreadySelectedOption) {
newSelectionState.push(element); }
        } else {

userClickedOnAlreadySelectedOption =
true;
        }


    });



    if
(userClickedOnAlreadySelectedOption &&
lastAdded != newElement.option_id) {
        //remove also last inserted, and
readded it
        $("#variants-area-" +
lastAdded).remove();
    }

    newSelectionState.push(newElement);
    previouslySelected =
newSelectionState;
    debugMe("Selection",
JSON.stringify(previouslySelected));
    setVariants();


}

function appendOptionValue(name, value,
enabled, option_id) {
    $('#variants-area-inside-' +
option_id).append('<label style="opacity:
' + (enabled ? 1 : 0.5) + '" class="btn
btn-outline-primary mb-2"><input
onchange="optionChanged(' + option_id +
',\'' + value + '\')"  ' + (enabled ? ""
: "disabled") + ' type="radio"
name="option_' + option_id + '"
value="option_' + option_id + "_" + name
+ '" autocomplete="off" />' +
js.trans(name) + '</label>')
}

function setVariants() {
    //1. Determine previously selected
variants
    // var previouslySelected=[];

    //HIDE QTY
    $('.modal-footer').hide();

    $('#exrtas-area-inside').empty();
    $('#exrtas-area').hide();

    //2. Get the new option to show
    var newOptionToShow = null;
    debugMe("previouslySelected length",
previouslySelected.length);
    newOptionToShow =
currentItem.options[previouslySelected.le
ngth];
```

```
    debugMe("newOptionToShow",
JSON.stringify(newOptionToShow));

    if (newOptionToShow != undefined) {
        //2.1 Add the options in the
table

appendOption(newOptionToShow.name,
newOptionToShow.id);


        var values =
(newOptionToShow.optionsiconv ?
newOptionToShow.optionsiconv :
newOptionToShow.options).split(",");
        var titles =
(newOptionToShow.options).split(",");

        for (let index = 0; index <
values.length; index++) {
            const theValue =
values[index];
            const theTitle =
titles[index];

            if
(checkIfVariableExists(newOptionToShow.id
, theValue)) {
                //Next variable exists
                console.log("Exists:
"+theValue);

appendOptionValue(theTitle, theValue,
true, newOptionToShow.id);
            } else {
                //Varaiable with the next
option value doens't exists
                console.log("Does not
exists: "+newOptionToShow.id);

appendOptionValue(theTitle, theValue,
false, newOptionToShow.id);
            }

        }

    } else {
        console.log("No more options");
        getTheDataForTheFoundVariable();
    }



    //3. Add the new option options
    //3.1 If new option is null, show the
variant price
}


function setCurrentItem(id) {


    var item = items[id];
    currentItem = item;
    previouslySelected = [];
    $('#modalTitle').text(item.name);
    $('#modalName').text(item.name);
    $('#modalPrice').html(item.price);
```

```
$('#modalTotalPrice').html(item.price);
    $('#modalID').text(item.id);
    $('#quantity').val(1);

    if (item.image !=
"/default/restaurant_large.jpg") {
        $("#modalImg").attr("src",
item.image);

$("#modalDialogItem").addClass("modal-
lg");
        $("#modalImgPart").show();


$("#modalItemDetailsPart").removeClass("c
ol-sm-6 col-md-6 col-lg-6 offset-3");

$("#modalItemDetailsPart").addClass("col-
sm col-md col-lg");
    } else {
        $("#modalImgPart").hide();

$("#modalItemDetailsPart").removeClass("c
ol-sm col-md col-lg");

$("#modalItemDetailsPart").addClass("col-
sm-6 col-md-6 col-lg-6 offset-3");


$("#modalDialogItem").removeClass("modal-
lg");

$("#modalDialogItem").addClass("col-sm-6
col-md-6 col-lg-6 offset-3");
    }


$('#modalDescription').html(item.descript
ion);


    if (item.has_variants) {
        //Vith variants
        //Hide the counter, and extrasts
        $('.modal-footer').hide();

        //Now show the variants options
        $('#variants-area-
inside').empty();
        $('#variants-area').show();

        setVariants();

//$('#modalPrice').html("dynamic");




    } else {
        //Normal
        currentItemSelectedPrice =
item.priceNotFormated;
        $('#variants-area').hide();
        $('.modal-footer').show();

    }
```

```
    $('#productModal').modal('show');

    extrasSelected = [];

    variantID = null;

    //Now set the extrast
    if (item.extras.length == 0 ||
item.has_variants) {
        console.log('has no extras');
        $('#exrtas-area-inside').empty();
        $('#exrtas-area').hide();
    } else {
        console.log('has extras');
        $('#exrtas-area-inside').empty();
        item.extras.forEach(element => {
            console.log(element);
            $('#exrtas-area-
inside').append('<label
class="container_check">' + element.name
+ '<span>+ ' + element.priceFormated
+ '</span><input
onclick="recalculatePrice(' + id + ');"
class="" id="' + element.id + '"
name="extra"  value="' + element.price +
'" type="checkbox"><span
class="checkmark"></span></label>');
        });
        $('#exrtas-area').show();
    }
}

function recalculatePrice(id, value) {
    //console.log("Triger price
recalculation: "+id);
    // console.log(items[id]);
    var mainPrice =
parseFloat(currentItemSelectedPrice);
    extrasSelected = [];

    //Get the selected check boxes

$.each($("input[name='extra']:checked"),
function() {
        mainPrice +=
parseFloat(($(this).val() + ""));

extrasSelected.push($(this).attr('id'));
    });

$('#modalPrice').html(formatPrice(mainPri
ce));

}




$(".nav-item-category").on('click',
function() {
    $.each(categories, function(index,
value) {
        $("." + value).show();


//$("#nav_"+value).removeClass("active");
    });

    var id = $(this).attr("id");
    var category_id =
id.substr(id.indexOf("_") + 1,
id.length);

//$("#nav_"+category_id).addClass("active
");

    //$("."+category_id).hide();

    $.each(categories, function(index,
value) {
        if (value != category_id) {
            $("." + value).hide();
        }
    });
});
```

cartFunction.js

```
"use strict";
var cartContent = null;
var cartContentMobile = null;
var cartTotal = null;
var cartTotalMobile = null;
var footerPages = null;
var total = null;

$('#localorder_phone').hide();



/**
 *
 * @param {Number} net The net value
 * @param {Number} delivery The delivery
value
 * @param {Boolean} enableDelivery
Disable or enable delivery
 */
function updatePrices(net, delivery,
enableDelivery) {
    net = parseFloat(net + "");

    delivery = parseFloat(delivery + "");
    var deduct = cartTotal.deduct;
    console.log("Deduct is " + deduct);

    var formatter = new
Intl.NumberFormat(LOCALE, {
        style: 'currency',
        currency: CASHIER_CURRENCY,
    });

    //totalPrice -- Subtotal
    //withDelivery -- Total with delivery

    //Subtotal
    cartTotal.totalPrice = net;
    cartTotal.totalPriceFormat =
formatter.format(net);

    if (enableDelivery) {
        //Delivery
        cartTotal.delivery = true;
        cartTotal.deliveryPrice =
delivery;
        cartTotal.deliveryPriceFormated =
formatter.format(delivery);
```

```
        //Total
        var ndd = net + delivery -
deduct;
        cartTotal.withDelivery = ndd;
        cartTotal.withDeliveryFormat =
formatter.format(ndd); //+"==>"+new
Date().getTime();
        total.totalPrice = ndd;



    } else {
        //No delivery
        //Delivery
        cartTotal.delivery = false;

        //Total
        var nd = net - deduct;
        cartTotal.withDelivery = nd;
        cartTotal.withDeliveryFormat =
formatter.format(nd);
        total.totalPrice = nd;
    }
    total.lastChange = new
Date().getTime();
    cartTotal.lastChange = new
Date().getTime();
    console.log("DAta");
    console.log(cartTotal)
    console.log("Total is " +
total.totalPrice);
    console.log("Total is " +
cartTotal.withDelivery);


}

function setDeduct(deduction) {
    var formatter = new
Intl.NumberFormat(LOCALE, {
        style: 'currency',
        currency: CASHIER_CURRENCY,
    });

    cartTotal.deduct = deduction;
    cartTotal.deductFormat =
formatter.format(deduction);
    total.lastChange = null;
    cartTotal.lastChange = null;
    getCartContentAndTotalPrice();
}

function updateSubTotalPrice(net,
enableDelivery) {

    updatePrices(net,
(cartTotal.deliveryPrice ?
cartTotal.deliveryPrice : 0),
enableDelivery)
}
/**
 * getCartContentAndTotalPrice
 * This functions connect to server to
get the current cart items and total
price
 * Saves the values in vue
//  */
function getCartContentAndTotalPrice() {
axios.get('/guest/cart').then(function(re
sponse) {

            cartContent.items =
response.data.data;



//cartContentMobile.items=response.data.d
ata;

updateSubTotalPrice(response.data.total.t
otal, true);
        })
        .catch(function(error) {

        });
};

$("#promo_code_btn").on('click',
function() {
    var code = $('#coupon_code').val();

    axios.post('/coupons/apply', { code:
code, cartValue: cartTotal.totalPrice
}).then(function(response) {
        if (response.data.status) {

$("#promo_code_btn").attr("disabled",
true);

$("#promo_code_btn").attr("readonly");

            $("#promo_code_war").hide();
            $("#promo_code_succ").show();


setDeduct(response.data.deduct);
            js.notify(response.data.msg,
"success");

            $('#promo_code_btn').hide();

            //$( "#coupon_code" ).prop(
"disabled", true );
        } else {
            $("#promo_code_succ").hide();
            $("#promo_code_war").show();

            js.notify(response.data.msg,
"warning");
        }
    }).catch(function(error) {

    });

});

$("#fborder_btn").on('click', function()
{

    var address = $('#addressID').val();
    var comment = $('#comment').val();

    axios.post('/fb-order', {
            address: address,
            comment: comment
        })
```

```
        .then(function(response) {

            if (response.status) {
                var text =
response.data.msg;

                var fullLink =
document.createElement('input');

document.body.appendChild(fullLink);
                fullLink.value = text;
                fullLink.select();

document.execCommand("copy", false);
                fullLink.remove();

                swal({
                    title: "Good job!",
                    text: "Order is
submited in the system and copied in your
clipboard. Next, messenger will open and
you need to paste the order details
there.",
                    icon: "success",
                    button: "Continue to
messenger",

}).then(function(isConfirm) {
                    if (isConfirm) {

document.getElementById('order-
form').submit();
                    }
                });

            }
        }).catch(function(error) {

        });
});

/**
 * Removes product from cart, and calls
getCartConent
 * @param {Number} product_id
 */
function
removeProductIfFromCart(product_id) {
    var action = "remove";
    axios.post('/guest/cart-action', {
id: product_id, action: action
}).then(function(response) {
        getCartContentAndTotalPrice();
    }).catch(function(error) {

    });
}

/**
 * Update the product quantity, and calls
getCartConent
 * @param {Number} product_id
 */
function incCart(product_id) {
    var action = "increase";
    axios.post('/guest/cart-action', {
id: product_id, action: action
}).then(function(response) {
        getCartContentAndTotalPrice();
    }).catch(function(error) {

    });
}


function decCart(product_id) {
    var action = "decrease";
    axios.post('/guest/cart-action', {
id: product_id, action: action
}).then(function(response) {
        getCartContentAndTotalPrice();
    }).catch(function(error) {

    });
}

//GET PAGES FOR FOOTER
// function getPages() {
//     axios.get('/footer-
pages').then(function(response) {
//             footerPages.pages =
response.data.data;
//         })
//         .catch(function(error) {

//         });

// };

function dineTypeSwitch(mod) {

    $('.tablepicker').hide();
    $('.takeaway_picker').hide();
    $('.qraddressBox').hide();
    $('#clientBox').hide();


    if (mod == "dinein") {
        $('.tablepicker').show();
        $('.takeaway_picker').hide();
        $('.qraddressBox').hide();
        $('#clientBox').hide();

        //phone
        $('#localorder_phone').hide();
    }

    if (mod == "takeaway") {
        $('.tablepicker').hide();
        $('.takeaway_picker').show();
        $('#clientBox').show();

        $('.c').hide();

        //phone
        $('#localorder_phone').show();
    }
    if (mod == "delivery") {
        $('.tablepicker').hide();
        $('.takeaway_picker').hide();
        $('.qraddressBox').show();
        $('#clientBox').show();

        //phone
        $('#localorder_phone').show();
    }
}
```

```
function orderTypeSwither(mod) {


    $('.delTime').hide();
    $('.picTime').hide();

    if (mod == "pickup") {

updatePrices(cartTotal.totalPrice, null,
false)
        $('.picTime').show();
        $('#addressBox').hide();
    }

    if (mod == "delivery") {
        $('.delTime').show();
        $('#addressBox').show();
        getCartContentAndTotalPrice();
    }
}

setTimeout(function() {
    if (typeof initialOrderType !==
'undefined') {


orderTypeSwither(initialOrderType);
    } else {

    }

}, 1000);

function chageDeliveryCost(deliveryCost)
{
    $("#deliveryCost").val(deliveryCost);
    updatePrices(cartTotal.totalPrice,
deliveryCost, true);

}

//First we beed to capture the event of
chaning of the address
function deliveryAddressSwithcer() {
    $("#addressID").change(function() {
        //The delivery cost
        var deliveryCost =
$(this).find(':selected').data('cost');


        //We now need to pass this cost
to some parrent funct for handling the
delivery cost change
        if (deliveryCost != undefined) {

chageDeliveryCost(deliveryCost);
        }

    });

}

function deliveryAreaSwithcer() {
    $("#delivery_area").change(function()
{
        //The delivery cost
        var deliveryCost =
$(this).find(':selected').data('cost');

        //We now need to pass this cost
to some parrent funct for handling the
delivery cost change
        chageDeliveryCost(deliveryCost);
    });
}

function deliveryTypeSwitcher() {
    $('.picTime').hide();

$('input:radio[name="deliveryType"]').cha
nge(function() {
        orderTypeSwither($(this).val());
    })
        //phone
    $('#localorder_phone').show();
}


function dineTypeSwitcher() {

$('input:radio[name="dineType"]').on('cha
nge', function() {
        $('.delTimeTS').hide();
        $('.picTimeTS').show();
        dineTypeSwitch($(this).val());
    })
}

function paymentTypeSwitcher() {

$('input:radio[name="paymentType"]').chan
ge(

        function() {
            //HIDE ALL
            $('#totalSubmitCOD').hide()

$('#totalSubmitStripe').hide()
            $('#stripe-payment-
form').hide()

            //One for all

$('.payment_form_submiter').hide()


            if ($(this).val() == "cod") {
                //SHOW COD

$('#totalSubmitCOD').show();
            } else if ($(this).val() ==
"stripe") {
                //SHOW STRIPE

$('#totalSubmitStripe').show();
                $('#stripe-payment-
form').show()
            } else {
                $('#' + $(this).val() +
'-payment-form').show()
            }
        });
}

window.onload = function() {
    //VUE CART

    cartContent = new Vue({
```

```
        el: '#cartList',
        data: {
            items: [],
        },
        methods: {
            remove: function(product_id)
{

removeProductIfFromCart(product_id);
            },
            incQuantity:
function(product_id) {
                incCart(product_id)
            },
            decQuantity:
function(product_id) {
                decCart(product_id)
            },
        }
    })
    console.log(cartContent.items)
    //VUE CART Mobile
    cartContentMobile = new Vue({
        el: '#cartListMobile',
        data: {
            items: []
        },
        computed: {
            items: function() {
                return cartContent.items
            }
        },

        methods: {
            remove: function(product_id)
{

removeProductIfFromCart(product_id);
            },
            incQuantity:
function(product_id) {
                incCart(product_id)
            },
            decQuantity:
function(product_id) {
                decCart(product_id)
            },
        }
    })

    //GET PAGES FOR FOOTER
    // getPages();

    //Payment Method switcher
    paymentTypeSwitcher();

    //Delivery type switcher
    deliveryTypeSwitcher();

    //For Dine in / takeout
    dineTypeSwitcher();

    //Activate address switcher
    deliveryAddressSwithcer();

    //Activate deliveryAreaSwithcer
    deliveryAreaSwithcer();
```

```
    // VUE FOOTER PAGES
    // footerPages = new Vue({
    //     el: '#footer-pages',
    //     data: {
    //         pages: []
    //     }
    // })

    // VUE COMPLETE ORDER TOTAL PRICE
    total = new Vue({
        el: '#totalSubmit',
        data: {
            totalPrice: 0
        }
    })

    //VUE TOTAL
    cartTotal = new Vue({
        el: '#totalPrices',
        data: {
            totalPrice: 0,
            deduct: 0,
            deductFormat: "",
            minimalOrder: 0,
            totalPriceFormat: "",
            deliveryPriceFormated: "",
            withDeliveryFormat: "",
            delivery: true,
        }
    })

    //VUE TOTAL Mobile
    cartTotalMobile = new Vue({
        el: '#totalPricesMobile',
        data: {
            totalPrice: 0,
            deduct: 0,
            deductFormat: "",
            minimalOrder: 0,
            totalPriceFormat: "",
            deliveryPriceFormated: "",
            withDeliveryFormat: "",
            delivery: true,
        },
        computed: {
            totalPrice: function() {
                return
cartTotal.totalPrice
            },
            deduct: function() {
                return cartTotal.deduct
            },
            deductFormat: function() {
                return
cartTotal.deductFormat
            },
            minimalOrder: function() {
                return
cartTotal.minimalOrder
            },
            totalPriceFormat: function()
{
                return
cartTotal.totalPriceFormat
            },
            deliveryPriceFormated:
function() {
                return
cartTotal.deliveryPriceFormated
```

```
            },
            withDeliveryFormat:
function() {
                return
cartTotal.withDeliveryFormat
            },
            delivery: function() {
                return cartTotal.delivery
            }
        },
    })

    //Call to get the total price and
items
    getCartContentAndTotalPrice();


    var addToCart1 = new Vue({


        el: '#addToCart1',
        methods: {
            addToCartAct() {


                axios.post('/guest/cart',
{
                        id:
$('#modalID').text(),
                        quantity:
$('#quantity').val(),
                        total:
$('#modalPrice').text(),
                        extras:
extrasSelected,
                        variantID:
variantID
                    })
.then(function(response) {

console.log(response.data.status)
                        if
(response.data.status) {

$('#productModal').modal('hide');

//$('#productModal').modal('close');

getCartContentAndTotalPrice();
                            //if
(TEMPLATE_USED == "defaulttemplate" ||
"luxe-template") {
                            openNav();
                            //}

//$('#productModal').modal('close');
                        } else {

$('#productModal').modal('hide');

//$('#productModal').modal('close');

js.notify(response.data.errMsg,
"warning");
                        }
                    })

.catch(function(error) {

console.log(error)
                    });
            },
        },
    });
}
```

APPENDIX H

Data Elements / Dictionary

Data Dictionary Outlining in Database on Amenities				
Field	**Data type**	**Size**	**Default**	**Description**
id	int	11	NOT NULL	Id of amenities
name	varchar	60	NOT NULL	Name of user
description	varchar	255	NOT NULL	Description of the amenites
for_all	tinyint	4	NOT NULL	For all of the amenities
status	tinyint	4	NOT NULL	Status of the amenities
icon	varchar	50	NOT NULL	Icon of the amenites
created_at	datetime		NOT NULL	Time created
updated_at	datetime		DEFAULT NULL	Time updated
Deleted_at	datetime		DEFAULT NULL	Time deleted

Data Dictionary Outlining in Database on cancellation_reason				
Field	**Data type**	**Size**	**Default**	**Description**
id	int	11	NOT NULL	Id of cancellation reason
name	varchar	100	NOT NULL	Name of cancellation reason
description	text		NOT NULL	Description of cancellation reason

Data Dictionary Outlining in Database on facilities				
Field	**Data type**	**Size**	**Default**	**Description**
id	int	11	NOT NULL	Id of the facilities
type_id	int	11	NOT NULL	Id of the facilities
name	varchar	60	NOT NULL	Name of the facilities
status	tinyint	4	NOT NULL	Status of the facilities

Data Dictionary Outlining in Database on facility_types				
Field	**Data type**	**Size**	**Default**	**Description**
id	int	11	NOT NULL	Id of the facility type
name	varchar	60	NOT NULL	Name of the facility type
image	text		NOT NULL	Image of the facility type

Data Dictionary Outlining in Database on mail_templates				
Field	**Data type**	**Size**	**Default**	**Description**
id	int	10	NOT NULL	Id of the mail templates
name	varchar	255	NOT NULL	Name of the mail template
Subject	varchar	255	NOT NULL	Subject of the mail template
content	text		DEFAULT NULL	Content of the mail template

Data Dictionary Outlining in Database on housekeeping				
Field	**Data type**	**Size**	**Default**	**Description**
Id	int	11	NOT NULL	Id of the housekeeping
Room	int	11	NOT NULL	Room of the housekeeping
Status	tinyint	4	NOT NULL	Status of the housekeeping
remarks	varchar	255	NOT NULL	Remarks of the housekeeping
User_id	Int	11	NOT NULL	User id of the housekeeping
Date_limit	Datetime		NOT NULL	Date limit of housekeeping
Date_finish	Datetime		DEFAULT NULL	Date finish of housekeeping
Deleted_at	Datetime		DEFAULT NULL	Time deleted

Data Dictionary Outlining in Database on news				
Field	**Data type**	**Size**	**Default**	**Description**
id	int	11	NOT NULL	Id of the news
Email	varchar	255	NOT NULL	Email of the news
Status	Tiny iny	4	NOT NULL	Status of the news
Created_at	Datetime		DEFAULT NULL	Time created

Data Dictionary Outlining in Database on orders				
Field	Data type	Size	Default	Description
Id	int	11	NOT NULL	Id of user
Order_no	varchar	255	NOT NULL	Order number of the orders
User_id	int	10	NOT NULL	User id of the orders
Room_type_id	int	10	NOT NULL	Room type of orders
Room_id	Int	11	NOT NULL	Room id of the orders
Check_in	Date		NOT NULL	Check in of orders
Check_out	Date		NOT NULL	Check out of orders
Adults	tinyint	3	NOT NULL	Adults of the orders
Kid	tinyint	3	NOT NULL	Kid of the orders
Request	varchar	255	NOT NULL	Request of the orders
Status	tinyint	1	NOT NULL	Status of the orders
Mode_payment	tinyint	1	NOT NULL	Mode payment of orders
Payment_proof	Varchar	255	NOT NULL	Payment proof of orders
Payment_ref	varchar	255	NOT NULL	Payment ref of orders
Payment_status	tinyint	1	NOT NULL	Payment status of orders
Additional_person	Tiny	4	NOT NULL	Additional person of orders
Additional_person_amount	decimal	10,2	NOT NULL	Additional Amount of orders

amount	Decimal	10,2	NOT NULL	Amount of orders
Taxamount	Decimal	10,2	NOT NULL	Tax amount of orders
Paid_service_id	varchar	255	NOT NULL	Paid service id of orders
Paid_service_amount	Decimal	10,2	DEFAULT NULL	Paid service amount of orders
Totalamount	decimal	10,2	NOT NULL	Total amount of orders
Advance_amount	Decimal	10,2	NOT NULL	Advance amount of orders
nights	tinyint	4	NOT NULL	Night of orders
Is_new	Tinyint	1	NOT NULL	new of orders
Is_cancel_by_guest	tinyint	1	NOT NULL	Cancel by guest of orders
Reason_id	Tinyint	1	NOT NULL	Reason id of orders
Feedback	varchar	255	NOT NULL	Feedback of orders
Created_at	datetime		NOT NULL	Time created
Updated_at	datetime		DEFAULT NULL	Time updated

Data Dictionary Outlining in Database on orders_services				
Field	**Data type**	**Size**	**Default**	**Description**
Order_id	Int	10	NOT NULL	Order id of orders services
Service_id	int	10	NOT NULL	Service id of orders services
amount	double		NOT NULL	Amount of the orders services

Data Dictionary Outlining in Database on orders_taxes				
Field	**Data type**	**Size**	**Default**	**Description**
Order_id	Int	10	NOT NULL	Order id of orders taxes
tax_id	int	10	NOT NULL	Tax id of orders taxes
amount	double		NOT NULL	Amount of order taxes

Data Dictionary Outlining in Database on paid_services				
Field	**Data type**	**Size**	**Default**	**Description**
Id	Int	11	NOT NULL	Id of paid services
Title	varchar	120	NOT NULL	Title of paid services
description	Varchar	255	NOT NULL	Description of paid services
Price	double		NOT NULL	Price of paid services
status	tinyint	4	NOT NULL	Status of paid services

Data Dictionary Outlining in Database on photos				
Field	**Data type**	**Size**	**Default**	**Description**
Id	Int	11	NOT NULL	Id of photos
Category	varchar	50	NOT NULL	Category of photos
image	Text		NOT NULL	Image of photos
Description	varchar	255	NOT NULL	Description of photos

Data Dictionary Outlining in Database on payment				
Field	Data type	Size	Default	Description
Id	Int	10	NOT NULL	Id of payment
Order_id	int	10	NOT NULL	Order id of payment
Date_time	Datetime	255	DEFAULT NULL	Date time
Amount	Decimal	10,2	NOT NULL	Amount of payment
invoice	int	10	NOT NULL	Invoice of payment
Is_main_amount	Tinyint	1	NOT NULL	Main amount of payment
Payment_method	varchar	255	DEFAULT NULL	Payment method of payment
Created_at	Date	4	DEFAULT NULL	Time created

Data Dictionary Outlining in Database on prices				
Field	Data type	Size	Default	Description
Room_type_id	Int	11	NOT NULL	Id of prices
Mon	Double	10	NOT NULL	Monday prices
Tue	Double		NOT NULL	Tuesday prices
Wed	Double		NOT NULL	Wednesday prices
Thu	Double		NOT NULL	Thursday prices
Fri	Double		NOT NULL	Friday prices

Sat	double		DEFAULT NULL	Saturday prices
sun	Double		DEFAULT NULL	Sunday prices
Created_at	datetime		DEFAULT NULL	Time created
Updated_at	Datetime		DEFAULT NULL	Time updated
Deleted_at	Datetime		DEFAULT NULL	Time deleted

Data Dictionary Outlining in Database on profiles				
Field	Data type	Size	Default	Description
id	Int	11	NOT NULL	Id of profiles
Firstname	varchar	50	NOT NULL	Firstname of profile
Lastname	varchar	50	NOT NULL	Lastname of profiles
User_id	Int	11	NOT NULL	User id of profiles
Email_sub	varchar	255	DEFAULT NULL	Email sub of profiles
Birthday	datetime		NOT NULL	Birthday of profiles
Gender	varchar	30	NOT NULL	Gender of profiles
Citizenship	varchar	50	NOT NULL	Citizenship of profiles
Contact	varchar	20	NOT NULL	Contact of profiles
Zip	varchar	20	NOT NULL	Zip of profiles
Region	varchar	50	NOT NULL	Region of profiles

Province	varchar	60	NOT NULL	Province of profiles
City	varchar	100	NOT NULL	City of profiles
Barangay	varchar	50	NOT NULL	Barangay of profiles
Street	varchar	50	NOT NULL	Street of profiles
Landmark	Varchar	60	NOT NULL	Landmark of profiles
avatar	varchar	255	NOT NULL	Avatar of profiles
Data Dictionary Outlining in Database on policies				
Field	Data type	Size	Default	Description
id	Int	11	NOT NULL	Id of policies
policy	text		NOT NULL	Policies
Data Dictionary Outlining in Database on promos				
Field	Data type	Size	Default	Description
id	Int	11	NOT NULL	Id of promos
Roomtype_id	int	11	NOT NULL	Room type of promos
Season	varchar	50	NOT NULL	Season of promos
Name	Varchar	30	NOT NULL	Name of promos
Description	varchar	255	NOT NULL	Description of promos
Discount	int	11	DEFAULT	Discount of promos

videoLink	text		NOT NULL	Videolink of promos
Thumbnail	text		NOT NULL	Thumbnail of promos
Expiration	Datetime		DEFAULT NULL	Expiration of promos
Status	tinyint	4	NOT NULL	Status of promos
Created_at	Datetime		NOT NULL	Time created
Updated_at	Datetime		DEFAULT NULL	Time updated
Data Dictionary Outlining in Database on reviews				
Field	Data type	Size	Default	Description
id	Int	11	NOT NULL	Id of reviews
service_id	int	11	NOT NULL	Service id of reviews
User_id	int	11	NOT NULL	User id of reviews
Item_id	Int	11	NOT NULL	Item id of reviews
Rate	Int	11	NOT NULL	Rate of reviews
Value_money_rate	tinyint	4	DEFAULT	Value money of reviews
Service_rate	tinyint	4	NOT NULL	Service rate of reviews
Content	Varchar	50	NOT NULL	Content of reviews
Comment	Text		DEFAULT NULL	Comment of reviews
Created_at	datetime		NOT NULL	Time created

Data Dictionary Outlining in Database on process				
Field	**Data type**	**Size**	**Default**	**Description**
id	Int	11	NOT NULL	Id of process
name	varchar	50	NOT NULL	Name of process
Data Dictionary Outlining in Database on rooms				
Field	**Data type**	**Size**	**Default**	**Description**
id	Int	11	NOT NULL	Id of rooms
type_id	int	11	NOT NULL	Type id of rooms
roomNumber	int	11	NOT NULL	Room number of rooms
Floor	Varchar	20	NOT NULL	Floor of rooms
Image	text		NOT NULL	Image of rooms
Status	int	11	NOT NULL	Status of rooms
Created_at	Datetime		DEFAULT NULL	Time created
Deleted_at	Datetime		DEFAULT NULL	Time deleted
Updated_at	varchar	20	DEFAULT NULL	Time updated
Data Dictionary Outlining in Database on settings				
Field	**Data type**	**Size**	**Default**	**Description**
id	Int	10	NOT NULL	Id of settings
Name	varchar	255	NOT NULL	Name of settings

Logo	Varchar	255	NOT NULL	Logo of settings
Address	Text		NOT NULL	Address of settings
Gmap_link	text		NOT NULL	Gmap link of settings
Email	varchar	255	NOT NULL	Email of settings
Phone	varchar	255	NOT NULL	Phone of settings
Timezone	varchar	255	NOT NULL	Timezone of settings
Check_in_time	Time		NOT NULL	Time check in
Check_out_time	time		NOT NULL	Time check out
Room block start date	Date		DEFAULT NULL	Start date of room block
Room block end date	Date		DEFAULT NULL	End date of room block
Maintenance mode	tinyint	1	NOT NULL	Maintenance mode of settings
Maintenance message	Text		DEFAULT NULL	Maintenance message of settings
Facebook_link	Text		DEFAULT NULL	Link of facebook
Footer_text	Text		DEFAULT NULL	Text of footer
General_police s	Text		DEFAULT NULL	General policies of settings
Cancellation_ Policy	Text		DEFAULT NULL	Cancellation policy of settings

Meta_keywords	Text		DEFAULT NULL	Meta keyword of settings
Meta_description	Text		DEFAULT NULL	Meta description of settings
Data Dictionary Outlining in Database on users				
Field	Data type	Size	Default	Description
id	Int	11	NOT NULL	Id of users
Email	varchar	100	NOT NULL	Type id of users
Password	Text		NOT NULL	Password of users
Usertype	Varchar	30	NOT NULL	User type of users
Status	tinyint	4	NOT NULL	Status of users
Created_at	Datetime		DEFAULT NULL	Time created
Deleted_at	Datetime		DEFAULT NULL	Time deleted
Updated_at	varchar	20	DEFAULT NULL	Time updated

APPENDIX I

Research/Capstone Project Evaluation Form

Survey Questionnaire

Title : QRIFY: A CLOUD-BASED HOTEL AND RESTAURANT MANAGEMENT SYSTEM FOR ANAHAW ISLAND VIEW RESORT

Direction: Respond to the following statement with regards to the proposed system for evaluation. Put check (/) in the column that corresponds to your answer using the scale below.

Name (Optional): ____

Respondent: ☐ Admin ☐ Staff ☐ Customer ☐ IT Expert

RATING

5 = Excellent **4** = Very Good **3** = Satisfactory **2** = Fair **1** = Poor

1. Functional Sustainability	5	4	3	2	1
1.1 How would you rate the system's ability to manage hotel and restaurant operations, such as reservations, check-ins, and order processing.					
1.2 The systems function delivers the intended result with its accuracy.					
1.3 The system features able to accomplish specified task and objectives.					
2. Reliability	5	4	3	2	1
2.1 How reliable is the system in terms of data accuracy and availability, especially during peak usage times.					
2.2 The system is operational and accessible whenever needs to use.					
3. Portability	5	4	3	2	1
3.1 Suitable for use on any device or platform, easily accessible.					
3.2 The system adapts well to varying network conditions, including slow or unstable connections.					

3.3 The system can be utilized with modest hardware specifications.					
4. Usability	5	4	3	2	1
4.1 Users can recognize whether the system is appropriate for their needs.					
4.2 The system can be used by specified users to achieve specified goals of learning to use the system with effectiveness, efficiency, freedom from risk and satisfaction in a specified context of use.					
4.3 The system is easy to operate and control.					
4.4 The system's interface and design prevent users from making errors.					
4.5 The system enables pleasing and satisfying user interface.					
4.6 The system can be use by all people, including those with disabilities or impairments.					
5. Performance Efficiency	5	4	3	2	1
5.1 The system manages hotel and restaurant operations, in terms of speed and resource utilization.					
5.2 Under specific conditions, the system exhibits satisfactory performance while managing resource utilization.					
5.3 The maximum limits of the system parameter meet its requirements.					
6. Security	5	4	3	2	1
6.1 How confident are you in the system's ability to protect sensitive data and maintain the privacy of users?					
6.2 The system authenticates both the identity of data and its origins.					
7. Compatibility	5	4	3	2	1
7.1 The system efficiently executes its functions without causing disruptions to other systems.					
7.2 System seamlessly work with other existing software or hardware systems within the hotel and restaurant environment.					

8. Maintainability	5	4	3	2	1
8.1 The system easy is to update, maintain, and make necessary changes to the system.					
8.2 System changes, error fixes, and environmental failures requires less effort.					
8.3 It is feasible to evaluate the effect on a product or system of a planned alteration to one or more of its components, or to identify issues or causes of failures in a product, or to pinpoint components that need to need to be altered.					
8.4 The system can be improved effectively and efficiently without causing new defects or reducing the current quality of the system.					
8.5 Test standards can be set for a system, product, or component, and evaluations can be conducted to assess if those standards have been met.					

Recommendations/Comments:

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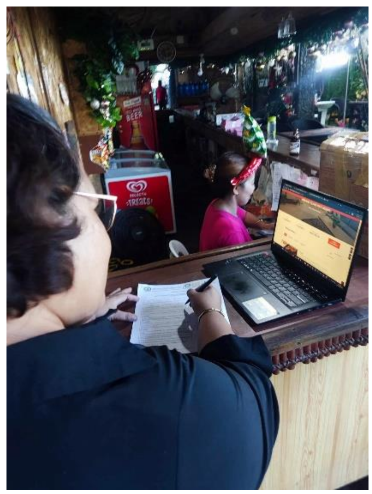
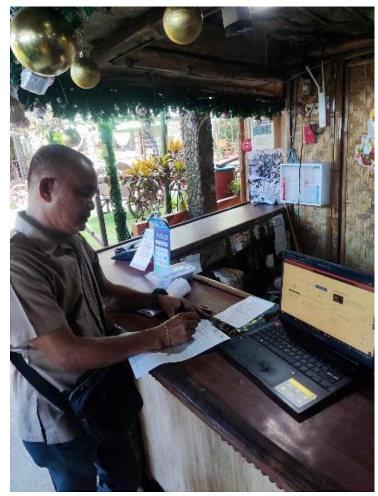
Signature

APPENDIX J

Pictures During Development

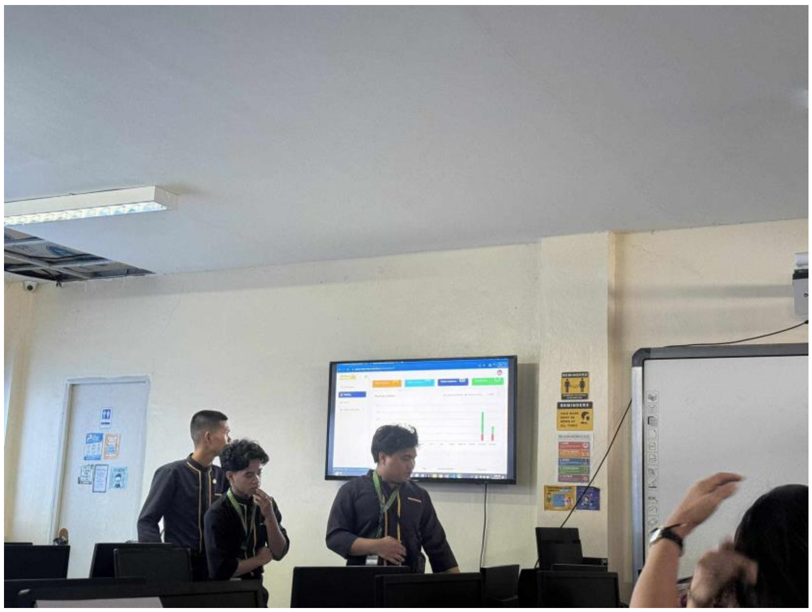
The photos below are taken during the deployment of the QRIFY.

APPENDIX K

Pictures During Final Defense

The photos below are taken during the capstone final defense.



APPENDIX L

Research/Capstone Project Waiver

RESEARCH/CAPSTONE PROJECT WAIVER

We, **James Samuel B. Ducusin, Arvin M. Divinagracia** and **Joshua S. Mercene**, of legal ages, the sole proponent of the Research/Capstone Project entitled **"QRIFY: A CLOUD-BASED HOTEL AND RESTAURANT MANAGEMENT SYSTEM FOR ANAHAW ISLAND VIEW RESORT"**, hereby authorize our Adviser, Course Facilitator and the College of Computer Studies – Research Unit to submit our Research/Capstone Project for paper presentation, publication and funding.

Whereas, if the research project will be accepted in any form of publication or presentation, the proponents shall be assisted by their Adviser, Course Facilitator and/or CCS Research Unit Head. However, if the sole proponents unable to attend, the Adviser, Course Facilitator and/or CCS Research Unit Head may represent the group.

Whereas, the Adviser, Course Facilitator and/or CCS Research Unit Head may improve the content of the manuscript in accordance with the requirements set forth by the conference and forum organizers. If improvement had been made by the Adviser, Course Facilitator and/or CCS Research Unit Head, they will be apart of the research group as Co-Authors. The original proponents shall be notified and included as one of the proponents of the Research/Capstone Project.

Whereas, our actions and decisions will be beneficial/advantageous for all of us, to the College of Computer Studies and its respective programs.

JAMES SAMUEL B. DUCUSIN
Research Proponent

Date: __________

ARVIN M. DIVINAGRACIA
Research Proponent

Date: __________

JOSHUA S. MERCENE
Research Proponent

Date: __________

APPENDIX M

Certificate of Oral Research Presentation

APPENDIX N

ACM Format

QRIFY: A CLOUD-BASED HOTEL AND RESTAURANT MANAGEMENT SYSTEM FOR ANAHAW ISLAND VIEW RESORT

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ABSTRACT

This research paper focused on the development of a cloud-based hotel and restaurant management system for Anahaw Island View Resort. The system's features included implementing an online platform for reservation, managing business transactions, and providing services for the resort in terms of accommodation, hotel and restaurant management, and other services. It also provide efficient and reliable services, and enhance the overall guest experience for the resort. The researchers used a descriptive and developmental method to fulfill the project. An SDLC Agile model was used to ensure its capability to achieve the aim of the project. The ISO 25010 standard was also used to evaluate the software's overall functionality.

Furthermore, findings showed that the use of the website greatly supported the resort's day-to-day transactions. The resort owner, staff, IT experts, and customers rated the project with an overall mean of 3.99 (Very Good), indicating that it greatly helped the users and met their expectations and requirements.

Keywords — cloud-based, QR-code technology, online reservation, food ordering system, e-reservation

1. INTRODUCTION

1.1 PROJECT CONTEXT

The primary objective of the hotel and restaurant management system was to ensure streamlined operations and enhance guest experience through an online web application. This system provided efficient and reliable services to customers, enabling them to book rooms and tables, view menus, place orders, and access information about the hotel and restaurant's amenities (Leung, R., 2019).

The website's overall quality played a crucial role in delivering the intended message to potential customers. The website's usability was also a significant factor that provided a better user experience, ensuring that users could easily navigate and use the website's features. The website was designed to be user-friendly, interactive, and accessible to all ages and genders (Pathak, et al., 2021).

The hotel and restaurant management system allowed users to interact with the website by creating, modifying, and making changes anytime needed. This meant the system was not static but provided real-time services, ensuring customer satisfaction (Buhalis, et al., 2019).

The development of the hotel and restaurant management system was systematic and relative to current technological advancements. The system's structure was based on its intended purpose: streamline operations and enhance guest experience (Law, et al., 2019).

This project aimed to create an online web application as a one-stop shop for all hotel and restaurant services. The system provided various features such as online reservations, menu viewing, order placement, and access to hotel and restaurant amenities. The system aimed to streamline operations, provide efficient and reliable services, and enhance the overall guest experience.

2. REVIEW OF RELATED LITERATURE/SYSTEM

According to Man et al. (2021); and Prasetyo et al. (2020), the more determined someone is to use a QR code, the more probable they will do so. QR codes are frequently used if the program offers a variety of benefits. Evidence from earlier studies suggests that behavioral intention to use can spur actual action.

Zhang et al. (2020) claim that patients who believe their COVID-19 condition is severe and vulnerable have a favorable attitude toward adopting technology. According to these results, Covid 19 can be harmful to a person's health by causing symptoms like fever, colds, coughing, asthma, and even death. Customers adopt an adaptive behavior by scanning a QR code to examine the menu to reduce the risk of transmission. A person can avoid transmitting Covid 19 since the app can lessen direct contact for patrons at eateries where patrons can virtually order menus without queuing.

Prasetyo et al. (2020) incorporated PMT and the Theory of Planned Behavior (TPB) to examine the efficacy of Covid 19 prevention measures. The result shows how perceptions of a person's fragility and seriousness can indirectly influence behavioral intentions and result in actual behavior. In Indonesia, in particular, no research has ever been done on the relationship between PMT and TPB in applying QR codes in the restaurant industry. Therefore, this study aims to evaluate consumer perceptions of the use of QR codes by integrating PMT and TPB theories to shed light on the restaurant industry, particularly in Indonesia.

Zhao and Bacao (2020), by fusing the UTAUT with the task-technology fit model and the expectancy confirmation model, extended the UTAUT in the setting of OFDS. The study findings demonstrated that performance expectancy, social influence, trust, and perceived task-technology fit were significant predictors of the continuous intention to use OFDS. Specifically, trust, perceived task-technology fit, and satisfaction were added to the original UTAUT. The three main factors from UTAUT, namely performance expectancy, effort expectancy, and social influence, are used in this study along with two additional constructs (i.e., trust and food safety risk perception) to improve the prediction of the factors influencing the intention to purchase OFDS.

Liang (2020) that hotel restaurants have been under pressure to draw in and keep guests due to intense competition, decreased expenditure per person, and rising prices. A global public health emergency's unexpected shock has worsened the situation. Non-essential human mobility and business activity were halted in China from late January to April 2020 due to the COVID-19 epidemic.

Parulis-Cook (2020), for opulent hotels and their restaurants, survival has taken precedence amid the pandemic. Offering an ordering and home delivery service with online-to-offline (O2O) food delivery companies like Meituan and Eleme—two O2O services heavyweights in China—has been one of the tactics used by luxury hotel restaurants in China to stay alive.

In the study conducted by Polizzi et al. (2020), to the best of our knowledge, no study has looked at upscale eating options in an O2O setting. Additionally, people's attitudes and behaviors have changed due to the COVID-19 outbreak. It is vital to look at O2O dining experiences with upscale hotel restaurants during the pandemic in light of the issues posed by these inquiries.

Ma and Hsiao (2020) state that these environmental signals are especially crucial for luxury restaurants since they promote favorable feelings in patrons and have a greater influence on patrons' behavioral intentions. Most researchers concur that ambient conditions, facility aesthetics, spatial arrangement, and seating comfort are all sub-dimensions of physical environment quality.

According to Cho et al., (2020), in a restaurant where illnesses like coughs and colds may be common, Covid 19 is highly contagious when in direct contact with another person. Exposure to COVID-19 results in a stronger impression of threat, which suggests that adaptive behavior is more likely.

According to Ong et al. (2020); Wu, 2020; Yang et al., (2019), scan QR codes to stop transmission when looking at menus. Numerous studies have shown that a person's attitude will be more upbeat the more serious something is perceived.

According to Kocaman (2020), more people are likely to visit because of the application's excellent performance in reducing direct contact and lowering the risk of transmission. Supply chain workers claim that by enhancing this service system, it is possible to raise sales and service quality standards.

Köhler and Pizzol (2020), point out that blockchain is a component of a larger system of technologies rather than a standalone technology and say that further research is necessary to determine whether blockchain-based technologies will improve food supply networks' sustainability.

According to Lee et al., 2019; and Zhao & Bacao (2020), recent studies in the OFDS literature have shown no significant correlation between effort expectancy and the continuous usage intention of OFDS, indicating that customers experience little difficulty using new apps, including OFDS, as the use of smartphones and apps has matured. The interface of smartphone apps has stabilized due to the development of information and communication technologies over time.

According to Ray et al. (2020) and Yeo et al. (2019), many clients are driven to O2O by lower pricing, as the platforms make it easier for price comparison, is another negative effect of using O2O meal delivery services. This is not good for upscale restaurants. This raises the question of whether upscale hotel restaurants and online marketplaces for food delivery can coexist harmoniously. Further difficulties affect this, such as whether upscale restaurants on O2O platforms target the wrong customers and whether the challenges brought on by the platforms negatively impact the restaurants, which then harms the restaurants' reputations and the hotels' reputations.

According to Pillai et al. (2019), "QR codes" are posted in the Kempinski Hotel in Ajman rooms so that visitors can place meal orders in four different languages. The kitchen receives these orders right away. Hotels like the Fontainebleau Hotel Miami Beach, the Mirage Hotel and Casino, and the Best Western Hotel Le

Montparnasse utilize "QR codes" as well. Using "QR codes" that contain PDFs with extensive descriptions of the amenities found in hotel rooms can enhance the guest experience.

Li et al. (2019), due to the maturity of the field and the requirement to keep track of its development, continual efforts to examine and assess the body of hospitality research are especially crucial. Despite the significance of the restaurant industry to the hospitality sector, the bibliometric approach has not yet been used in a study in this area. By addressing the need for more information on the state of the art in restaurants and illustrating its thematic evolution, the current study aims to close this gap in the academic literature. The study uses a bibliometric approach using co-word analysis and science mapping to analyze the scientific literature on restaurants in-depth, focusing on the years 2000–2019. It examines the scientific articles on restaurants included in the "Hospitality, Leisure, Sports, and Tourism" category of the Web of Science (WoS) citation index.

Ray et al. (2019) state that online food delivery services (OFDS) connect customers with partner restaurants via their websites or mobile applications for meal ordering and delivery. The COVID-19 pandemic, proclaimed in March 2020, increased the OFDS market's quick expansion (EHL Insights, n.d.). The OFDS market has been expanding significantly over the previous few years.

According to Lilyquist (2019), "QR codes" are a useful tool for improving interactive marketing and streamlining various operational tasks. With interactive marketing, businesses hope to increase customer and business interaction, increasing demand for their goods and services. Interactive marketing is also known as trigger-based marketing, which occurs when a consumer activity triggers the addition of more marketing initiatives.

In the study conducted by Kim et al.and Maimaiti et al. (2019), OFDS has helped restaurants by giving them a new way to serve customers. However, because OFDS adds delivery to the standard restaurant service process, there are potential risks for restaurant owners and clients regarding temperature control during delivery, delivery drivers' hygiene, and even food tampers. In addition, because customers using online/mobile platforms frequently struggle with interface design, communication speed, privacy, and security of the service interfaces, including payment processors, OFDS customers are not free.

In the study conducted by Roh & Park (2019 and Troise et al. (2020), research explained that customers are more likely to believe that adopting new technology will boost their life's productivity the greater their effort expectations are. Most researchers who conducted OFDS-related studies discovered that once customers experience ease in using OFDS, they tend to consider the service useful. A recent study in the context of OFDS argued that ease of OFDS no longer had a meaningful impact on discovering the service's usefulness in case of repeated usage.

According to Nguyen et al. (2019), specifically, it was found that customers' trust in an online food shopping website influences their intention to use the site favorably. Similarly, one of the main factors influencing the continued use of online banking services is trust in the service. Customers are more likely to use OFDS when they think the service will go smoothly, according to existing OFDS studies.

According to Xiao et al. (2019), it is a growing e-commerce paradigm that combines offline consumption of goods and services with online marketing, sales, and assessment. O2O systems have become increasingly important in various consumer life scenarios, particularly for locally focused life services items like food delivery, tickets, and auto rentals.

According to Kapoor and Vij (2019); and Cho et al. (2019), the Stream of research focuses on understanding the variables that affect consumers' opinions of food delivery services. It has been proven that app features like convenience, design, dependability, and the diversity of food options have a significant impact on how valuable a meal delivery app is, which increases the likelihood that users would use the service.

According to Zhang et al. (2019) and He et al. (2019), the quantity of reviews and total ratings is especially important for high-sale restaurants to boost sales volume, whereas the delivery service effect is more significant for low-sale restaurants. The adaptive behaviors of restaurants have also been investigated using an agent-based methodology, with the results showing that a crucial element in boosting sales volume is the timely modification of food quality to account for changes in consumer preferences.

Suhartanto et al. (2019), food quality, including presentation of flavors, variety, and accessibility of healthy options, attitude of the delivery team, order conformance, packaging, and food condition upon receipt, as well as delivery costs, have a significant impact on customer loyalty.

Kiatkawsin and Han (2019), some comments also revealed that when the restaurants failed to deliver as expected, their brand credibility was damaged. Many customers complained about the ordinariness of the food, noting that the taste was just "too normal and not distinctive enough." When there are discounts or special offers, these criticisms become more serious. A probable explanation is that lowering the price is viewed as a problematic promotion because a high price is one of the defining characteristics of upscale dining establishments.

Gagarin (2019), restaurant staff members from Spain, Germany, France, Italy, and Japan range from managers and owners to chefs. Employees from a variety of various food establishments, including fully functional restaurants, cafes/bars, fast food restaurants, canteens, and even commercial kitchens that solely serve take-out ready meals, were questioned at the same time.

According to ProHotelia (2019), the Henn administration thinks that some technologies are developing too quickly and others are becoming obsolete. For almost four years, the hotel has employed many robots; during that period, several technologies have "gone forward," like Siri or Google Assistant. The hotel administration understood that the complex still needs people but did not abandon the idea of an autonomous hotel.

Yang et al. (2019) that people are deterred from using QR codes if their use is hard and full of difficulties. They will, however, accept the technology if mastering QR codes doesn't take a lot of time, effort, or money. The results showed that people are less likely to adopt new technologies when the response cost increases.

3. **METHODOLOGY**

3.1 System Architecture

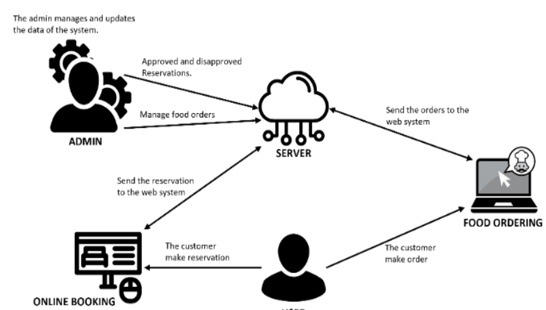

Figure 1. System Architecture

This figure displayed the system architecture of the Hotel and Restaurant Management System. It illustrated the flow of the system, where the server or system was the center of all transactions. The customer used the system for reservations and ordering food. On the other hand, the admin managed the customer's reservations and orders, being responsible for all the data saved in the system.

3.2 Use Case

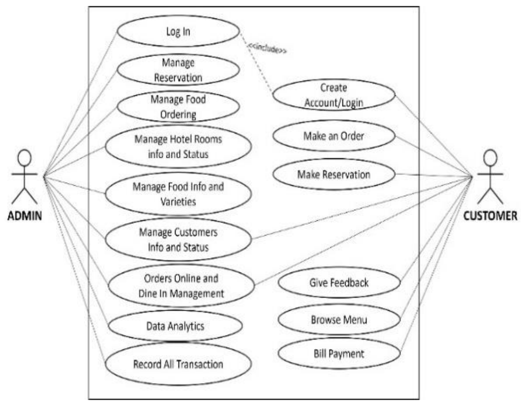

Figure 2. Use Case of the System

This figure illustrated the process to be done by the users in the developed website. It presented the users' perception of the functionality provided by the proponents of the project.

3.3 Agile Model

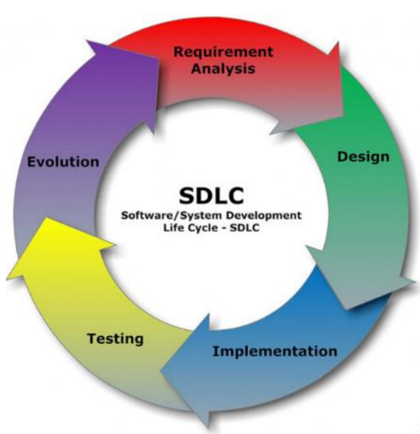

Figure 3. Agile Model

The sequential phases in Agile model are:

1. **Requirement Analysis.** In the requirements phase of developing QRify, an agile approach was adopted to meticulously plan and research the objectives of creating a cloud-based hotel and restaurant management system for Anahaw Island View Resort. Project prioritization was a key focus during this stage, involving strategic planning executed through stakeholder meetings, requirements collection, and market research. To achieve a comprehensive understanding of the system's scope, identification of key stakeholders was crucial, and their needs and potential challenges were nuancedly explored. The outcome of this phase was a high-level project plan that outlined tasks, timelines, and resource requirements. Moving into the Analysis phase, a deeper exploration of requirements unfolded. This encompassed a thorough physical analysis of tangible elements, logical analysis defining the system's structure and data models, and architectural considerations outlining the high-level system architecture. Technologies, frameworks, and platforms were prioritized to align with the overarching goals of creating a cloud-based management system for Anahaw Island View Resort. The process ensured that QRify would be finely tuned to meet the specific needs and aspirations of the resort.

2. **Design.** In the design stage of QRify, we created a user-friendly graphical interface for Anahaw Island View Resort's cloud-based hotel and restaurant management system. This involved two key phases: UI Concept and UI Designing. In the UI Concept stage, we focused on refining project goals, ensuring alignment with the resort's requirements, and defining specific user interactions and aesthetics. Visual aids like wireframes were prepared to facilitate effective communication. Moving to UI Designing, an iterative process was adopted for collaboration among designers, developers, and stakeholders. The goal was to prioritize usability, consistency, and a cohesive user experience. Interactive prototypes were crafted for stakeholder feedback, ensuring QRify's interface met the resort's unique needs.

3. **Implementation.** In the implementation stage of QRify, the researcher executed the development of a cloud-based hotel and restaurant management system for Anahaw Island View

Resort. This phase encompassed coding the front-end user interface, writing back-end code, and integrating third-party APIs. The implementation phase was structured into smaller tasks or sprints, each focused on delivering specific features. Agile methodologies were embraced, ensuring adaptability and regular feedback loops. Collaboration tools like Jira facilitated task management, while version control systems like Git ensured seamless code integration. This iterative approach allowed for continuous refinement, addressing issues promptly and aligning the implementation with the evolving needs of Anahaw Island View Resort. The result was a system that met the specific requirements of the resort's hotel and restaurant operations, providing a tailored and efficient solution.

4. **Testing.** In the testing stage of QRify, our focus was on ensuring the reliability and performance of the cloud-based hotel and restaurant management system designed for Anahaw Island View Resort. Various testing methodologies were employed, including functional, performance, and security testing. Collaborating closely with QA specialists, we executed functional tests to verify that QRify's features met the specified requirements. Performance testing tools like JMeter were employed to assess system responsiveness and scalability. Security testing addressed potential vulnerabilities, ensuring data protection. An iterative testing approach allowed for prompt identification and resolution of issues. Regular feedback loops with stakeholders ensured QRify's testing phase aligned with the resort's expectations. The result was a thoroughly tested system, meeting high-quality standards and ready to deliver a secure and seamless experience for Anahaw Island View Resort.

5. **Evolution.** In the evolution stage of QRify, our team continues to enhance and refine the cloud-based hotel and restaurant management system to meet the evolving needs of Anahaw Island View Resort. This phase involves incorporating user feedback, fixing bugs, and adding new features to ensure the system remains up-to-date and aligned with the latest industry standards. Regular feedback loops with resort stakeholders help identify areas for improvement and guide the development team in prioritizing enhancements. The agile development methodology is maintained, allowing flexibility and responsiveness to changing requirements. The evolution phase is about fixing issues and actively seeking ways to enhance QRify's functionality. New features and improvements are planned and implemented, ensuring the system grows in tandem with the resort's operational requirements. This ongoing process ensures QRify remains a dynamic and efficient solution for Anahaw Island View Resort's hotel and restaurant management needs.

4. RESULTS AND DISCUSSION

4.1 Result and Discussion

In the implementation plan of the website, an evaluation was conducted to test the quality standard of the website using ISO 25010 in terms of functional suitability, reliability, performance efficiency, usability, security, compatibility, maintainability, and portability. This was evaluated by IT experts, admin, staff, and customers.

4.2. Summary Results of the Evaluation

Items	Mean	Rank	Verbal Interpretation
1. Functional Suitability	3.97	5.5	Very Good
2. Reliability	3.85	8	Very Good
3. Portability	4.07	1	Very Good
4. Usability	3.97	5.5	Very Good
5. Performance Efficiency	3.94	7	Very Good
6. Security	4.04	2.5	Very Good
7. Compatibility	4.01	4	Very Good
8. Maintainability	4.04	2.5	Very Good
Overall Mean	**3.99**		**Very Good**

Table 1. Summary Results of the Evaluation

The table displayed the results of the website evaluation, along with compliance with ISO

25010 standards as determined by the admin, the staff, the customers, and the IT expert. The obtained mean of the following characteristics in terms of functional suitability was (M=3.97), reliability (M=3.85), portability (M=3.07), usability (M=3.97), performance efficiency (M=3.94), security (M=4.04), compatibility (M=4.01), and maintainability (M=4.04) with an overall mean of 3.99, indicating that users found the system effective to use.

5. SUMMARY, CONCLUSION AND RECOMMENDATION

5.1 Summary

The study aimed to develop Qrify: A Cloud-Based Hotel and Restaurant Management System for Anahaw Island View Resort. The researchers focused on the design, user-friendly user interface, and technical innovation material that supported and complemented the project's ability to function and meet the needs and expectations of users.

The data collected from every respondent was used to efficiently evaluate the system's functional suitability. It operated without displaying any unusual errors and functioned in accordance with user requirements. The system was beneficial and functional, according to the data acquired, suggesting that the website could continue to operate and manage updated errors.

The website was evaluated in compliance with the ISO 25010 standards. The system was evaluated by 50 respondents, including the IT Experts, Staff, Admin, Client, and Customers. The system's performance was thoroughly tested and evaluated using ISO25010, and it was rated based on specific criteria. The evaluation results demonstrated the system's effectiveness, as it operated in accordance with the proposed and suggested features. Functional Suitability got an overall mean of 3.97, which was described as “Very Good”, Reliability with an overall mean of 3.85, which was described as “Very Good”, Portability with an overall mean of 4.07, which was described as “Very Good”, Usability with an overall mean of 3.97, which was described as “Very Good”, Performance Efficiency with an overall mean of 3.94, which was described as “Very Good”, Security with an overall mean of 4.04, which was described as “Very Good”, Compatibility with an overall mean of 4.01, which was described as “Very Good”, and lastly, Maintainability with an overall mean of 4.04, which was described as “Very Good”.

In general, the Qrify system proved effective in managing hotel and restaurant operations, as evidenced by the acquired data. It demonstrated significant assistance to users in disseminating information and efficiently tracking documents, empowering them to manage information seamlessly.

5.2 Conclusions

Upon the culmination of the study on Qrify: A Cloud-Based Hotel and Restaurant Management System for Anahaw Island View Resort, several key conclusions and observations were identified:

1. The successful development of a user-friendly web application with these features is expected to enhance efficiency and convenience for both staff and guests significantly.

2. Implementing personalized services is anticipated to lead to an enhanced overall guest experience, potentially resulting in increased guest satisfaction and loyalty.

3. Integrating various departments is expected to streamline operations, fostering better communication and coordination. This could reduce errors, improve workflow, and increase overall operational efficiency.

4. Assessing the web application's effectiveness in QRify operations and enhancing the guest experience will provide valuable insights. Positive outcomes, such as improved staff productivity, faster service delivery, and positive guest feedback, suggest the system's success in meeting its objectives.

5.3 Recommendations

1. For future researchers, they should include a notification to the admin dashboard to alert staff about newly received data.

2. For future researchers, a more attractive and responsive website is recommended.

3. Future researchers in real-time chat systems should enhance user experience, prioritize security measures, and explore innovative features like advanced encryption.

4. The owner is advised to secure a reliable internet connection to support the system's optimal performance.

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APPENDIX O

Curriculum Vitae

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